

Cover Images



FRONT COVER: A U.S. Army paratrooper assigned 2nd Battalion, 503rd Infantry Regiment, 173rd Airborne Brigade fires towards the enemy while conducting a joint, chemical, biological, radiological and nuclear (CBRN) training in Pabradė, Military Operation Urban Terrain (MOUT), with French soldiers, Lithuania, May 21, 2025, during exercise Swift Response 2025, as a part of the broader Defender Europe 25 framework. The training highlights the high level of cooperation, readiness, and interoperability among allied forces operating in complex urban environments. (U.S. Army photo by Sgt. Jose Lora)



FRONT LEFT INSET: An unarmed Minuteman III Intercontinental Ballistic Missile launches during an operational test at 12:01 a.m. Pacific Time, May 21, 2025, at Vandenberg Space Force Base, Calif. ICBM test launches demonstrate that the U.S. ICBM fleet is ready, reliable and effective in leveraging dominance in an era of strategic competition. (U.S. Space Force photo by Senior Airman Kadielle Shaw)



FRONT MIDDLE INSET: Tennessee Army National Guard Soldiers with Alpha Battery, 1-181st Field Artillery Regiment conduct a training exercise using the M142 High Mobility Artillery Rocket System (HIMARS) at Camp Shelby, Mississippi, June 9. The unit's annual training enhances battalion readiness, focuses on mission-essential tasks and ensures Soldiers are proficient in critical skills. (U.S. Army National Guard photo by Sgt. Grayson Cavaliere)



FRONT RIGHT INSET: Operation Redwing, Mohawk Nuclear Weapons Test on July 2, 1956 at Eniwetok Atoll. Los Alamos National Laboratory, Redwing-Mohawk - (CN67-1477) LA-UR-10-07957.



BACK COVER: A Tunisian Armed Forces member prepares to enter a chemical-exposed bunker to assess, decontaminate, and process a fixed chemical attack site at 61st Officers Mess, Tunisia, April 23, 2025 during African Lion 2025 (AL25). Service members from Wyoming National Guard's 84th Civil Support Team, the 173rd Combat Engineer Company, 173rd Airborne Brigade and members of the Tunisian Forces conduct chemical, biological, radiological, and nuclear (CBRN) training to prepare Soldiers for dealing with CBRN threats. (U.S. Army Reserve Photo by Lt. Col. Dale D. Barnes.)

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Notes from the Director

By: Col. Jarrod Gillam



Greetings, and welcome to Issue 30 of the *Countering WMD Journal*.

I am Col. Jarrod Gillam, and I assumed the position as Director of the United States Army Nuclear and Countering Weapons of Mass Destruction Agency (USANCA) in May 2025. I'm very proud to be leading this great organization, and just as proud to be bringing you this thought-provoking and timely issue of the *Countering WMD Journal*.

Before I introduce the theme of this issue, I would like to recognize Col. Tina M. Schoenberger's outstanding leadership as the Director of USANCA and her contributions in publishing the *Countering WMD Journal* during the past two years. Under her leadership, the Agency grew stronger as an organization and became well-known for its ability to ably advise senior leaders while leading change across the nuclear, countering weapons of mass destruction, and biodefense communities.

One of the many ways in which she expanded USANCA's reach was through leveraging publications such as this one as a means of amplifying new and provocative ideas from throughout the enterprise. Under her guidance, and under the expert hand of editor Maj. Josiah Hennig, the *Countering WMD Journal* has reformed its layout, aligned its content against broad topical themes, and expanded its reach to a wide network of partners with equities in the world of nuclear deterrence, WMD education, chemical, biological, radiological, nuclear defense, and biological resiliency.

It only takes a quick browse through our archives to see the tremendous change that has taken place in just a few years. As I look at what the *Countering WMD Journal* was, and what it has become, it is immediately obvious to me that the reformed journal delivers tremendous value to the community thanks to a Director's vision and an editor's talent.

That insight is in itself a lesson, and that brings us to the theme of this issue. This edition is about the manner in which our community chooses to act on—or to ignore—the lessons of the past.

As the Department has gradually transitioned from a strategic focus on stability and regional threats in order to pivot towards great power competition and large-scale combat operations, most of us have, at one point or another, heard senior leaders remind us that strategic deterrence and global competition are not new missions. Seventy years of hard-won Cold War experience produced innumerable lessons about how deterrence is achieved, tested, and maintained.

A perfect example of this is the concept of Conventional-Nuclear Integration. Although this is, in fact, a new term codified within Joint Publication 3-72 just last year, numerous studies within the past few years have shown that throughout the Cold War, nuclear and conventional capabilities were inextricably linked across force design, doctrine, and operational practice. For an excellent explanation of how each Service accomplished this integration, I highly recommend a forthcoming Defense Threat Reduction Agency-sponsored study *Conventional Nuclear Integration: Cold War Lessons for Today's Warfighter*. An extract of this study was published in Issue 29 of the *Countering WMD Journal*.

Clearly there are many lessons that we can glean from studying the history of the past century. You'll find a case study in this issue that points to the parallel between U.S. grand strategy of the early 1900s and the strategic outlook of the second Trump administration. You'll also find an argument for nuclear testing as a means of demonstrating both national capability and political will in order to enhance the credibility of a nation's deterrence posture. You'll see discussion of how the lack of epidemic models within wargames and the lack of nuclear literacy in officer education permit leaders and staffs to fall victim to false assumptions about the resiliency of their units—and their plans—to biological or nuclear threats.

All of these are things that the Department once knew but either forgot or chose to ignore. Other missions took priority and, over time, leaders with Cold War experience became harder and harder to find.

It's important to note that lessons from the past cannot be simply dusted off and thoughtlessly applied to contemporary problems. The geostrategic environment has changed. The nature of warfare, and how it will be fought, has also changed. While the lesson might be enduring, the solution that it informs must align closely with current and future problems.

In this vein, you'll read within this issue an argument for returning nuclear capabilities to the land component as a means of creating more rungs on the escalation ladder. The author doesn't argue that the Army should seek the same capabilities it once had—only that the lessons from the Army's long experience with land-based nuclear weapons should be considered as senior leaders look for future options.

Similarly, the basic elements of deterrence that were applied in a bipolar world need to be re-examined as the United States considers three-way deterrence against Russia and the People's Republic of China. Many of the ideas of Schelling and Kahn have yet to be tested in a three-way global competition. The stakes are too high to let bad assumptions guide future policy.

It is absolutely worth taking a very hard look at the lessons from the past to ensure that we understand them fully. Only once we've done this can we responsibly apply them to the future.

Speaking of the future, our next edition of this journal, Issue 31, will be focused on Transformation. I encourage all of you who have a part in shaping the future of the Joint Force and the Department to contribute your thoughts and keep this conversation going. As always, you can find instructions on how to submit your work on the inside of the back cover.

Thanks to all of you for everything that you and your organizations do for both the Department and the nation. I'm really excited to have the opportunity to guide this publication and I intend to use it to reflect the incredible work that each of you do every day.

Without further ado, read on and enjoy! ■

Integrating Risk from Weapons of Mass Destruction: Defining Strategic and Emerging Threats

By: Lt. Col. James C. Bowen

Summary

Weapons of Mass Destruction (WMD) threats are not properly evaluated against other strategic risks, and the tools to deter and respond to disruptive technology, emerging threats, and strategic threats are not functionally aligned to be developed, messaged, and integrated properly. Therefore, the Department of Defense (DOD) should define disruptive technologies, emerging threats, and strategic threats with their associated risk category to clarify which DOD Operations, Activities, and Investments (s) should be aligned against each threat and which communities should be responsible for managing and implementing s.

PROPOSED THREAT DEFINITIONS

DISRUPTIVE TECHNOLOGY: Technologies or scientific discoveries that are expected to have a detrimental effect on the military's ability to meet current and projected future strategic and operational mission requirements.

EMERGING THREAT: New or evolving risks with lack of intelligence on their attack methods, strategies, or techniques that could threaten the military's ability to accomplish operational mission objectives or cause unacceptable loss.

STRATEGIC THREAT: Threats that would prevent the DOD from maintaining force balance or performing established functions.

Introduction

When the 2014 *Countering Weapons of Mass Destruction (CWMD) Strategy* was written the United States military posture was drastically different from where it is today. The interagency had succeeded in demilitarizing the Syrian chemical weapons program in 2013, the Russian–US reduction of nuclear weapons from the Cold War had been relatively successful, and the United States was steeped in a war on terrorism in Iraq and Afghanistan.¹ The strategy focused primarily on developing the ability to keep malign actors from obtaining weapons of mass destruction by, with, and through the interagency while maintaining the ability of the United States (US) military to fight through Chemical, Biological, Radiological, and Nuclear (CBRN) weapons effects.²

The major players in the game became U.S. Special Operations Command in 2016 when they were designated the DOD Coordinating Authority, and the Chemical and Biological Defense community which was responsible for acquiring and fielding detection and protection equipment. To the Joint Force, WMD efforts were driven by the risk of non-state actor acquisition and use of WMD, and risks from the operational force not being able to fight in and through CBRN-contaminated environments. There were many successes involved in interdiction efforts, developing capabilities, and building partner capacity to detect and remediate CBRN weapons effects.

In 2022 the *National Defense Strategy* was written to align with contemporary priorities. The DOD shifted focus from terrorism to peer conflict, identifying China and Russia as a pacing and acute threat.³ This shift was followed by an alignment of documents

attached to the *National Defense Strategy* including the *Nuclear Posture Review* and the *Missile Defense Review*, then by WMD related strategies to include the *Biodefense Posture Review* and the 2023 *Strategy to Counter Weapons of Mass Destruction*. The 2023 CWMD Strategy included deterrence of WMD attack from peer adversaries against the homeland, allies, and partners along with the long-standing pathway defeat and CBRN response lines of effort.⁴

The 2023 CWMD Strategy added complexity and confusion rather than simplicity and clarity. Although the strategy included a focus shift towards integrated deterrence, the communities involved are dedicated to efforts outside of deterrence. The US Special Operations Command (SOCOM) has a central role in unconventional warfare space while the CBRN defense community continues development of defensive and response capabilities to post-event environments of chemical and biological attacks. In addition, the primary backstop for strategic deterrence (nuclear weapons) remains the responsibility of Strategic Command, which does not have CWMD as part of their current mission space. Further, geographic combatant commanders do not have defined roles against peer adversaries concerning weapons of mass destruction and there is a void in mission space at the operational level.⁵

SOCOM and the CBRN Defense communities have done a phenomenal job in degrading threats, assuring ally and partner ability to operate in a WMD effects environment, and building new capabilities to protect soldiers from chemical threats. To keep pace with contemporary strategy against peer adversaries, the DOD should establish a framework that defines a hierarchy of new and existing threats that are or have potential to become strategic military threats based on risk. This framework should enable evaluation of WMD threats properly against other strategic risks and ensure responsibilities to deter and respond to a strategic threat are developed, messaged, and integrated properly.

A Better Way to Assess Risk

The current definition of WMD in the DOD is “chemical, biological, radiological, or nuclear weapons capable of a high order of destruction or causing mass casualties, excluding the means of transporting or propelling the weapon where such means is a separable and divisible part from the weapon.”⁶ This definition is problematic as it defines WMD based on materials they are composed of instead of what

WMD could do to the Joint Force. From instructional experience teaching conventional nuclear integration in Professional Military Education, the US Army Nuclear and CWMD Agency observed that students have a hard time comprehending the scope and scale of nuclear weapons and escalation ladders are drastically misunderstood. Additionally, risk analysis is primarily focused on the tactical effects of weapons of mass destruction and extrapolating operational and strategic impacts instead of comparing similar risks. Defining WMD by materials and evaluating risk based on effects is confusing to operational planners attempting to evaluate WMD against other threats on the battlefield. If the DOD centers definitions of strategic threats on risk, it could clarify how WMD risk analysis is integrated holistically for operational commanders.

The first obstacle to achieving a useful risk-balancing framework is the grouping of C-B-R-N (chemical, biological, radiological, and nuclear) is very misleading when considering risk because each letter of the acronym is very different in categories of risk. Chemical attacks, although devastating to unprotected populations, represent a relatively tactical or operational risk to military formations. Enemy tactics focus on tactical attrition or denial of terrain destructive or terrain denying as seen in mass chlorine attacks in World War I or current riot control agent attacks in Ukraine.⁷ The tactical risk of a chemical attack is comparable to the risk of any other battlefield shaping capability (e.g., improvised explosive devices or mine fields) where it is limited and, when detectable, can be mitigated. Operationally and strategically, chemical weapons can have follow-on effects that are not related to their tactical impacts, especially if they affect host nation civilians and spur international recourse when terror or fear tactics are involved.

In contrast, biological threats are a risk that have a daily presence around the world, can spread relatively easily, and are difficult to contain. Humans take vaccines, track diseases, and build force protection measures to prevent the spread of disease. In this way, the risk of biological attacks is more like cyber attacks. The world uses computers daily and there are inherent risks from existing on the internet. Hackers, malware, and natural computer problems limit Americans ability to function in digital space. The military purchases anti-virus software, mandates cyber awareness training, and has physical control measures around digital systems to defend digital networks. The effects of a biological incident as we saw during the coronavirus disease (COVID-19) are hard to contain and can rapidly shut down supply chains, infrastructure, and the ability

to project forces, like a network being shut down. COVID-19 had operational origins in minimal force degradation, but elevated to an institutional risk where the Joint Force no longer could function.

Finally radiological and nuclear threats are vastly different from each other but are grouped together in classroom instruction and training. Prompt effects of nuclear attacks are unique in their massive kinetic scale, and nuclear weapons are legal for nuclear weapons states. Legality makes their development and positioning an overt deterrent to nuclear weapons states engaging in conflict. Although most concerns of nuclear weapons are large-scale and devastating, the CBRN defense community is responsible for their protection responsibilities. This makes the Services' focus less about prompt effects and deterrence, and more on radiological contamination which is more like chemical contamination but easier to detect and mitigate. The risks of a radiological environment are relatively tactical while positioning and use of nuclear weapons is a risk to joint-force operations. By centering threats around the risk instead of the effects on the battlefield, the WMD community might better help Joint Force commanders understand problems facing a WMD armed adversary.

Risk Framework

The following proposed framework evaluates strategic military risk in a linear fashion with the simple question of "what would prevent the Joint Force from acting as an instrument of national power?" Evaluating risk in a linear fashion allows senior military leaders to prioritize operations, activities, and investments depending on the threat of marginalization or obviolation of the military arm of national power. A framework like this attempts to keep tactical and operational problems from becoming strategic by building a hedge against

potential emerging unknown technology applications, building operational capabilities to control escalation and deter strategic capability use, and building response options against strategic threats. Risk is defined using the Joint Risk Assessment Methodology (JRAM) instead of by levels of war, and on strategic and emerging threats to the application of military force and not to the nation. Centering definitions of threats on risk bounds the framework and enables decision makers to reestablish deterrence (bringing conflict back to the operational military domain).

Figure 1 outlines the spectrum of disruptive technologies to strategic threats aligned with different category of risk. By aligning each with a category of strategic risk, it gives policy makers the ability to determine the category of risk and what investments they should prioritize.

For option development this framework uses hedging options, deterrence options, or response options. Each option is elaborated on below:

HEDGING: According to Ben FitzGerald and Scott Cheney-Peters in their article for War On the Rocks, "a hedging strategy prioritizes investments in those processes that enable the Department of Defense to delay full-scale production decisions until requirements gain greater fidelity, while at the same time learning from the small-scale batches of prototypes."⁸ Hedging options mitigate vulnerabilities through force design and prudent capability and requirements development. They can involve workforce development, development of non-material investments (i.e., training, doctrine, and education), physical procedures, policy changes, or integration of new technologies.

DETERRENCE: In Joint Publication (JP) 5-0, deterrence actions are described as actions that seek to prevent the threat's undesired action. Deterrence is a



FIGURE 1: Threats Aligned with Strategic Risk Categories

state of mind brought about by the threat's perception of three factors: the likelihood of being denied the expected benefits of action, the likelihood of having excessive costs imposed for taking the action, and the acceptability of restraint as an alternative."⁹ The goal of deterrence option development in this context is to develop, posture, and demonstrate resilience to threats while developing the capability to impose cost for threat activities. Two ways to deter are deterring by denial of benefit (convincing enemies that the benefit of using this capability will not have the intended outcome) or deterring through degrading the enemies' confidence in their capabilities to achieve their outcomes.

RESPONSE: In JP 5-0, Flexible Response Options are operational-to strategic-level concept of operation that is easily scalable, provides military options, and facilitates rapid decision making by national leaders in response to heightened threats or attacks against the US homeland or US interests.¹⁰ Response options for the military are offensive and punitive in nature. Development of offensive response capabilities reinforces the last method of deterrence which is through cost imposition (convincing enemies the cost associated with their action is greater than the outcome achieved).

In the rest of this paper, threats are defined and categorized according to their alignment with joint risk categories.

Category 1: Disruptive Technologies

The North Atlantic Treaty Organization (NATO) Science & Technology Trends 2020-2040 report defines disruptive technologies as "Those technologies or scientific discoveries that are expected to have a major, or perhaps revolutionary, effect on NATO defense, security or enterprise functions in the period 2020-2040."¹¹ This category includes biotechnology and quantum computing.

Although many military uses of disruptive technologies are developing, it is necessary to build hedges against them to ensure future battlefield risks are accounted for and the technologies do not disrupt planned Joint Force capabilities. The NATO definition of disruptive technology aligns well with "Future Challenges Risk" from the JRAM which is "a function of the probability and consequence of a known harmful event that causes failure to meet current and projected future strategic and operational mission requirements."¹²

With these two definitions in mind, disruptive technologies for the purpose of assessing risk can be defined as *technologies or scientific discoveries that are expected to have a detrimental effect on the military's ability to meet current and projected future strategic and operational mission requirements.*

Because disruptive technology employment strategies are either unknown or not integrated, DOD policy makers should develop hedging strategies to ensure the joint force accounts for future threats. An example of a modern disruptive technology is biotechnology. Security of biological data, future human enhancing biotechnology, limitations of detection systems, and development of bioweapons are all concerns of NATO.¹³ Similarly, the Department of Defense is already hedging against threats posed by the accelerating development of biotechnology through policy updates (such as the 2023 *Biodefense Posture Review*) to guide investments in modernization efforts, non-material education on signs and symptoms, and doctrinal solutions. These are hedging options that increase Joint Force resilience against a potential unknown use of a disruptive technology. Increasing the Joint Force ability to operate aligns with the *National Defense Strategy* techniques to deter through resilience.¹⁴ Such options defend United States forces against unknown challenges, help us understand possible employment scenarios, and support the development of international controls to limit proliferation of technology or integration with military capabilities.

The acquisition and modernization authorities at the level of the Secretary of Defense should be responsible for policy and requirements against disruptive technology. The Office of the Under Secretary of Defense (OUSD) for Intelligence and Security is responsible for oversight of defense intelligence organizations and functions. This is a starting point to see what new technology is up and coming and driving requirements for acquisition priorities to the OUSD for Acquisition and Sustainment. Keeping responsibilities for disruptive technologies at the DOD secretariat level ensures appropriate prioritization in modernization efforts, to include equipment survivability requirements, and the defense industrial base for all services who are the capability developers for the Joint Force. The ability to operate on the battlefield can not be stressed enough, and the joint force inability to do so is in itself a strategic threat. Building equipment that is resilient and testing it against potential weapons effects, cyber intrusions, and other emerging threats will determine if service members

SIDE NOTE: MILITARY OPTION DEVELOPMENT

As this paper spans from secretariat level responsibilities to operational theaters, it is important to understand military options. Military options can be set apart into six areas shown in Table 1 below. In a simplistic manner, the military has force structure, tactics, techniques, and procedures (TTPs), or capabilities (to include military construction) that are offensive or defensive in nature. Developing, projecting, or employing these options are actions the military can take to influence other nation states, and therefore make them instruments of national power. This simplistic view of military power categorizes almost anything the military can do.

A reasoning for differentiating options by type (offense or defense) and action is to clarify what is used strategically versus operationally. Strategically, options do not deter unless they are credible, capable, and known. According to a recent RAND paper on escalation, “the challenges in distinguishing offensive and defensive capabilities may reduce the deterrent value of deploying offensive capabilities and lead an adversary to undertake undesired actions.”¹⁵ However, there are benefits to having unknown response abilities. Revealing all capabilities up front could reduce the effectiveness of their employment. Therefore, clarifying options by not just their intent (offensive or defensive) but what those options are doing as shown in Table 1 points at what aspect of integrated deterrence they’re helping.

Force, TTP, or Capability	Develop	Project	Employ
Defensive	Hedging	Deterrence	Deterrence
Offensive	Deterrence	Deterrence/Response	Response

TABLE 1: Progression of Military Strategic Options

can survive in their own equipment, and the requirements to do so should be held at the OUSD level.

Category 2: Emerging Threats

In cybersecurity, an emerging threat is “a new or evolving cybersecurity risk that is particularly challenging to defend against due to the lack of intelligence on its attack methods, strategies, and techniques.”¹⁶ Emerging threats are known capabilities with unknown methods and strategies. Although the line between disruptive technologies and emerging threats is murky, once technical feasibility is demonstrated, threats are aligned to force structure, there potential integration into military

TTPs, or threats are fielded, they can be classified as emerging threats. Aligning emerging threats with a strategic risk category, under risk to mission in the JRAM is the category of operational risk. Operational risk is “a function of the probability and consequence of failure to achieve mission objectives while protecting the force from unacceptable losses.”¹⁷

Combining the definitions of emerging threat from cybersecurity with operational risk from the JRAM gives the proposed definition for emerging threats of *a new or evolving risk with lack of intelligence on its attack methods, strategies, or techniques that could impact the military’s ability to accomplish operational mission objectives or cause unacceptable loss*. An example of an emerging threat is artificial intelligence (AI). AI is currently a dual-use technology being used for a variety of applications and being integrated into military functions. Although uses for AI are not fully understood, AI paired with military systems is revolutionizing warfare with unmanned autonomous systems, cyberwarfare, and accelerating decision support processes. The Joint Force is starting to hedge against it by integrating AI into decision support tools and experimenting with AI in autonomous platform development.¹⁸ Integration of new technology helps acquisition programs

understand both the capabilities involved and potential measures adversaries can take. Additionally, to build resilience, DOD personnel already have air-gapped isolated systems and developed cyber-awareness training requirements. The Joint Force is adjusting acquisition priorities watching developments from the war in Ukraine and the broader government community is limiting the People’s Republic of China access to technology involved in AI development.

Against emerging threats, the Joint Force can add on to previous hedging efforts by developing deterrence options against emerging threats. Investments made against peer emerging threats include capabilities to promote resilience and sustain force projection, attribution capabilities to assist the international community

in maintaining established norms, or capabilities to degrade enemy confidence or capabilities. Deterrence options work to deter enemy development, implementation, and use of emerging threats.

Geographic Combatant Commands (GCCs) are in the best position to drive requirements for deterrence options. GCCs are looking directly at threat integration of technology and can evaluate if disruptive technology has the potential to disrupt their operational objectives, turning it into an emerging threat. The ability to deter specific adversary action is best fitted at the regional level, and the placement or employment of forces can deter adversary pursuit of technology development.

Category 3: Strategic Threats

Strategic threats are defined vaguely across different documents and generally focus on threats to the nation that the military can prevent. Strategic military risk on the other hand, is defined in the JRAM as a risk to national interests, while military risk is defined as risk to executing the National Military Strategy including risk to mission (achieving objectives) and risk to force (military resources).¹⁹ Alignment of strategic threats with the two categories of risk to force, force management risk and institutional risk, is a logical path.

Force management risk is “a function of the probability and consequence of not maintaining the appropriate force generation balance (breaking the force).”²⁰ It deals with the ability of the military to meet planned mission requirements with current forces. Force management risk balances operational requirements with currently planned modernization efforts for prioritization.

The JRAM defines institutional risk as “a function of the probability and consequence of the DOD failing to perform established functions.”²¹ The time horizon for institutional risk is from current problems to long-term planning goals. Institutional risk includes subsets of the acquisition process, force development, the defense industrial base, recruiting, and the ability to conduct national defense activities.

For strategic threats, a better definition would be *threats that would prevent the DOD from maintaining force balance or performing established functions*. For different threats that could be categorized here, anything that prevents the joint force from projecting power in a region, attrits at a rate higher than is replicable, or hampers the DOD from using its strategic tools is a strategic threat. Threats that can

affect the industrial base, reduce support for the military in a region (foreign or homeland), or affect joint force projection capabilities are strategic and keep the military from aiding as an instrument of national power. A few examples of military strategic threats are nuclear weapons, cyber-attacks, and biological agents. The inability to deploy, use technology, and control nuclear escalation can prevent the military from influencing a region.

Shifting to the Indo-Pacific future problem set, access, basing, and overflight, referred to as ABO, is threatened by other countries national will to partner with the United States. High order damaging long range weapon system positioning, biological threats, or cyber threats paired with a good information campaign can disrupt the ability to project power into a region. Force projection capabilities like ships and planes can be held at risk by future long-range standoff capabilities and artificial intelligence synchronized drone swarms. Each of these systems individually might not be a strategic threat, but a combination of TTPs, capabilities, and forces can absolutely prevent the Joint Force from operating in a region.

Against strategic threats, the Joint Force can develop a balanced combination of resiliency activities, deterrence activities, and response options. As the previous two categories noted, enabling the ability to act in a region, deterring use of strategic options, and degrading state actors' confidence or ability to use their systems is crucial. The addition is once a threat is strategic, the joint force needs aligned and communicated strategic options to control escalation and provide decisions to the President. This gives the President military options other than pointing nuclear weapons to influence state actors in crisis and conflict. As threats are defined by the DOD, strategic military options are those that can hold adversaries' ability to employ their militaries at risk or impact their militaries' ability to act as an instrument of state power.

There are a few successes in strategic option alignment among functional commands. One such effort is the Special Operations Forces (SOF)-Space-Cyber Triad looking at using Cyber and Space capabilities under Special Operations access and authorities to develop options for decision makers in campaigning which Maj. Brian Hamel wrote about for *Military Review*.²² Maj. Hamel's article explains how SOF is attempting to leverage cyber and space to have mutually beneficial effects contributing to deterrence. A separate effort is integration of nuclear and non-nuclear options called “Conventional Nuclear Integration”

which is now defined in Joint Doctrine (JP 3-72, *Joint Nuclear Operations*).²³ Strategic Command is responsible for nuclear options which are labeled as the backstop of the Joint Force in deterrence activities.²⁴ Conventional-Nuclear Integration is an attempt to integrate nuclear and non-nuclear planning efforts to synchronize options across the force.

To properly deter and respond to strategic threats, it would be beneficial to have a single integrator of strategic military options defined in the Unified Command Plan. This makes a clear separation between operational conflict in regional fights and strategic threats that prevent the military from influencing regional conflicts. This could potentially be done under Strategic Command, with support from the other functional commands and integrated, when necessary, with the geographical combatant commands. The DOD already has a strategic integrator for nuclear weapons, so it is not too much of a stretch to take other strategic threats and bin them under a unified planning and action authority. Having a unified planning and action arm allows the military to integrate capabilities with the interagency for unified messaging and gives regional commanders the flexibility to have strategic options that are from a single source against peer competitors.

Conclusion

Clarifying definitions of threats that can eventually become strategic threats to the military helps align modernization efforts, s, and option development. New and upcoming disruptive technology that is a potential future challenge but is not integrated in military formations is something to consider and track but should be held at the OUSD level, so warfighters are not exhausting resources chasing moving targets. Once those technologies are integrated in enemy formations as emerging threats, the GCCs are in the best position to account for them as an operational risk and track their development to a potential strategic threat. As soon as threats are determined to be strategic, it would be beneficial to have a single functional command that executes presidential authorities for response options and integration with the interagency. Clarification of strategic military threats is a first step in integrating WMD risks with other risks to the Joint Force ■

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Notes

1. Brian Feeney, "Army Marks 10-Year Anniversary of Syrian Chemical Agent Destruction," *U.S. Army*, August 15, 2024, https://www.army.mil/article/278975/army_marks_10_year_anniversary_of_syrian_chemical_agent_destruction; Amy Woolf, "25 Years of Nuclear Security Cooperation with Russia and Other Newly Independent States," *Russia Matters*, February 28, 2017, <https://www.russiamatters.org/analysis/25-years-nuclear-security-cooperation-us-russia-and-other-newly-independent-states>.
2. U.S. Department of Defense, *Strategy to Counter Weapons of Mass Destruction*, (Department of Defense, 2014).
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Should the Army Go Nuclear?

The Case For Returning Nuclear Weapons to the Ground Force

By: Rohin Sharma

In 2024, the Department of Defense reported that the People's Liberation Army (PLA) “may have deployed a long-range DF-27 ballistic missile” that also contained a “Hypersonic Glide Vehicle (HGV) for... conventional land attack and conventional anti-ship.”¹ The DF-27 has a range of 5000-8000 km, putting targets as far as Alaska and Hawaii at risk.²

If true, this would add significant hypersonic range for Chinese surface to surface missiles. Such an advanced capability will compromise airfields throughout the Pacific and could potentially embolden Chinese activities in the region. Runways, pockmarked with craters, could be rendered unusable, with hangars and repair facilities also being devastated. Even if aircraft were to get off the ground, China has plans to develop missiles that could engage them—particularly slow-moving tankers that would be needed to support short range fighters and bombers.³ Given the speed and trajectory of hypersonic missiles, they would be very difficult for traditional air defense assets (MIM-104 Patriot, Terminal High Altitude Area Defense [THAAD], and RIM-161 Standard Missile 3 [SM-3]) to intercept.⁴

In this scenario, delivering non-strategic nuclear weapons with air assets would be almost impossible. The US would likely rely on low yield W-76 warheads from submarines, which may not be on station and could be confused to be a strategic nuclear attack.⁵ In the future, the Submarine Launched Cruise Missile-Nuclear (SLCM-N) might be an option, however, it is not slated for deployment until 2034 and may still suffer from stationing and escalation issues.⁶

Another possibility is to return nuclear weapons to the Army inventory. This would give the US the ability

to effectively deploy non-strategic nuclear weapons in a conflict. Given advances in Anti-Access/Area Denial (A2AD) technologies, increases in the nuclear capabilities of US adversaries, and the survivability of Army platforms, military and political leaders must consider returning nuclear weapons back to the land component. The number of weapons need not be on the same scale as it was during the Cold War, but it should be large enough to deliver non-strategic nuclear weapons if air platforms are compromised.

This paper will make the case for reintroducing nuclear weapons into the Army. It will begin by outlining the differences between strategic and non-strategic weapons and explain why the US needs to accelerate its non-strategic nuclear weapons numbers and capabilities. It will then cover the strategic environment, showing how US adversaries are expanding their capabilities, while US theater capabilities are languishing. The next part of the paper will look at key advantages that nuclear Army systems would bring to the table. The paper will conclude by assessing the counterarguments for the Army not having nuclear weapons.

The Advantage of Non-Strategic Nuclear Weapons

The crux of the issue is the difference between strategic and non-strategic weapons. While there is no wide agreed definition between the two, strategic weapons are designed for inflicting maximum damage on high level political and military targets. They are largely confined to ballistic missiles (both submarine and land based) and long-range

bombers. Given their target sets, strategic weapons tend to be higher yield and are more likely to be used if there is existential threat to the nation.⁷

Strategic weapons usually have an intercontinental range, although there are some exceptions. Strategic weapons between Russia and the United States are governed by various intercontinental agreements. Non-strategic weapons are generally considered for battlefield applications. They are targeted dynamically and tend to focus on conventional military targets. Given the desire to avoid collateral damage, non-strategic nuclear weapons could be developed as low yield..⁸

While militarily the differences may seem semantic, politically they are quite significant. Planners have tried to divorce the two, for fear that a tactical battlefield weapon could be interpreted as a more escalatory strategic strike. Major nuclear powers use different platforms for strategic and non-strategic weapons, largely due to the different employment options. Strategic weapons are limited and accounted for under the Strategic Arms Reduction Treaty (START), while there are currently no treaty limitations on non-strategic weapons.⁹

The main difference, however, is strategic nuclear weapons are viewed as an existential threat to a potential adversary of the United States. Spinning up ICBMs, deploying B-52s, or positioning nuclear capable submarines close to their shores could be viewed by China and Russia as the first step to regime decapitation. This in turn may increase the likelihood of a retaliatory strike. Therefore, it remains vital that non-strategic nuclear weapons use be “compartmented” from strategic weapons.¹⁰ As a corollary, it is important that any political leader be given a range of options for deploying nuclear weapons, and not be forced to rely on strategic systems that could be perceived as escalatory.

Non-strategic weapons offer other advantages. In a scenario where there is conventional overmatch, low yield nuclear weapons can mitigate this unbalance. Pakistan has developed non-strategic weapons to offset India conventional superiority.¹¹ The DPRK has also likely invested in non-strategic nuclear weapons to mitigate the more capable South Korean military.¹² Furthermore, during a potential conflict, even sophisticated militaries will likely face a conventional imbalance before larger forces can flow into theater.

For the US, low yield weapons can also serve as a credible signal during an escalation phase of a conflict.

The appearance of a visible nuclear capable system can indicate that the US is willing to stop aggression, even employing the use of nuclear weapons. In addition, this asset can reassure allies, showing that the US is committed to their defense. As opposed to non-strategic systems, strategic systems are far harder to deploy, less visible (submarines), and are far less credible than tailored theater capabilities.

The Strategic Environment: How the US is Lagging Behind Its Adversaries

Since 2014, US adversaries have been far more dominant in respect to non-strategic nuclear weapons numbers and capabilities. Russia currently has an active stockpile of up to 2,000 non-strategic nuclear weapons.¹³ These items include a wide variety of weapons like “short range ballistic missiles, gravity bombs, depth charges for medium-range bombers, tactical bombers, and naval aviation, as well as anti-ship, anti-submarine, and anti-aircraft missiles and torpedoes for surface ships and submarines, and nuclear antiballistic missile system.”¹⁴ In the event of any conflict, Vladimir Putin will have an array of non-strategic nuclear options.

China is going through a significant nuclear expansion, looking to expand its total number of warheads to 1,000 by 2030.¹⁵ As mentioned above, China is also developing multiple delivery systems, most of which are designed to defeat US missile defense assets. More relevant, China maybe expanding its low yield capabilities. Chinese writings have noted “the introduction of new precise small-yield nuclear weapons could allow for the controlled use of theater-range nuclear weapons for warning and deterrence.”¹⁶ In addition, Department of Defense reporting indicated that the DF-26 intermediate range ballistic missile (IRBM) could “hot swap” between nuclear and conventional weapons.¹⁷ If true, this would give China a precise theater nuclear capability that the US does not have.

Even the DPRK may have advantages in non-strategic nuclear weapons. Pyongyang is currently investing in a suite of theater missile capabilities to augment their deteriorating conventional capability. In 2021, Kim Jong Un publicly called for “producing smaller and lighter nuclear weapons for tactical use,” although it is unclear how far they have developed the actual warheads.¹⁸ Finally, DPRK’s relationship with Russia may lead to greater cooperation within their nuclear program.¹⁹

In comparison, the US is largely limited in its non-strategic nuclear weapons employment. According to the Center for Arms Control and Non-Proliferation, the current US tactical nuclear arsenal is comprised of B61 gravity bombs dropped from F-15, F-16s, and Tornados (with the F-35 likely coming online at a later date).²⁰ These assets are very vulnerable to adversary A2AD capabilities when they are on the ground. While they are airborne, they can be targeted by integrated air defense assets. The US also has the low yield W-76 warhead on a submarine launched ballistic missile. However, since it's on a strategic platform, this likely blurs the distinction between strategic assets and theater nuclear capabilities.²¹

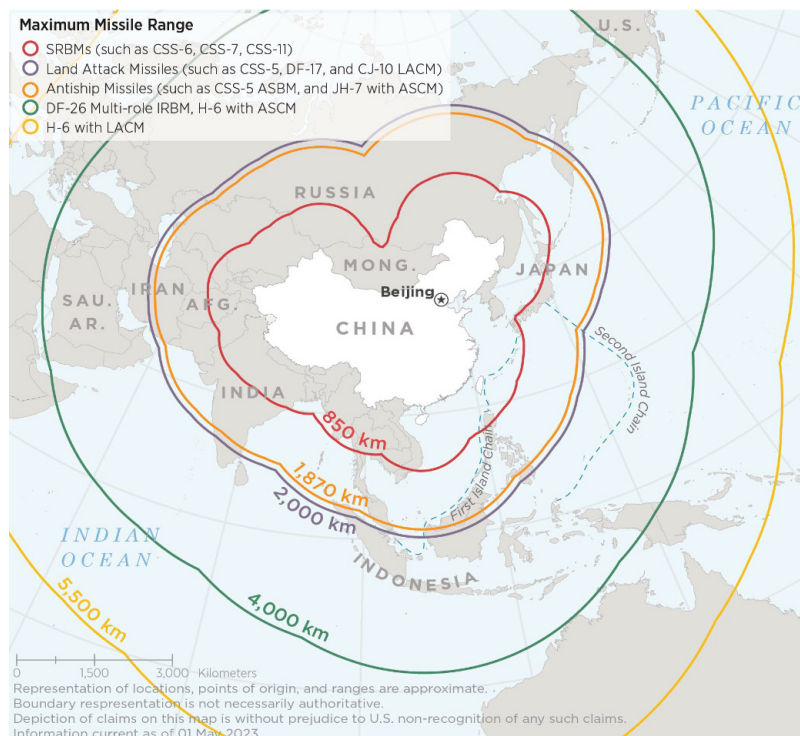


FIGURE 1: Chinese Missile Ranges.²²

Why the Army Needs Nuclear Weapons

VARIED DELIVERY OPTIONS:

First, and foremost, a nuclear Army would give the Joint Force another option for the deployment of non-strategic weapons. Currently, non-strategic weapons would only be delivered by gravity bombs from vulnerable F-15s and F-16s (with the F-35 coming online in the next few years).²³ As previously mentioned, if A2AD weapons deny access to these assets, the President could only direct the use of strategic platforms (B-52s, nuclear submarines) if there was a need to employ nuclear weapons. This jumps several rungs up the escalation ladder, making a catastrophic nuclear exchange much more likely. If the Army has nuclear capabilities, the President would have this vital tool without taking the necessary risk of using the strategic arsenal.

SURVIVABILITY:

Unlike air platforms that operate from fixed airfields, mobile Army nuclear weapons would be much harder to identify and more difficult to neutralize. History has shown that mobility has vastly increased the resiliency of platforms. During the Vietnam War,

advanced ISR and satellite technology did not spot the mobile surface-to-air missiles that proved so devastating.²⁴ Likewise, during the Gulf War, the US could not completely destroy the mobile SCUD missiles that were in an open desert.²⁵

Even in the current era, the issue of concealment is still an issue. Second-rate Houthi rebels have hidden their missiles within a small "box" and frustrated superior Saudi forces.²⁶ Lebanese Hezbollah has missile capabilities that Israel, even with the use of advanced imagery and artificial intelligence, cannot successfully target.²⁷ Despite significant effort, Russia only located and targeted one Army High Mobility Artillery Rocket Systems (HIMARS) in Ukraine.²⁸ Therefore it's unlikely that a potential adversary could completely identify and target US Army long range platforms that had a nuclear capability.

CONVENTIONAL IMBALANCE:

In the early stages of a conflict, the joint force is likely to face a significant conventional shortfall. In a generic conflict with either the PRC or Russia, US adversaries have a significant advantage of strategic interior lines allowing for more rapid conventional

deployment as compared to NATO. In the interim, small, nuclear armed Army formations might provide significant delaying capability, serving as either a deterrent to further aggression or even as a counterweight in the event of a conflict. In a cross straight scenario, low yield nuclear weapons would be effective against “concentrated and large-scale landing operations... and could be executed without appreciable collateral damage ashore.”²⁹ According to the Atlantic Council, if “the United States was to destroy the Chinese amphibious landing force (with low yield nuclear weapons) before China could seize major ports and airfields, the invasion would fail.”³⁰

Low yield nuclear weapons can also address the conventional imbalance in Eastern Europe. In a Baltic scenario, where Russian conventional strength would likely overmatch NATO’s in the early part of a conflict, Army low yield weapons could be employed. While likely less effective than in Taiwan, low yield nuclear weapons could be used to strike Russian troop concentrations and logistics centers, making a Baltic invasion much less likely.

PARTNER ASSURANCE:

US nuclear policy has “sought to extend the US ‘nuclear umbrella’ over allies in Europe and Asia,” providing them with the protection of the US nuclear deterrent.³¹ This policy serves two purposes. First, it allows easier deployment of nuclear weapons, providing a forward weapons system in the event of conflict. Secondly, it ensures that partners do not need to develop an indigenous nuclear capability, serving a key US nonproliferation goal.

A nuclear capable Army would advance extended deterrence in the region. While much of extended deterrence relies on assets that might not be in theater (submarines, F-35 aircraft), Army nuclear capable assets would likely be much closer. These will serve as tangible element in extending the deterrent, as opposed to submarines which will not be visible to an ally.

Furthermore, as mentioned above, aircraft (whether on the ground or in the air), are much more susceptible to adversary A2AD capabilities, undermining their credibility as an extended deterrent asset.

STRATEGIC AMBIGUITY:

On the flip side, a nuclear armed land force could provide strategic ambiguity in the run up to the conflict. If there were conventional theater assets, it would be unclear to an adversary which units were nuclear and which were conventional only. Therefore, they would have to assume that all systems are nuclear capable, forcing them to take appropriate actions (limit movement or not concentrate assets).

NEGOTIATING TACTIC:

During the latter stages of the Cold War, the deployment of Pershing II missile system in Europe was a factor in bringing Russia to negotiating table. The fear of having nuclear weapons on their

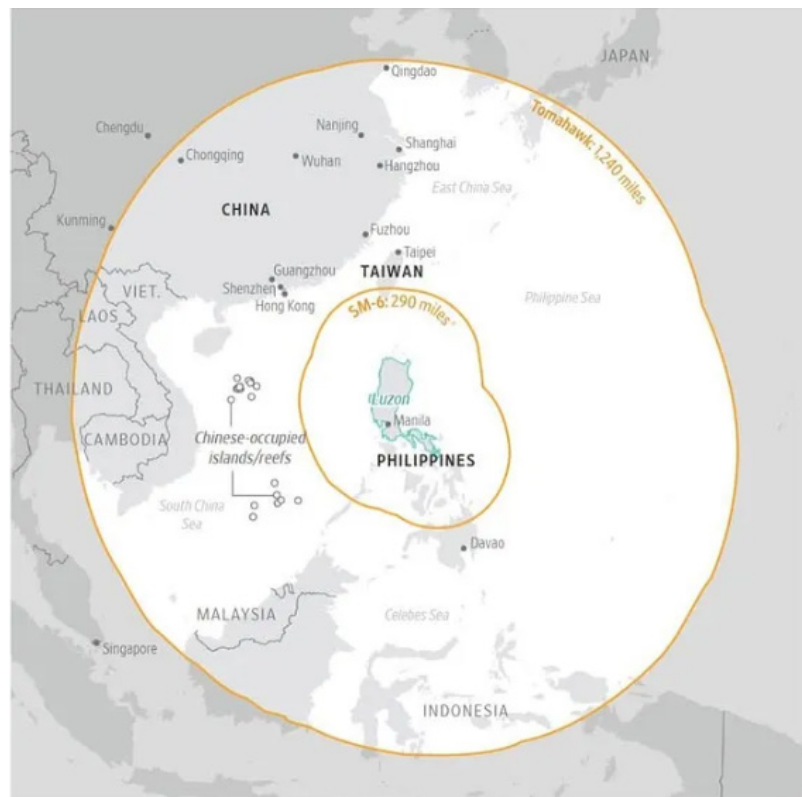


FIGURE 2: Typhon Ranges.³³

border arguably lead the Soviet Union to negotiate Intermediate Nuclear Forces treaty in 1987.

Currently, a similar dynamic is present in the Indo-Pacific. In April 2024, the Army deployed the conventional Typhon Mid-Range Capability (MRC) to the Philippines.³² From the northern islands of the Philippines, MRC could hold at risk targets near Taiwan as well as much of the Eastern China coast. This was the first real deployment of an intermediate missile since the end of the Cold War.

The Chinese response, even to a conventional missile, was significant. Chinese defense spokesperson said that the deployment had “intensified geopolitical confrontation and escalated tensions in the region.”³⁴ Foreign Ministry spokesperson Lin Jian told a press briefing on September 19 that the deployment “seriously threatens the security of regional countries and intensifies geopolitical confrontation.”³⁵ The Chinese have also pressured the Philippines into expelling the MRC system, an effort that has proven fruitless as of March 2025.³⁶

If this was a response to a conventional capability, one might expect an actual nuclear armed missile would trigger an even sharper reaction. While China would likely be more severe in their condemnation, there is a potential they would be willing to make concessions for the removal of the MRC. While it’s unclear what China would be willing to concede, it’s important to note that previous negotiating attempts have not borne fruit.³⁷ The deployment of other military assets to region have also not pressured China to the negotiating table. A nuclear capable missile system may alter this dynamic.

Counterarguments

While this paper makes the argument for the Army to return as a nuclear force, there are several good faith arguments for not bringing nuclear weapons to the ground force:

COST OF STORAGE, SURETY, AND SECURITY:

Given that the Army has not possessed nuclear weapons since the early 1990s, there would need to be significant investments in personnel, storage, and security facilities. These will have to be developed from the ground up, starting with investments in human capital to develop and employ a nuclear operating force. Given the current

resource constrained environment, the associated costs might be greater than perceived benefits.

While these investments are expensive, the thousands of nuclear weapons that the Army had during the Cold War would not be necessary due to improved ranges and accuracy of current delivery systems. Furthermore, the Army could tap into the expertise of the Air Force and Navy for storage capacity as well as trainers, until the Army can develop and indigenous surety capability.

NON-PROLIFERATION RISK:

Adding more nuclear weapons will undermine non-proliferation goals. The 1991 Presidential Nuclear Initiatives withdrew nuclear weapons from Europe, including eliminating them from the Army inventory. By adding nuclear weapons back to the Army inventory, America would undermine decades of non-proliferation work that has slowed the spread of nuclear weapons.

While there are some merits to this argument, the Army would not need the thousands of nuclear weapons that they had during the Cold War. More importantly, if non-proliferation is the goal, nuclear capable Army systems may limit the spread of nuclear weapons. South Korea, Taiwan, Poland, and other countries that are likely to be victimized by aggression would be less likely to develop their own nuclear weapons if they were under a nuclear umbrella that would be strengthened by ground based nuclear systems.

Conclusions and Way Ahead

The Army must consider a return to nuclear weapons. American adversaries have developed a myriad of non-strategic weapons. In addition, the PRC and Russia have made significant advances in A2AD capabilities, which may neutralize the current air delivered theater nuclear weapons. Army nuclear assets far more survivable than fixed airfields, preventing decapitation by a potential adversary. Army nuclear assets may deter a potential conventional invasion and may resist the invasion if deterrence fails. Finally, Army nuclear assets provide another “rung” on the theater escalation ladder, without resorting to more aggressive strategic nuclear assets. ■

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The Gunfighter's Analogy: Tripolar Nuclear Deterrence

By: Maj. Ty Thompson

Introduction

Accompanied by Ennio Morricone's iconic "The Trio," three gunfighters distance themselves from one another in a dilapidated cemetery, each waiting on another to draw his weapon and fire first, and each knowing that whoever fires his weapon last will be the victor. Clint Eastwood's unnamed character in Sergio Leone's *The Good, the Bad, and the Ugly*, however, had perfect information after unloading Tuco's weapon while the rivals of Eastwood's character did not, and thus, Eastwood's character won the contest.¹ Similar to both movie examples, a deterrence environment in a tripolar world may not look so different, given that no Great Power has a distinct information advantage over another. During the Cold War, the United States and the Union of Soviet Socialist Republics (USSR) engaged in bipolar deterrence, which for this analogy more closely resembles a standoff between two gunfighters vice three. In a bilateral standoff, the gunfighter that draws first has the advantage. In a nuclear context, the US and USSR spent decades developing exquisite platforms and capabilities designed to negate a decapitation or "bolt out of the blue" attack—mitigating the gunfighter's "first move" advantage in a bipolar environment.

This paper will seek to elucidate the recent ascendancy of China's nuclear program to near-parity with the US and Russian Federation. Nuclear triads, arms control histories, and the current treaty environment will all be analyzed to effectively frame the potential effects of adding a third player to global nuclear deterrence. China's unchanged policy of "No First Use" will also be discussed, as its current nuclear

modernization efforts appear to be at least a strategic reinterpretation of its original, unchanging nuclear strategy. Acknowledging that there are other nuclear weapons states besides those with emerging or functional globally capable nuclear triads, this paper will only examine the potential dynamics between the US, the People's Republic of China, and the Russian Federation. Some nuclear weapons states, such as India, maintain a triad, but have only a regional capability.² Others, such as the United Kingdom and France, are globally capable, but do not have a triad.³ The outliers, or the potential rogue states, particularly North Korea and Iran, have either an ambiguous nuclear policy or a limited nuclear capacity, indicative of an inability to influence global nuclear deterrence.

Nuclear deterrence theory, or the theory that nations or powers with nuclear weapons will not attack other nuclear powers out of fear of retaliation, is fraught with nuance and risk. In the Cold War's bipolar nuclear deterrent environment between the global hegemonic powers, the US and the USSR, close calls, escalations, regional and proxy wars, and security dilemmas occurred at both a frightening scale and pace. What would the Cold War have been like if there were three global powers, all culturally and philosophically opposed to one another, and all vying for increased global influence and power? With the advent of the People's Republic of China's (PRC) modernization under Xi Jinping, the world is entering a new era of strategic deterrence: Tripolar deterrence. How will deterrence dynamics take shape with three peer nuclear powers? Even though the PRC's nuclear triad is not as advanced as the US' or the Russian Federation's, it still stands to benefit from

the element of existential deterrent that comes from being a major nuclear power. To adapt and remain in a position of strategic influence, the US must continue its own modernization efforts while reinvigorating arms control efforts and establishing avenues for strategic dialogue with both the PRC and Russia amidst deteriorating strategic relationships. The US must also understand that the emergence of the PRC as a nuclear peer will afford the PRC a measure of influence in diplomatic and economic spheres as well.

Modern Nuclear Triads and Differing Deterrence Philosophies

Although initially developed during the Cold War by the US and the USSR, nuclear triads still afford a nation an incredibly effective measure of existential deterrence. Formed through the combined delivery platforms of strategic, nuclear-capable bombers, sea (primarily submarine) launched nuclear weapons, and ground-launched nuclear weapons, nuclear triads afford a nation credible, variable, and scalable strategic weapons delivery options. Aerial delivery platforms offer recallability to a nuclear state, missiles responsiveness and rapidity, and submarines a guaranteed “second strike” capability. A complete and functional nuclear triad guarantees that a nation can endure a nuclear attack, and with a variety of nuclear platforms, conduct an effective counterattack, thus achieving nuclear deterrence with regionally and globally capable nuclear weapons states. Even in the proverbial “dawn” of the space and cyber domains of warfare, nuclear weapons provide a nation existential deterrence, or the ability to deter even conventional, asymmetric, or hybrid threats.⁴ The Russian Federation and the PRC are the only adversary states capable of ranging the entirety of the continental US with Intercontinental Ballistic Missiles (ICBMs), with North Korea’s emergent capability capable of ranging the eastern seaboard.⁵ Of the world’s nuclear powers, only two have complete, functional, and global nuclear triads: The U.S. and the Russian Federation.⁶ Over the last five years, the PRC began rapidly expanding their nuclear arsenal and capabilities in both fissile material production and platform diversification. Both the U.S. Department of Defense and the RAND corporation agree that the PRC is working to adopt a launch on warning (LOW) policy, akin to policies employed by the U.S. and the Russian Federation, a significant step towards nuclear modernization.⁷

Launch on warning policies, a deterrence strategy developed shortly after the advent of ICBMs, are an

aggressive deterrence posture with their own sordid history of close calls and accidents. As the name suggests, if a state has adopted a LOW policy, it will launch a retaliatory strike against an adversary’s nuclear attack before any missiles have impacted, or as soon as it is warned of an impending nuclear attack. Though the US did away with its LOW policy in 1981 and adopted a “launch under attack” strategy, the Russian Federation employs a holistic LOW strategy while the PRC is incrementally adding platforms to LOW status.⁸ The US deterrence strategy now relies on dual-phenomenology, or early warning concurrence between ground- and space-based sensors before seeking a decision from national leadership—a more prudent, albeit constrained, approach to deterrence than US adversaries.

The Russian Federation employs an additional grim strategy that is perhaps the most extreme interpretation of modern existential deterrence: Perimeter, or the “Dead Hand.” Developed during the Cold War, the Dead Hand system was designed to guarantee a Soviet nuclear counterstrike in the event of a nuclear decapitation attack by the US.⁹ Originally intended as a completely autonomous system, the Soviets fielded a modified version with human firewalls in 1985.¹⁰ The system remains functional and is still in use in the post-Soviet Russian Federation, with some Western assessments suggesting that the Perimeter system has been modernized from its original Cold War form.¹¹

Chinese deterrence philosophy appears to be at odds with its capability and platform diversification. As explained by Li Bin and Tong Zhao, Chinese policy and nuclear disarmament experts, Chinese Communist Party (CCP) leaders and policymakers “consider nuclear deterrence and compellence to be indistinguishable in most cases.”¹² In nuclear thinking, deterrence is designed to prevent action while compellence is designed to instigate action. While the US and Russia have long maintained nuclear dialogue even under periods of extreme tensions, the PRC appears to interpret any interference in nuclear capability developments or policy as adversarial, further dividing two of the three tripolar actors amidst deteriorating strategic relationships, and perhaps pushing the PRC to further nuclear modernization and diversification efforts.

Chinese Nuclear Philosophy

Chinese defense policy analysis yields even less clarity, as China’s stated nuclear policy has remained unchanged since its first nuclear test while Chinese



ABOVE: PACIFIC OCEAN (June 15, 2022) – An unarmed Trident II D5 Life Extension, also known as a D5LE missile, launches from the Ohio-class ballistic missile submarine USS KENTUCKY (SSBN 737) during a Commander Evaluation Test (CET) off the coast of Southern California. As the nuclear triad modernizes, the new Columbia-class SSBN—which is designed to provide SBSD capability into the year 2084—will be initially outfitted with the D5LE SWS. Each new Columbia-class boat will complete a Demonstration and Shakedown Operation (DASO) as part of the process to certify the boat for the strategic deterrence mission. (U.S. Navy Photo/Released)

conventional military policy, force structure, and leadership undergoes frequent periods of change. This is partly in fact because, as M. Taylor Fravel describes, the Chinese military has a generally custodial relationship with its nuclear weapons while policy, development, and employment are functions centralized within the highest levels of the CCP.¹³ Fravel continues to illuminate Chinese nuclear thinking and strategy by illustrating that Chinese leaders view nuclear weapons as having little utility besides “preventing nuclear coercion and deterring a nuclear attack on China.”¹⁴ The PRC’s leaders, now and at the advent of its nuclear weapons program, “did not envision either engaging in nuclear warfighting or using nuclear weapons to deter conventional threats.”¹⁵ Thus, Chinese leaders settled for a force large enough to conduct a nuclear counterstrike, but small enough to remain unprovocative throughout the Cold War.

Western, and particularly American, interpretations of Chinese nuclear policy and doctrine fail to account for why China has so rapidly developed and expanded its nuclear arsenal. Some current and popular Chinese military theorists have grown to appreciate the existential deterrence provided by a robust nuclear arsenal, as noted by the RAND Corporation in its 2017 analysis of PRC nuclear strategy.¹⁶ In a tripolar deterrence environment, existential deterrence may play a central role in the PRC’s global ambitions, particularly within the South China Sea.

Chinese nuclear ambitions are further fueled by doubts surrounding global and regional rival deterrence programs.¹⁷ Hallmarks of U.S. power, Conventional Precision Global Strike (CPGS) and Ballistic Missile Defense (BMD) are critical to U.S. national defense. Though both CPGS and BMD are framed and employed as defensive capabilities, an unintended consequence of developing exquisite

defenses is causing rival nations to develop equally exquisite offensive platforms. Under Xi Jinping, the existence and stated perceptions of CPGS and BMD are used to justify the PRC's modernization efforts.¹⁸ Though Western eyes may see a significant divergence between Chinese stated nuclear policy and capability development, as far back as 2013, the PRC's Central Military Commission reconciled "no first use" with nuclear diversification to address a "global complex nuclear security environment."¹⁹

The Current Nuclear Treaty Environment

Nuclear treaties are the bedrock of international order and agreement as it pertains to the construction, deployment, and international accountability of nuclear weapons. According to the Office of the Deputy Assistant to the Secretary of Defense for Nuclear Matters, the US has either been a signatory on, or abided to terms set by, twenty-three nuclear treaties since the 1950s.²⁰ Of those, the US has formally withdrawn from three: the Joint Comprehensive Plan of Action in 2018, the Intermediate-Range Nuclear Forces Treaty in 2019, and the Anti-Ballistic Missile Treaty in 2002.²¹ In the post-Cold War era, the US' most significant nuclear obligations and constraints fall under the New START Treaty.

Signed jointly by the United States and the Russian Federation, New START took effect on February 5th, 2011, and is slated to remain in effect until February 4th, 2026.²² New START places limits on several US and Russian offensive, strategic nuclear capabilities: 700 or fewer deployed ICBMs, submarine-launched cruise missiles, and nuclear-capable bomber aircraft.²³ Additionally, the US continues to abide by the Nuclear Nonproliferation Treaty and the Comprehensive Test Ban Treaty (CTBT), although the US is not legally bound to the CTBT. In a deterrence context, the US treaty process is more exacting than its adversaries but fosters international trust. Once a resolution for treaty ratification is approved by the US legislative branch, the treaty is codified into US law, requiring an additional sequence of debates and voting to overturn the treaty.²⁴ In authoritarian or centrally-governed states, however, treaties can be withdrawn from rapidly and unilaterally, as seen with Russia's recent withdrawal from the CTBT in 2023.²⁵ The US, however, faces challenges to existing nuclear norms in the treaty environment, from old and new adversaries alike.

In February 2023, the global deterrence environment fundamentally changed. After months of

delaying or outright cancelling routine New START inspections, Vladimir Putin formally suspended the Russian Federation's participation in the New START program.²⁶ This, of course, comes after the complete breakdown of the Intermediate-Range Nuclear Forces Treaty between the US and Russian Federation in the mid- to late-2010s.²⁷ Russia's actions in 2023 continued to degrade nuclear stability with Putin's unilateral withdrawal from the CTBT.²⁸ Putin's stated reasoning for the Russian withdrawal are perceived imbalances with both the US and China, though opportunities abound for an escalatory demonstration in light of its struggles in the Ukraine.²⁹ As the world enters a more challenging deterrence environment, treaties that cooled global tensions have disappeared, creating an environment of unconstrained nuclear development by at least one of the three centers of gravity in tripolar deterrence.

If the PRC continues its current trajectory, it will be a nuclear peer to the Russian Federation and the US within the next decade.³⁰ Adding to global complexities and arms control challenges, the overwhelming majority of arms control agreements limiting specific capabilities or arsenals are historically limited to bilateral agreements between the US and Russian Federation. To navigate a tripolar deterrence framework in a season of frustrated arms control, the US must remain clear-eyed about its adversaries and their goals, intentions, and ambitions while simultaneously working to create strategic dialogue among the three Great Powers.

The US should also look to its past, acknowledging the shifting dynamics between bipolar and tripolar environments, to guide and shape nuclear policies, dialogue, and efforts to prevent miscalculations in a more complex environment. The addition of the PRC to the nuclear roundtable may not necessarily induce destabilizing effects, as indicated by the PRC's willingness to dissuade Russian nuclear aggression.³¹ The US will need to place more emphasis on cooperative actions within competition while acknowledging that its own capability advancements may continue to alter international perceptions.

Deterrence Dynamics in Tripolar Environments

Recalling the gunfighter's analogy, perhaps the most significant shift from bipolarity to tripolarity lies in where the dominant strategy exists. In a bipolar framework, the first player to take a turn has a distinct advantage over the other. In a tripolar environment, the player

that moves last gains a fundamental advantage, given there are no information advantages, hidden alliances, and all players are honest. Tripolar nuclear deterrence will not be so simple for the US, though the basic framework of tripolarity favors defensive (blocking an adversary's shot) or deterrent (dissuading an adversary from moving first) behaviors and technologies.

Deterrence dynamics are set to fundamentally change as China navigates its way from nuclear state to nuclear power in the coming years. To maintain effective deterrence, the US must seek avenues for strategic dialogue with the PRC, messaging stability and prudence to mitigate China's adversarial views of international deterrence. Compounding to those emergent challenges, the US must now deter not just one, but two peer nations with a history of violating or ignoring arms control agreements.³² Simultaneously, the US must also acknowledge and deter the reinvigoration of nuclear weapons as part of Russia's overarching military strategy. Accomplishing both, in a simultaneous and enduring fashion, will pose a new set of challenges for the US Department of Defense and Department of State.

The US has one key advantage over the Russian Federation and the PRC in the coming era: stable international partnerships and alliances. The US may need to expand its "by, with, through" defense doctrine to encompass nuclear deterrence by allowing regional nuclear allies in the United Kingdom and France to deter Russian nuclear bullying, while engaging the PRC in diplomatic centers of gravity to maintain nuclear stability in the geopolitically fluid environment of the Indo-Pacific Region. When considering arms control agreements and nuclear treaties, as soon as the PRC is a nuclear peer to the US and Russia, bilateral agreements will only constrain two of the three centers of gravity. Unless an agreement is trilateral (or greater) in nature, and all signatories allow for independent inspection and verification that all rules are followed, any bilateral agreement will disincentivize participation for risk of ceding advantage to the non-signatory. Multiple bilateral nuclear agreements may also arise, in situations like India-Pakistan or Israel-Iran, offering the US opportunities to demonstrate credible leadership while simultaneously generating nuclear stability. To circumvent a more complex deterrence environment, US partners and allies can provide the US an avenue to deter Russian nuclear aggression without simultaneously raising tensions with the PRC.

Behavior Accompanies Status

Compared to the previous Chairmen of the CCP, Xi's vision and strategy for a dominant China was the impetus for changed behavior. Achieving a nuclear triad is an extension of Xi's vision. With the existential deterrence provided by a robust nuclear arsenal and detection matrix, questions abound for PRC ambitions and global ambitions. Is a functional nuclear triad a strategic trigger for the PRC to conduct military operations to seize Taiwan? Will the PRC use their nuclear umbrella to further efforts in Military-Civil Fusion or the Belt and Road Initiative? Based on their history of self-imposed nuclear constraint, not necessarily. However, the PRC does not ethically constrain itself to the rule of law as the US does, frequently agreeing to historical arms control agreements while furthering its own capability development.³³ Under Xi Jinping, the CCP is the most centralized it has been since at least the 1980s, fueled by Xi's national desire to create a "truly world class military."³⁴ The behavior has already changed. Xi's military resurgence accompanied an annual economic growth of ~10% per year from 2000-2022 – it should come as no surprise that a leader determined to bring his country to the world's stage would like the military capability to both project influence and to protect it.³⁵

As the US looks to navigate a tripolar deterrence environment, diplomats, policymakers, and defense experts should expect China to continue its policies of global expansionism and garnering influence in economically underdeveloped nations. Policymakers should, however, understand that the PRC will soon be protected by nuclear capabilities at least as capable as the one that has served as the backbone of US policies since the Cold War.

Balance or Bandwagon: Adversary Behavior in Integrating the Third Peer

The second and most enduring adversary for the US in the upcoming gunfighter's dilemma is the Russian Federation. Russia and China have, at the surface level at least, mutually fulfilled their "partners without limits" agreements, though the latter is quickly becoming the former's elder brother in both "soft" (diplomatic) and "hard" (military or economic) power.³⁶ Given China's economic rise since 2000, it should come as no surprise that two authoritarian regimes would choose to create strategic alliances, in both economic and military spheres. After all, Russia's economy is dwarfed by

both the PRC and the US, with the US gross domestic product approximately 14 times larger than Russia's.³⁷ In building strategic partnerships, Russia demonstrated eagerness to partner with the PRC as authoritarian regimes tend to seek authoritarian allies as frequently as democracies seek out allies in other democracies.³⁸

With the PRC's emergence as a peer, and with both the PRC and Russian Federation viewing the US as the threat, theorist Stephen Walt provides insight into their burgeoning partnership. Vladimir Putin, and by extension the Russian Federation, faced a decision in how to relationally approach the modernized PRC. In the "bandwagoning" theory, Russia could have sought an alliance with the US (the threat) to tip the balance of power in favor of the US.³⁹ Conversely, by "balancing" the threat (the US), Putin found a strategic ally in Xi Jinping's centralized, economically viable, and militarily ambitious PRC.⁴⁰ Russia's balancing behavior should be expected to continue as long as both the PRC and Russia view the US as the preeminent threat – until the US either radically alters the strategic paradigm or is no longer considered a threat to PRC and Russian interests, or until the PRC-Russo partnership erodes. This may work in favor of global security and nuclear stability. To maintain its strategic relationship with the PRC, Vladimir Putin may be forced to cool nuclear rhetoric and adopt more non-threatening nuclear policies to assuage PRC concerns and prevent damaging the PRC's global reputation. After all, the PRC considers any sort of nuclear interference as adversarial, and Vladimir Putin is prone to voice his willingness to employ nuclear weapons.⁴¹

Revisiting the Gunfighter's Analogy

The US, with its decades of experience in close calls, bipolar deterrence, and mitigating miscalculations, may soon face a new dilemma. The PRC, despite its unchanging nuclear doctrine, now has the fastest growing nuclear arsenal in the world with global ambitions to match. The Russian Federation rounds out our tripolarity. In the gunfighter's analogy, the US is faced with an adversary with superpower ambitions that will soon enjoy the protection only a functional nuclear umbrella can provide, and another volatile adversary with an appetite for invasion and unjust war. To navigate the new tripolar world, the US must remain clear-eyed in deterrence, seek to reinvigorate dialogue even through globally deteriorating strategic relationships, and understand that the deterrence framework will fundamentally change. During the Cold War, the US successfully conducted bipolar deterrence,

where the gunfighter who draws and shoots first has a dominant advantage over the opposite gunfighter. The advantage will soon lie with the gunfighter who is patient enough to avoid miscalculation, talks before drawing his weapon, keeps nuclear temperatures cool, and uses the most robust defenses to mitigate or deter an accidental or miscalculated first shot. ■

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Lessons from An Earlier 'Golden Age': 1900-1922

By: Col. Asheleigh O.J. Gellner

America's victory over Spain in 1898 had two significant impacts on American politics and foreign policy. First, it ushered in a new era of American hegemony in the Western Hemisphere, which neatly aligned with America's isolationist tendencies and the pivotal 1823 U.S. policy known as the Monroe Doctrine. This doctrine, which remained influential throughout the 20th century, warned European powers against further colonization or interference in the Americas and threatened U.S. intervention if they did not adhere.¹ Second, the Spanish-American War victory propelled the U.S. into world politics with its new insular possessions, including the far eastern territories of Guam, the Philippines, and Samoa. More than 120 years later, there is growing consensus that the U.S., under a second Trump administration, is returning to strategies from the early 1900s including a new era of grand strategy that prioritizes the Homeland and the Pacific.² In the first half of the twentieth century, U.S. Naval leaders routinely used the Monroe Doctrine and the Open Door to China policies as justifications for an expanded Naval fleet and rationale for advanced bases across the Pacific.³ The Navy acknowledged these were American policies that carried no international legal basis. Still, Navy leaders thought integrating the Monroe Doctrine and the Open Door to China policy into naval strategy could further America's foreign policy, defend its interests, and in doing so, expand the U.S. Naval fleet.

This article draws from larger historical research concerning inter-service rivalries over Guam and the Philippines in the early 1900s in which the U.S. Navy's thoughts about advanced basing and the U.S. Army's lessons from the 1904 Russo-Japanese War

strongly influenced each services' concerns over land and coastal defense.⁴ Together, both services, while entrenched in their doctrines, struggled to reach a joint solution to advanced basing in the Pacific before the 1922 Washington Conference treaties redirected naval investment and innovation elsewhere. The repercussions left both the Philippines and Guam vulnerable. Today, as current military leaders pivot their services to align with President Donald Trump and Secretary of Defense Pete Hegseth's visions focused on the Homeland and the Pacific, the inter-service rivalries, poor civil-military relations, and fiscal constraints of the early 1900s may provide a window into hard lessons about readiness and lethality and the services' alignment with the Executive Branch.

The U.S. Navy's endorsing of an American-focused doctrine and advocating for an expanded fleet through the first two decades of the twentieth century did not translate to support for advanced basing options, resources, or fortifications abroad. Although policymakers supported acquisition of the Philippines and Guam and viewed the expansion as a gateway to China, many leaders did not desire to advance the military's role in the Pacific nor establish European-style imperialism.⁵ Furthermore, policymakers did not see an international or political policy role for the military and viewed statecraft and diplomacy as civilian matters. The work of various historians such as William R. Braisted and Richard D. Challener showcased inefficiencies with civilian oversight of the Navy's proposals.⁶ This was especially evident during the Woodrow Wilson Administration, with Secretary Josephus Daniels heading the Navy. For example, in 1913 when heightened tensions between Japan and

U.S. led the General Board and the Army Navy Joint Board to urge the Wilson Administration to mobilize the fleet and begin war preparations in the Pacific. Secretary Daniels failed to support the military recommendations.⁷ Many critics, including then Vice Admiral Williams Sims also cited Daniels' aggressive preoccupation with corporate profiteering, including his brazen seizure of oil reserves and refusal to pay oil companies during World War I, as an example of how his tenure created civilian-military friction and failed to prepare the Navy with the needed modernization and investment.⁸ Nevertheless, research does not specify if the Navy understood the complexities of their civil-military interactions and that military decisions and rivalries with the U.S. Army could have unintended consequences on subjects, such as resource allocation and investment.

The General and Joint Boards

To forge naval consensus, the Secretary of the Navy John D. Long established the General Board in 1900 with the purpose of preparing the Navy fleet for war and for naval defense of the coast. In 1903, the Secretaries of War and the Navy established a Joint Board comprised of Army and Navy board members with a charter that required members reach "common conclusions for all matters calling for the cooperation" between the Army and Navy.⁹

Although the Navy's General Board had free reign to determine probable enemies, committees worked independently versus collaboratively in the early years.¹⁰ The Army and Navy Joint Board's work also did not seek guidance or consult with the State Department, the executive branch (aside from the service secretaries or occasional engagement with the president), or the legislative branch. Instead, officers from both boards drew their understanding of foreign affairs from observation. For example, the General Board argued for a strong fleet and naval capability. General Board members viewed conflict as inevitable due to trade rivalries among great powers, and, guided by the Mahanian philosophy, believed that "nations must expand if they are to achieve greatness."¹¹ Furthermore, many officers who served on the General Board from 1900-1917 were veterans of or knew shipmates who served in Manila Bay during the Spanish-American War. Therefore, a "decisive battle" at sea and the need for a powerful battle fleet shaped much of the veterans' early recommendations.¹² Recommendations were often incompatible with U.S. Army priorities and

offered legislators insufficient appropriation options which ultimately sidelined Far East basing decisions.

Two trends became apparent in the early days of the General Board and the Army and Navy Joint Board. First, Guam was of secondary importance and primarily considered a coal depot. Second, no consensus existed where to install a fortified fleet base in the Philippines although board members agreed on the need for one and that both Guam and the Philippines required defense. From the onset, Admiral George Dewey, a hero of the Spanish-American War who was responsible for the victory at Manila Bay and headed the Navy's General Board, favored the Philippines and specifically Subic Bay.¹³ The Army's defense assessments favored the larger area of Manila Bay. In 1906 the Taft Board recommended additional build-up of the insular possessions. Dewey actively endorsed its findings by arguing that the territories "cannot be held unless the principal ports, naval bases, and coaling stations are fortified before the outbreak of war."¹⁴ In 1907, the General Board directed the Naval War College to study air-raid defenses needed for Guam. This study resulted in defense proposals; however, in the same year, the Army and Navy Joint Board concluded it was not practical to assemble sufficient defense forces in Guam.¹⁵

Perhaps more importantly, despite military experts' conversations regarding the need to protect the new American Far East territories, lawmakers prioritized domestic needs and declined defense spending elsewhere. For example, Congress' Naval Committee cut the Department's recommendation from 16 ships (consisting of destroyers, submarines, colliers, mine layers, and repair support) in half, suggesting it unlikely advanced basing anywhere would have been approved.¹⁶ Meanwhile, the General Board and the Army and Navy Joint Board's conflicting recommendations over location continued, weakening their analytical credibility. The disagreements proved particularly damaging given the impact of reduced budget proposals created by marred joint service work and minimal congressional support overall. Together, these factors resulted in stagnation regarding Far East advanced basing initiatives.

Entrenched Doctrines

The services continued asking for military spending often using narrow and service-specific justifications. One of the main catalysts for this wrangling occurred because of lessons from the Russo-Japanese War.

In 1908, Major General Lenard Wood successfully encouraged President Theodore Roosevelt to halt build-up plans for Subic Bay. Major General Wood presented analysis that compared the geography between the Philippines' Subic Bay and Port Arthur where the Russian Navy was decimated in 1904. Major General Wood convinced the President of Subic Bay's "utterly indefensible" geography that provided for a barrage of gunfire from the mountains and rear. He added that Subic Bay's natural geographic structure provided operational advantages for executing an amphibious assault and could result in the adversary's defeat similar to the slaughter of the Russian Navy in 1904.¹⁷ Wood's persuasion worked, but Roosevelt was furious about spending the political capital on Subic Bay and chastised the Joint Board for the lack of coordination on a joint matter, for which he had spent years advocating.¹⁸ Admirals and Generals of the period frequently sacrificed civil-military credibility for an inter-service victory, which ultimately paid zero long-term dividends. In this case, the president put his political efforts toward Pearl Harbor, Hawai'i instead.

The Russian defeat at Port Arthur revealed a crucial difference in how the Army and Navy thought about advanced bases, coastal defenses, and naval stations. The Army viewed these through the lens of defensibility. Leaders focused and based decisions on the cost of mortars, direct gun fires, and men to defend a harbor, port facility, or advanced base and these views remain fundamental in current Army doctrine.¹⁹ On the contrary, several of Mahan's writings stressed the subordinate nature of defensive operations to naval offensive capabilities. "The offensive element in warfare is the superstructure, the end aim for which the defensive exists."²⁰ Mahan later wrote that the main military requirement of a naval station should be that its "useful in war."²¹ Port Arthur's attack presented a significant problem for Navy and Army strategists. Each service saw the *front*, aligned its forces, supplies, and offensive and defensive positions differently in times of war.²² This dilemma eventually led to a stagnant, inter-service rivalry over the future of Subic and Manila Bays. The discord also unconsciously extended to Guam with the General Board and the Joint Board's inability to select an advanced base that excluded the Philippines. Between 1907 and 1909, President Roosevelt directed the Joint Board to re-evaluate Subic Bay, Manila, and Cavite, a province of the Philippines located in the Calabarzon region (see Figure 1). Serious consideration of Guam as a permanent Far East fleet base option may have generated a definitive solution between those options, but board members did not consider Guam. Efforts were instead narrowly and consistently

focused on the Philippines and arguments about the two services' doctrinal strengths. To be sure, building and fortifying Guam would have been expensive.²³ Still, San Luis D'Apra was the only location possible for a fleet base in the region, and board members did primarily agree upon Navy and Army requirements in board meetings. However, the General Board and Joint Board leadership did not consider Guam a viable option until 1913. Those studies became overshadowed by a world war and resulted in inaction. Finally, the 1922 Washington Conference article XIX expressly prohibited the build-up of any Far East location west of Pearl Harbor, Hawai'i. This permanently eliminated the opportunity to invest in the Philippines or Guam entirely. Therefore, the broad indecisiveness and arguments over the location of any Far East bases crippled naval progress and led to conflicting recommendations from the General Board and Joint Board. As a result, policy-makers abandoned all future build-up opportunities.

The Navy experienced some successes in their civil-military interactions. Pearl Harbor, Hawai'i was a popularly supported naval project, well-funded, and able to be built in a relatively short time.²⁴ Additionally, World War I pressed President Woodrow Wilson to order service Secretaries to develop a comprehensive plan for national defense. However, military planners turned their focus to threats in the Western Hemisphere with Germany's potential attack of the Caribbean. Nonetheless, the General Board responded with a five-year, \$100 million shipbuilding proposal.²⁵ Congress approved the Naval Appropriations Bill in 1916, which consisted of ten battleships, six battle cruisers, and fifty destroyers. Still, passing the legislation required considerable pressure from Secretary of the Navy Daniels and General Board President Dewey.²⁶ The difficulty in which the Navy had to shore up support and budgetary spending for the U.S. despite a world war affecting its interests showcased the broader struggles of the General Board in convincing lawmakers that the U.S. Navy needed real investment. Far East basing priorities, therefore, took a backseat.

Conclusion

By the turn of the century, with new territories in America's possession and Admiral Mahan's ideas about sea power taking hold, the U.S. Navy had a formidable fleet from which to project its influence. However, America also found itself politically naïve to the challenges before it. From 1900 onward, the General Board made consistent recommendations to undertake a major shipbuilding program and build critical

defense, fortifications, and fleet infrastructure. The Navy argued this would enable a two-ocean Navy, which was necessary to protect the U.S. and adequately project its influence abroad. Unfortunately, the Navy was less than effective in its advocacy for naval investment, especially in the newly acquired territories of the Pacific despite integrating executive branch policies. Board recommendations indicate the debate over suitable Far East locations for fleet bases was further exacerbated by an Army-Navy rivalry, which undercut the military's position during interactions with the president, Congress, and other civilian policymakers. Furthermore, the independent Navy Bureaus, who reported directly to the Secretary of the Navy, were often resistant to General Board recommendations.²⁷ These factors delayed or sometimes ended Far East naval expansion as initially envisioned and routinely resulted in Guam being considered of nominal importance.

Likewise, the contemporary military struggles with a unified focus on integrating its main mission and operations into the Pacific theater, which is naval and airpower dominant. Yet, the U.S. Army's current voice and advocacy, especially with a prior Army officer as Secretary of Defense, is strong and forceful. Given the U.S. defense priorities are once again returning to Monroe Doctrine isolationism, the services would do well spending their energy on developing true joint objectives and proposals that align directly with current Department of Defense and Presidential priorities, rather than on those that support their unique doctrines. Joint initiatives and cost-saving proposals that support and defend the Homeland and the first island chain in the Pacific are likely the only way for the services to collectively preserve and garner investment and financial support from a Congress and President focused on severe federal budget cuts and governmental reforms. ■

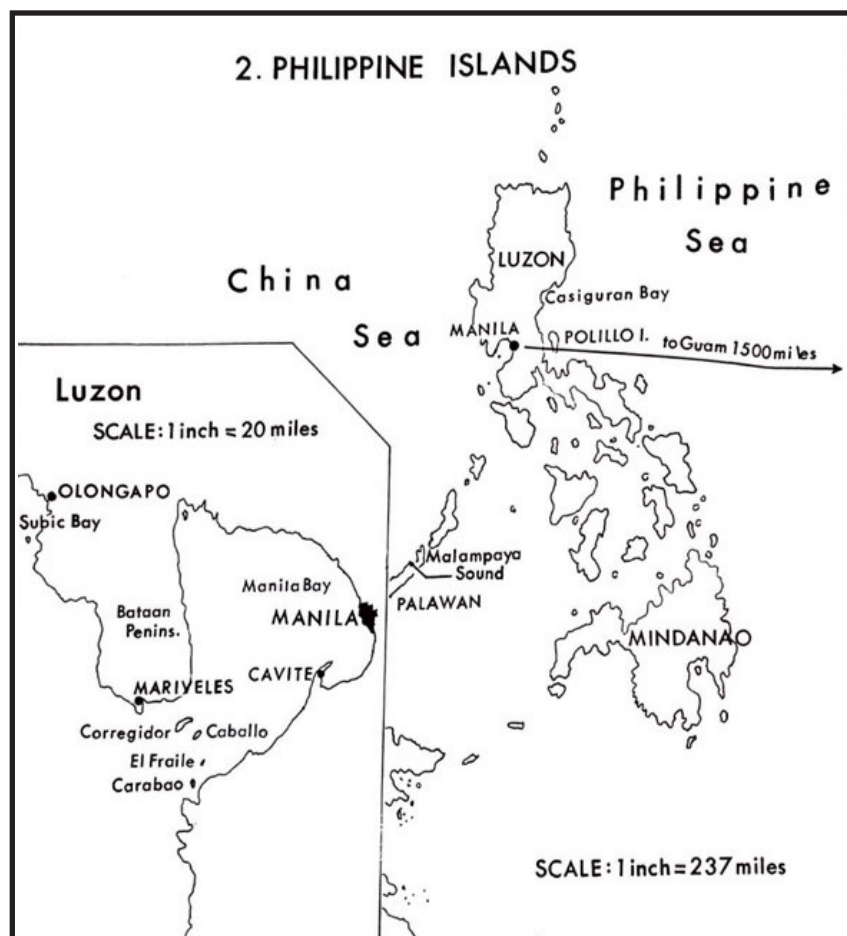


FIGURE 1: Map of Philippines.²⁸

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Nuclear Weapons Testing as an Instrument for Advancing a State's Political Objectives

By: Lt. Col. Christopher Mihal

Introduction

Nuclear weapons tests are traditionally viewed as a means by which a state ensures the utility and capability of its nuclear weapons. However, nuclear weapons tests, when used for demonstrative rather than scientific purposes, are a public and unquestionably dramatic means by which a state can reinforce deterrence and demonstrate its technical capability and power. As such, it is surprising that there has been little research into the political effects of nuclear weapons tests. A cornerstone of nuclear deterrence is credibility, and one means by which a state can achieve credibility is by conducting a nuclear weapons test. Demonstrating a state's willingness to detonate a nuclear weapon reinforces a state's credibility and potentially effects the state's political outcomes.¹

Frankel, Scouras, and Ullrich, along with other defense community experts, debate the utility of the US resuming nuclear weapons testing.² Since February 2022, Russian officials and their allies have threatened nuclear strikes against the West 87 times.³ Russia has un-ratified the Comprehensive Test Ban Treaty (CTBT), while the dissolution of the Intermediate Nuclear Forces and Open Skies Treaty have further decayed arms control between the United States (US) and Russia. Policy experts in the US have begun debating whether a resumption of nuclear weapons testing aligns with the US' strategic interests.

Authors including Peters and O'Brien focused their debate on the technical need of testing new nuclear weapons systems, how a nuclear test will benefit US security, and the risks associated with other

countries resuming nuclear testing.⁴ Opponents to resuming nuclear testing note that states with nascent nuclear capabilities—e.g., North Korea—are much more likely to test to prove their systems work, whereas for the US, “the US nuclear arsenal's ability to inflict devastating damage on Moscow or Beijing is unquestionable to anyone with a pulse.”⁵

Policymakers proposed reconvening nuclear weapons testing to demonstrate resolve, though it is not always clear that nuclear weapons testing is effective at signaling resolve.⁶ There may be times where conducting nuclear weapons testing signals a state's resolve, and this paper analyzes several case studies of when states potentially used nuclear weapons tests for political rather than scientific purposes.

Background

Ten countries have developed nuclear weapons since the nuclear age began in 1945, and of those, only one, South Africa, chose to relinquish its weapons. The list of 10 countries excludes countries that partnered with other countries for storing weapons at their sites. For example, the US stationed nuclear weapons in the homeland of several North Atlantic Treaty Organization allies and Russia placed tactical nuclear weapons in Belarus in 2023.⁷ Ukraine, Belarus, and Kazakhstan authorized storing of Soviet nuclear weapons in their territories following the dissolution of the Soviet Union, but all three states chose to return the weapons to Russia.⁸ States have chosen different means by when and how they conduct nuclear weapons tests, with two states choosing not to conduct any nuclear weapons

testing. Figure 1 shows the timeframe for when each nuclear weapons state acquired and ceased to test nuclear weapons.

Figure 1 illustrates that the US and Union of Soviet Socialist Republics (USSR) conducted the majority of the nuclear tests, and they possessed approximately 90% of the total nuclear weapon stockpile.¹⁰ While some US and Russian nuclear weapons are stationed outside their respective countries, the nuclear weapons remain under the US and Russian control. By analyzing a selection of cases demonstrating significant moments in the history of nuclear weapons testing, this study seeks to answer the research question: *do states use nuclear weapons tests to advance political objectives?*

Literature Review

Research by Ramesh, Jacobs, Smith, Toshihiro, and others examining the politics of nuclear weapons testing specifically is surprisingly scant. Most research into nuclear weapons tests investigate either the sociopolitical aspect of site selection for the tests or the health effects on residents near such sites.¹¹ Given the continued impact and health effects of radioactive fallout on a population that had no real say in conducting nuclear tests on their territory—including Kazakhstan, Algeria, French Polynesia, and the Marshall Islands—it is understandable that a variety of human rights (and anti-nuclear) activists place great emphasis on this research. As individuals in states where nuclear testing occurred against the will of the local populace are still dealing with claimed health issues relating to fallout from nuclear testing to this day, the impact of so-called nuclear colonialism is a sensitive and easily studied topic for a wide range of academic fields according to Sheriff, Cooper, Schalk, and others' research.¹² However, broader research on the political, rather than human, impact of nuclear testing is minimal.

Studies, analyses, and interpretations of data and evidence currently existing on nuclear weapons tests is almost entirely technical, with little information on the effect nuclear weapons testing has on international

State	Year Nuclear Weapons Fielded	Year Ended Nuclear Testing	# of Tests
US	1945	1992	1,054
USSR/Russia	1949	1990	715
United Kingdom (UK)	1952	1991	45
France	1960	1996	210
China	1964	1996	45
India	1974	1998	3
South Africa	1979	No explosive testing	0
Pakistan	1998	1998	2
North Korea	2008	2017 / testing may be ongoing	6

FIGURE 1. Dates When States Acquired Nuclear Weapons, Ended Nuclear Weapons Testing, and Total Number of Tests by Country

relations and world politics. Important technical studies regarding nuclear weapons testing includes Yang, North, Romney, and Richards' worldwide nuclear explosions databases, a continually updated tally from Arms Control Today, as well as US-specific information from the Nevada Test Site (NTS).¹³ The NTS is now called the Nevada National Security Site (NNSS). The NNSS provides an overview of atmospheric nuclear weapons testing and the Catalog of Worldwide Nuclear Testing.¹⁴ Figure 2 provides a graphical representation of nuclear tests from 1945 to 2022.

Implementation of treaties such as the 1974 treaty between the US and the USSR known as the Treaty on the Limitation of Underground Nuclear Weapon Tests, limited tests to a maximum yield of 150 kilotons.¹⁶ Informal testing moratoriums between the US and USSR impacted the number of nuclear tests from 1958 to 1961. The 1961 moratorium ended when the USSR launched its largest test series ever and detonated the Tsar Bomba, the largest nuclear weapon known to date.¹⁷ The US announced a unilateral nuclear weapons testing moratorium in nuclear testing 1992.¹⁸ Current research omits examining nuclear weapons testing as an instrument for advancing a state's political objectives such as nuclear deterrence.

When it comes to nuclear deterrence theory, however, Mehta and Whitlark; Powell; Sagan; Walt; and Hopf provided a considerable body of research, beginning in the 1940s, that compares nuclear latency to nuclear acquisition and addresses how nuclear proliferation impacts deterrence. Walt, Hopf, and other researchers

identified how countries' decisions to enter alliances supported their security objectives and served as a form of nuclear deterrence.¹⁹ However, research by scholars such as Alexander Wendt and Scott Sagan identified a gap in information about states' use of nuclear testing for advancing political objectives. Academic scholars analyzed nuclear weapons acquisition more frequently than nuclear weapons testing and identified factors that influenced states' acquisition of nuclear weapons.²⁰ Wendt's constructivism theory explains how "five hundred British nuclear weapons are less threatening to the US than five North Korean ones, despite the material disparity, because of the different meaning and identities attached to both states." Wendt's theory reinforces the role a state's perception of another state plays when categorizing a state as a rival versus friend. Based on Wendt's theory, North Korea owning and testing a nuclear weapon is perceived and treated as a threat versus France's ownership and testing of a nuclear weapon.²¹

Research into states' decisions regarding nuclear weapons focuses specifically on acquisition of nuclear weapons, rather than the actual testing of the nuclear weapons.²² Works by Schilling and Rosenberg detail the logic behind acquiring a hydrogen bomb, the most advanced type of nuclear warhead yet developed. Deterrence research emphasizes that a state's nuclear deterrence must be credible—that is, if a state makes a nuclear threat, the state's opponent must believe that the state will deploy an operationally capable weapon.

Nuclear Weapons Testing Timeframes

A Presidential Decision Directive from the Department of Energy states that the timeframe it would take to renew nuclear weapons testing at the NNSS is 24-36 months.²³ This is significantly longer than historical tests, with data also from the NNSS indicating that a US test during the Cold War could take place anywhere from one to four months from the announcement of testing, and from 1961 to 1992 nuclear weapons tests were conducted on at least

Number of nuclear weapons tests, 1945 to 2022

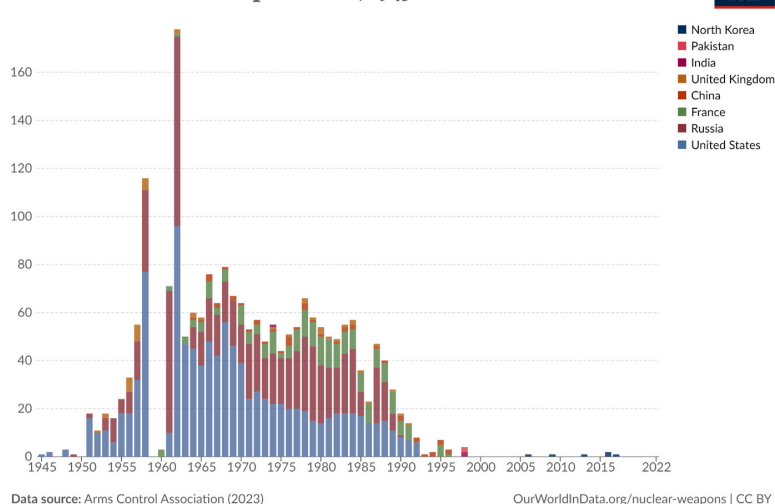


FIGURE 2. Global Nuclear Weapons tests by country.¹⁵

an annual basis.²⁴ Following the Soviet breaking of the three-year bilateral nuclear testing moratorium in September 1961, the US conducted 100 nuclear tests in just 14 months.²⁵ Even the US' first test after World War II, *Operation Crossroads*, required less than six months from conception to execution.²⁶

Evidence from nations other than the US points to potentially even shorter setup times for nuclear testing. In March 1998, India decided to conduct a nuclear weapons test and conducted the test a month later during May 1998. India claimed that they conducted the test to counter Pakistan's nuclear testing preparations. However, Pakistan claimed their decision to test on May 30, 1998, was due to the Indian testing earlier in the month according to Sublette and Khan's research. Another example of timetable acceleration is France's *Gerboise Verte* nuclear test on April 25, 1961. During a military coup in Algeria that began on April 21, 1961, France's government accelerated the final detonation of its *Gerboise* series of nuclear tests by several weeks. India, Pakistan, and France's nuclear weapons tests demonstrated states' ability to conduct nuclear weapons tests within weeks of the planning phase.

The timeframe for conducting a nuclear test can be indicative of its purpose. A very short time frame is a potential indicator that a nuclear weapons test is political or demonstrative in nature, while a lengthy lead time is more an indicator that a test is scientifically motivated.

Scientific Purposes of a Nuclear Weapons Test

The US Department of Energy lists six purposes for nuclear weapons tests – Joint US-UK tests, Plowshare, Safety Experiment, Vela Uniform, Weapons Effects, or Weapons Related.²⁷ Figure 3 shows the number of nuclear tests by purpose conducted by the US, along with the definition from the Department of Energy for each term.

Note that none of these official DoE nuclear weapons testing purposes explicitly state a political reason for conducting a test. However, there is evidence that some states designed their nuclear tests for demonstrative purposes, possibly for deterrence, to potential adversaries rather than for scientific reasons. The following case studies support how states used nuclear weapons tests to advance political objectives.

Case Studies: Potential Politically-Motivated Nuclear Weapons Tests

The US conducted *Operation Crossroads* in 1946 to advance political objectives. *Operation Crossroads* demonstrated the effect of nuclear weapons on surface vessels. Secretary of State James Byrnes contended that a public nuclear weapons demonstration might make the USSR “more manageable.”²⁹ J. Robert Oppenheimer himself, lead scientist for the

Manhattan Project, noted that the data from *Operation Crossroads*’ two nuclear detonations could have been obtained much more accurately and cheaply in a lab.³⁰ Foreign officials from Australia, Brazil, Canada, China, France, Egypt, the United Kingdom, Mexico, the Netherlands, Poland, and the Soviet Union were all invited to the tests, which occurred in the remote Bikini Atoll in the South Pacific. Bringing high-level officials from such disparate parts of the globe to such a geographically isolated area was clearly to demonstrate American might.³¹ The Soviet press downplayed the significance of the tests and referred to the destructive power demonstrated at Bikini Atoll as “disappointing.” The Soviets likely used the propaganda to shape the narrative with the Soviet populace until the Soviets built their own nuclear weapon.³²

The race between India and Pakistan to conduct their nuclear weapons tests in 1998 was illustrative, with both India and Pakistan blaming the other for being “forced” to do a nuclear test. P.K. Iyengar, a leading scientist from India, conducted India’s first nuclear test, codenamed the “Smiling Buddha,” in 1974. P.K. Iyengar decried that the 1998 Indian nuclear tests were conducted in haste because of intelligence information that Pakistan was preparing their own nuclear test.³³ Pakistani Prime Minister Mohammad Nawaz Sharif stated that Pakistan’s nuclear tests were because “our hand was forced by the present Indian leadership’s reckless actions [conducting a nuclear test].”³⁴ Both India and Pakistan cited intelligence that the other was

Nuclear Weapons Test Purpose	Definition	Number of Tests Conducted
Joint US-UK	A nuclear test conducted jointly by the US and UK under a cooperative agreement that has remained in effect between the two countries since August 4, 1958	28
Plowshare	A test in the Plowshare Program intended to explore the application of nuclear explosives to peaceful uses	36
Safety Experiment	An experiment designed to confirm that a nuclear explosion would not occur in case of an accidental detonation of the chemical explosive associated with the device	88
Vela Uniform	A US Department of Defense program designed to improve the capability to detect, identify, and locate underground nuclear explosions	7
Weapons Effects	A type of atmospheric test conducted to evaluate the civil or military effects of a nuclear detonation on various targets, such as military hardware	100
Weapons Related	A type of atmospheric test conducted for the purpose of testing a nuclear device intended for a specific type of weapon system	890
Total Nuclear Tests		1,149

FIGURE 3. US Department of Energy nuclear weapons testing purposes and their definitions.²⁸

preparing for a nuclear test as an excuse for conducting their own nuclear tests.³⁵ The signing of the CTBT in 1996 worried India, who felt that the treaty was biased against it, and so India began taking steps to solidify its nuclear program.³⁶ After becoming Prime Minister of India in 1998, Atal Bihari Vajpayee accelerated nuclear weapons testing, leading to India's Pokhran-II series of tests in May 1998, stating that nuclear weapons would demonstrate India's might and garner respect.³⁷ The Prime Minister of India's comments reinforced India's use of nuclear weapons testing for impacting political outcomes. Shortly after the Indian test, The Deputy Prime Minister of India, L.K. Advani, publicly warned Pakistan that it should tread carefully regarding the disputed Kashmir region.³⁸ Parthansarathi reported that the nuclear devices tested did not match the Indian government's claimed yield, further reinforcing potential links between nuclear testing and states' attempts to advance political outcomes.³⁹

Finally, France's *Gerboise Verte* nuclear weapons test, conducted in April 1961, offers two potential political reasons for the test. It is unknown whether France accelerated the test to prevent a nuclear device from falling into the hands of rebel generals attempting a coup in Algeria at the time, or if it was a signal to the world that the French government maintained control despite the putsch.⁴⁰ Evidence points to the French government deciding to test the device on the morning of April 24, 1961, and, despite adverse weather, conducting the test on April 25, 1961. The weapon's substantially low yield supported speculation that the French rushed testing but were able to collect useful data on the design of the implosion device used in the test.⁴¹ The coup failed the following day. This public display of a nuclear weapon may have also been a means to deter other nations from recognizing or even supporting the coup.

Conclusion

While examples certainly exist of states using nuclear weapons testing for demonstrative purposes to possibly influence political outcomes rather than for scientific purposes, these instances are significantly rarer than the primary reason for nuclear weapons testing, which is scientific in nature. When states perceive a significant threat, as in the case of Pakistan in 1998, or perceive a vulnerability, as France did in 1961, a state can use a nuclear weapons test to demonstrate resolve, to serve as a deterrence to adversaries, and to advance political objectives. During the deterioration of relations between the US and

USSR after World War II, the US chose to use nuclear weapons to signal to the conventionally superior Soviet Union that any military action would be unwise.

The only country to test in the 21st century has been North Korea. Although the CTBT is not legally binding, almost all nations abide by the testing moratorium imposed by it. States' willingness to abide by the CTBT underscores states' reluctance to use nuclear weapons testing for responding to threats in the last 25 years. North Korea conducted its nuclear tests with no significant military disputes nor outside provocation, to universal condemnation.

This article identified examples of how states use nuclear weapons tests to advance political objectives. Countries' willingness to conduct nuclear weapons tests reinforces their credibility as nuclear powers, serves as a deterrent to adversary states, and advances states' political objectives. The US' previous nuclear weapons tests solidified the US' willingness to use nuclear weapons and reinforced the US' global position as a nuclear power. With the US' nuclear posture and credibility to employ nuclear weapons, the US can continue to advance its political objectives in its role as a nuclear power without conducting future nuclear weapons tests. The US' current position in conducting nuclear tests while still advancing the US' political objectives may serve as an example for other nuclear armed states. ■

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Tracking the Tiny Bugs: Epidemic Models for Wargaming

By: Dr. Jeffrey Rolfes

Executive Summary

The United States (US) Army faces a critical vulnerability due to its insufficient epidemic modeling capabilities. This deficiency hinders the accurate assessment of disease outbreak impacts on force readiness, operational timelines, and strategic objectives. Without robust modeling, resource allocation for medical needs becomes reactive, and the ability to evaluate mitigation strategies is critically limited.

This white paper analyzes the urgency of integrating advanced epidemic modeling into Army planning, training, and operations. It examines various epidemic models, including compartmental, agent-based, network, and hybrid models, highlighting their strengths and limitations for different scenarios.

KEY RECOMMENDATIONS:

- Establish an Epidemic Modeling Center of Excellence (EMCE): This centralized hub would develop, integrate, and operationalize epidemic modeling across all Army levels.
 - Focus on developing advanced modeling capabilities, integrating models into doctrine and wargaming, fostering collaboration and data sharing, and recruiting/training specialized personnel.

- Adapt Existing and Develop Future Wargames:

- Modify current wargames to incorporate realistic disease outbreaks and their operational impacts.
- Design future wargames with epidemic models as core components, incorporating expertise from epidemiologists and public health experts.

Developing robust epidemic modeling capabilities is crucial for maintaining military readiness and strategic advantage in an increasingly complex world. By establishing the EMCE and transforming wargaming methodologies, the Army can proactively address the threat of infectious diseases, ensuring force readiness, and global leadership in the face of biological threats.

Problem Statement

The US Army, despite its dedication to readiness and anticipating future threats, currently lacks a sufficiently robust and integrated epidemic modeling capability. This deficiency presents a critical vulnerability in an increasingly complex and interconnected world where infectious disease outbreaks have the potential to disrupt operations, degrade combat power, and undermine national security.

This lack of sophisticated epidemic modeling hampers the Army's ability to accurately assess the impact of disease outbreaks on force readiness, operational timelines, and strategic objectives.¹ Without data-driven

projections, resource allocation for medical personnel, equipment, and vaccines becomes reactive rather than informed, potentially leading to shortages and delays. Furthermore, the Army's ability to comprehensively compare the effectiveness of various mitigation measures like quarantine, vaccination, and treatment in different operational environments is severely limited, hindering its ability to minimize the impact of outbreaks on mission success.² Critically, the lack of integration between epidemic models and existing wargames prevents commanders from adequately preparing for the challenges of operating in a disease-affected battlespace.

Without a dedicated effort to develop and integrate advanced epidemic modeling capabilities, the Army risks being caught flat-footed in the face of a future pandemic or deliberate biological attack. This lack of foresight could lead to degraded readiness and ultimately mission failure due to unforeseen disease outbreaks impacting troop strength and logistical capabilities. Ultimately, inadequate preparation and response could result in increased casualties among service members and the populations they are sworn to protect. Such vulnerabilities risk diminishing global influence and credibility as the US military is perceived as susceptible to a threat that transcends traditional warfare. The Army must prioritize the development and integration of robust epidemic modeling into its planning, training, and operational doctrines to effectively counter this emerging threat and maintain its global leadership role in an increasingly uncertain world.

Background

Modern warfare extends beyond traditional battlefields. The rise of global connectivity and the potential for devastating pandemics necessitate incorporating biological threats into strategic planning and wargaming. Epidemic models provide the tools to simulate disease outbreaks, analyze their potential impact, and evaluate the effectiveness of various intervention strategies.

Despite significant advances in epidemiological modeling, wargaming has yet to fully incorporate these sophisticated tools into their simulations. While the need to address biological threats in military planning is increasingly recognized, wargames remain largely focused on traditional kinetic warfare. This disconnect stems from several factors, including the inherent complexity of integrating data-intensive epidemic models into existing wargame structures, concerns about data security and sharing, and a lack

of interdisciplinary collaboration between modelers and wargame developers. Furthermore, there's a reluctance to deviate from familiar, combat-centric scenarios, overlooking the potential strategic disruptions posed by disease outbreaks. This lag in integrating realistic disease dynamics into wargames represents a critical vulnerability, leaving military planners inadequately prepared for the complex challenges of operating in a future shaped by pandemics and bio-threats.

Ironically, while sophisticated epidemic models struggle to find a place in military wargames, the gaming community has embraced the complexities of disease dynamics. Games like "Plague Inc." and "Pandemic" have captivated millions with their realistic portrayals of disease spread, resource management, and strategic decision-making during outbreaks.³ These games demonstrate a public fascination with epidemiological challenges and highlight the potential for interactive simulations to educate and engage audiences on complex scientific concepts. This stark contrast underscores a crucial point: while military wargaming lags behind in embracing advanced epidemic models, the gaming industry demonstrates that engaging and informative simulations of disease outbreaks are not only possible but highly sought after. This disparity highlights a missed opportunity for the defense community to leverage similar approaches for training, planning, and public outreach in an era where biological threats are increasingly prominent.

Options for Epidemic Models

Today, as the world grows increasingly interconnected, understanding the dynamics of disease transmission is more critical than ever. Epidemic models provide a powerful lens through which the Army can study how diseases spread, predict their potential impact, and evaluate strategies to mitigate their effects. From simple compartmental models to complex agent-based simulations, each approach offers a unique tool on the dynamics of disease transmission and provides crucial information for training, exercises, and informed decision-making in the face of outbreaks.

COMPARTMENTAL MODELS:

The classic Susceptible-Infected-Recovered (SIR) model divides the population into three compartments, tracking the flow of individuals between them based on infection and recovery rates.⁴ Simple to implement, it provides a basic understanding of epidemic dynamics.

While easy to understand and computationally inexpensive and rapid, it assumes a homogeneous population and constant transmission rates, limiting its realism. The Susceptible-Exposed-Infected-Recovered (SEIR) model adds an “Exposed” compartment, acknowledging the delay between exposure and infectedness or disease onset, offering a more realistic representation of diseases with incubation periods. Numerous variations exist, incorporating factors like age structure, spatial heterogeneity, waning immunity, and different transmission pathways.

AGENT-BASED MODELS (ABM):

These models simulate individual “agents” with unique characteristics and behaviors.⁵ These agents interact within a defined environment, allowing for complex and emergent patterns of disease spread. ABMs provide a high level of detail and realism, capturing individual behaviors and interactions, making them suitable for heterogeneous populations and complex environments. However, they are computationally expensive and require extensive data and parameterization.

NETWORK MODELS:

Network models represent individuals as nodes and their interactions as links.⁶ This approach allows researchers to move beyond individual risk factors and analyze how social structures and contact patterns influence transmission. The strength of network models lies in their ability to capture the complexities of social dynamics, identifying high-risk individuals and informing targeted interventions. For instance, analyzing contact networks could help predict influenza outbreaks within Army bases and logistical chains. However, building these models presents challenges. Obtaining detailed data on individual interactions can be difficult, and analyzing large, complex networks demands significant computational power. Additionally, accurately representing the dynamic nature of real-world networks and estimating transmission probabilities across diverse interactions requires careful consideration. Despite these challenges, network models are crucial for understanding disease spread and will become increasingly important in shaping Army health policies as data availability and computational capabilities advance.

HYBRID MODELS:

Hybrid models combine elements from different approaches to leverage their respective strengths.⁷ For example, an ABM can be integrated with a network model to simulate disease spread within a specific structure, such as logistical nodes or maneuver elements. This offers increased realism and flexibility, allowing for tailored solutions. However, they can be complex to develop and implement.

Analysis

The selection of an appropriate epidemic model for wargaming is based on the specific objectives and constraints of the exercise. While using a more realistic and complex model is preferred, the optimal choice hinges on aligning model complexity to addressing the objectives and scope of the exercise.

For high-level strategic planning, where the focus lies in broad trends and long-term impacts on force posture and resource allocation, simpler compartmental models, such as SIR or SEIR, often provide sufficient insight. These models offer a valuable but generalized perspective on potential force readiness and strategic mobility constraints imposed by an epidemic. However, when transitioning to the operational level, where the emphasis shifts to planning specific campaigns and evaluating the efficacy of intervention strategies within a defined theater, the need for greater model sophistication is required to account for the unique factors at this level that can have a large impact on disease spread. In these cases, incorporating age structure, spatial components, or even metapopulation dynamics into SEIR models becomes essential for assessing the effectiveness of measures like quarantine zones, targeted vaccination campaigns, or medical surge capacity.

At the tactical level, where simulating the intricate interplay between disease propagation and on-the-ground decision-making is needed, agent-based models (ABMs) and network-based models are the program of choice. ABMs and network models excel in capturing the granular details of individual behavior among soldiers, enemy combatants, and civilians, while accounting for the influence of terrain, infrastructure, and unit-level interventions. However, this granularity comes at the cost of computational expense, calculation speed, and data requirements, so an evaluation of available resources and time constraints must be conducted.

Ultimately, the most effective model is not necessarily the most complex, but rather the one that strikes an optimal balance between realism, practicality, and the specific objectives of the wargame. A thorough assessment of data availability, computational resources, and the need for clear communication of model outputs to decision-makers should guide this selection process. By navigating these considerations, wargame designers can ensure their chosen model yields the most relevant and actionable insights for navigating the complexities of disease outbreaks within the Army.

Recommendation

To address the critical vulnerability posed by the lack of robust epidemic modeling capability, the US Army should establish a dedicated EMCE. This centralized hub would serve as the driving force for developing, integrating, and operationalizing epidemic modeling across all levels of Army planning, training, and response.

The EMCE would focus on the following key tasks:

1. Develop and Maintain Advanced Modeling Capabilities:

- b. Consolidate existing modeling efforts and expertise under a unified command structure.
- c. Invest in cutting-edge modeling tools and technologies, including artificial intelligence and machine learning, to enhance predictive accuracy and scenario analysis.
- d. Create a library of adaptable disease models tailorable to specific geographic regions, current threat scenarios, current intelligence reporting, and operational environments.

2. Integrate Epidemic Modeling into Army Doctrine and Wargaming:

- a. Modify existing wargames and training exercises specifically designed to incorporate realistic disease outbreaks and their impact on operations.
- b. Develop standardized doctrine and training materials to educate commanders and staff on interpreting and utilizing epidemic modeling outputs for decision-making.

- c. Establish clear lines of communication between modelers and military planners to ensure relevant and timely integration of modeling insights.

3. Foster Collaboration and Data Sharing:

- a. Build strong partnerships with civilian institutions, including universities, research labs, and public health agencies, to leverage existing expertise and data resources.
- b. Establish secure data-sharing agreements to facilitate access to real-time epidemiological data, both domestically and internationally.

4. Recruit and Train Specialized Personnel:

- a. Utilize appropriate Functional Areas (FAs) and Military Occupational Specialties within the Army for epidemic modeling experts, including opportunities for advanced education and research. These experts include FA52 Nuclear and Countering Weapons of Mass Destruction (CWMD) Officers, FA57 Simulation Officers, FA59 Strategists, 61G Infection Disease Officers, 71A Microbiology Officers, 71B Biochemistry Officers, and 740A Chemical, Biological, Radiological, and Nuclear Warrant Officers)
- b. Implement a comprehensive training program for existing personnel to develop a cadre of officers and analysts proficient in applying epidemic modeling to military planning and operations.

By establishing the EMCE, the Army can proactively address the growing threat of infectious disease outbreaks, ensure the readiness of its forces to operate effectively in a range of biological threat environments, and maintain its strategic advantage in an increasingly complex world.

Conclusion

The integration of epidemic models into the wargaming environment necessitates a multifaceted approach encompassing the adaptation of existing frameworks and proactive engagement in the development of future systems. Current wargames, often built around traditional kinetic and maneuver warfare architectures, need to be modified to accommodate the complexities

of disease propagation. This includes not only the technical integration of chosen models, ensuring data exchange and synchronization with existing simulations, but also the development of user interfaces that render model outputs readily interpretable for planners accustomed to conventional wargaming outputs.

Beyond simply grafting epidemic models onto existing structures, a more impactful approach lies in embedding these models as core components of future wargame design. This requires a paradigm shift within the wargaming community, recognizing epidemiologists, data scientists, and public health experts as essential stakeholders in the design process. By engaging these organizations from the outset, future wargames can move beyond rudimentary representations of disease outbreaks to incorporate nuanced factors such as pathogen evolution, network dynamics, and the interplay between disease and human behavior on the battlefield. This collaborative approach, bridging the gap between biodefense experts at United States Army Nuclear and CWMD Agency (USANCA) and the Center for Army Analysis community, holds the key to developing wargaming environments that accurately reflect the multifaceted challenges of warfare defined by biological threats. Such a transformation in wargaming methodology is essential not only for maintaining military readiness in the face of pandemics but also for informing broader responses to global health security challenges. ■

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The Imperative of Nuclear Literacy in Military Education

Insights from the NSERC Nuclear Deterrence Staff Ride

By: Maj. Dave Fobar and Maj. Tyler Halbert

The Nuclear Science and Engineering Research Center (NSERC), a Defense Threat Reduction Agency (DTRA) office located at West Point, NY, annually sponsors a three-week Nuclear Deterrence Staff Ride for select upper-class cadets and midshipmen attending either the three United States (US) Service Academies or an Reserve Officers' Training Corps (ROTC) program. During that time, students undergo an innovative and immersive education on the critical role nuclear weapons serve within US strategic deterrence policy. The Nuclear Deterrence Staff Ride presents a series of lessons and site visits across the US, each one punctuated with senior leader engagements and hands-on technical discussions and demonstrations. These experiences promote a robust understanding of nuclear deterrence as a component of modern US military strategy, as well as justifying its current and future requirement. As such, the Nuclear Deterrence Staff Ride provides an important opportunity for sowing the seeds of nuclear literacy within the next generation of Joint Force officers, thereby helping DTRA to cultivate future military leaders possessing a greater awareness and appreciation for the breadth and scope of the US nuclear enterprise.

The Structure and Purpose of the Staff Ride

This past year marked a significant evolution in the structure of the Nuclear Deterrence Staff Ride, introducing a dual-path itinerary that broadened the scope of exposure for all participants. Traditionally, the Nuclear Deterrence Staff Ride focused on Department of Defense (DOD) delivery platforms. However, this

iteration incorporated a parallel focus on the National Nuclear Security Administration (NNSA) design and manufacturing enterprise. This past summer, twenty future Joint Force military officers, drawn from a diverse mix of military academies and ROTC programs, were divided into two teams. One team focused on DOD sites and the other on Department of Energy (DoE) sites. This bifurcated approach enriched the student learning experience by providing a comprehensive view of both the operational and infrastructural requirements and challenges associated with modern nuclear deterrence.

The participants, representing various academic disciplines from Nuclear Engineering to Political Science, brought a broad range of perspectives to the Nuclear Deterrence Staff Ride. This diversity proved crucial, as the complexity surrounding nuclear deterrence encompasses technical, strategic, and policy dimensions. Thus, by engaging with both DOD and DoE components, as well as one another, the interns gained a holistic understanding and appreciation of the US nuclear enterprise, one that better prepared them to make informed decisions throughout their future military careers.

Building a Foundation in Nuclear Deterrence

The Nuclear Deterrence Staff Ride commenced at Kirtland Air Force Base (KAFB), leveraging the Defense Nuclear Weapons School to establish a foundational understanding of the US nuclear enterprise through formal classes on policy,

strategy, deterrence, delivery platforms, and weapons operation. Tours of the Nuclear Weapons Instructional Museum (NWIM) supplemented classes which provided a tangible context to the abstract classroom instruction. The NWIM offered a historical perspective on nuclear weapon development and the evolution of deterrence strategies, grounding the interns in the legacy and ongoing relevance of US nuclear deterrence. Prior to the commencement of these classes, students underwent a pre-assessment to measure their initial understanding of nuclear deterrence, policy, and strategy. Use of the pre-assessment helped establish a baseline for evaluating the educational impact of the Nuclear Deterrence Staff Ride.

During their time at KAFB, participants also visited the Trinity Test Site, where the first nuclear detonation occurred. This visit was particularly poignant, as it connected the theoretical discussions to the real-world implications and historical significance of nuclear weapons development. Tours of the Kirtland Underground Munitions Maintenance and Storage Complex, Manzano Mountain, and interactions with the 21st Explosive Ordnance Disposal and Sandia

National Laboratories, provided a comprehensive view of the operational, logistical, and research and development aspects of the US nuclear enterprise.

Diverse Experiences Across DoE and DOD Sites

Following these experiences at KAFB, the students were divided into DoE and DOD teams to allow for more specialized learning experiences. The DoE team visited Los Alamos National Laboratory (LANL), Kansas City National Security Complex, Y-12 National Security Complex, Oak Ridge National Laboratory (ORNL), and Pantex Plant. At LANL, discussions with weapons design experts offered insights into strategic deterrence and the physics of nuclear weapons. Tours of LANL facilities, such as the Los Alamos Neutron Science Center, highlighted the ongoing research and stockpile stewardship efforts essential for maintaining a safe and reliable nuclear arsenal.

In Kansas City, the students learned about the manufacturing and testing of critical components, emphasizing the rigorous standards and safeguards



ABOVE: The DOD team at the outdoor Submarine Launched Ballistic Missile display at Kings Bay, Georgia.

in place to ensure the functionality and security of US nuclear weapons. At Y-12 and ORNL, the group explored the processes involved in nuclear weapon manufacturing, uranium enrichment, and the operation of high-performance computing resources like Summit and Frontier, which are pivotal modeling and simulation tools for nuclear weapon development.

At the Pantex Plant, the final DoE team site visit, tours were conducted of the assembly and disassembly bays, as well as the high explosive manufacturing complex. These visits underscored the meticulous processes involved in handling special nuclear materials and the stringent safety protocols that govern these operations. Understanding the technical and procedural aspects of nuclear weapons manufacturing deepens student appreciation of the complexity and importance of maintaining the US nuclear arsenal.

Conversely, the DOD team explored the operational side of nuclear deterrence through visits to Naval Submarine Base Kings Bay (NSBKB), Whiteman Air Force Base (WAFB), US Strategic Command (USSTRATCOM), and Malmstrom Air Force Base (MAFB). At NSBKB, instruction and tours, including a visit to the Trident Training Facility and aboard the USS Wyoming, highlighted the role of ballistic missile submarines as the assured second-strike component of the US nuclear triad.

At WAFB, home to the B-2 Spirit bombers, the students received briefings on the visible and recallable component of the US nuclear triad, including the unique capabilities and mission profiles of these aircraft. Their discussions with senior officers provided insight into the strategic considerations and operational challenges of maintaining a credible bomber force. The tour of the weapons loading facility and interactions with Air Force Security Forces further illustrated the intricate logistics and robust security measures that underpin the bomber leg of the triad.

The student site visit to USSTRATCOM offered a comprehensive overview of the command and control structures that govern the US nuclear forces. Briefings on nuclear targeting, vulnerability assessments, and the role of the nuclear command, control, and communications highlighted the strategic and operational complexities of US nuclear deterrence. Interactions with senior leaders provided a strategic perspective on the challenges and imperatives of maintaining a credible deterrent in a dynamic global security environment.

At MAFB, the students toured missile silos and mission control units, gaining a firsthand understanding of the intercontinental ballistic missile component of the US nuclear triad. Lessons there on security and safeguards emphasized the critical importance of protecting these assets from various threats. These experiences collectively provided a comprehensive view of the



ABOVE: The entire cohort of students and mentors at the Department of Energy Headquarters, Washington, District of Columbia (DC).

infrastructure, operations, and strategic considerations that underpin the daily functioning of the most responsive component of the US nuclear deterrent.

Recombination and Cross-Pollination in Washington, D.C.

The Nuclear Deterrence Staff Ride culminated in Washington, DC, which facilitated the recombination of the DoE and DOD teams, allowing for cross-pollination of information and experiences. In DC, the students engaged in high-level discussions with senior military and civilian leaders, toured the Pentagon, participated in a tabletop exercise (TTX), which simulated various nuclear crises, and completed a post-assessment. These interactions and exercises not only reinforced the knowledge gained during the site visits, but they also highlighted the importance of interagency and DOD cooperation in maintaining and executing US nuclear deterrence strategies.

More specifically, the students toured the Pentagon, meeting with senior leaders such as Gen. Eric Smith, Commandant of the Marine Corps, and Lt. Gen. Laura Potter, Director of the Army Staff. These engagements provided a platform for students to ask questions and discuss the roles of different military branches in US strategic deterrence. The students gained insights into the broader strategic considerations that inform US national security decisions. The discussions also delved into the preparedness of the US Armed Forces for operations in a nuclear environment, underscoring the importance and multifaceted nature of US nuclear deterrence.

At the NNSA headquarters, the group engaged with senior leaders from NA-10 and NA-20, gaining insights into the policy and administrative aspects of the US nuclear enterprise. These discussions highlighted the collaborative efforts between the DoE and DOD in maintaining a credible deterrent and emphasized the importance of interagency coordination in addressing the complex challenges of global nuclear security.

A culminating TTX was conducted at the DTRA headquarters building, which presented the students with a series of realistic scenarios involving nuclear-armed adversaries. Tasked with developing potential courses of action to maintain or regain nuclear deterrence, the students were challenged with applying the knowledge and skills acquired during the Nuclear Deterrence Staff Ride to formulate reasonable strategic options. This

exercise underscored the strategic and operational challenges of nuclear deterrence and highlighted the critical importance of well-informed decision-making in maintaining US national security.

The Strategic Imperative of Nuclear Literacy

The primary objective of the NSERC Nuclear Deterrence Staff Ride is to enhance the nuclear literacy of future Joint Force leaders. Given the evolving nature of global threats and the critical role of nuclear weapons in US strategic deterrence, it is imperative that military officers at all levels possess a fundamental understanding of US nuclear policy, nuclear deterrent, and nuclear enterprise. Therefore, the insights and experiences these students gained from the Nuclear Deterrence Staff Ride are critical to ensuring the next generation of Joint Force leaders possess the knowledge and understanding to prepare their units and make informed decisions that will shape the future of national security.

Nuclear deterrence is a complex and multifaceted concept that requires a thorough understanding of policy, strategy, technology, and operations. Therefore, the comprehensive nature of the Nuclear Deterrence Staff Ride ensures that participating future Joint Force officers develop a well-rounded perspective on these issues, preparing them to navigate the challenges of modern military leadership. Most importantly, however, by deliberately promoting nuclear literacy, the Nuclear Deterrence Staff Ride contributes meaningfully to the future preparedness of the US Armed Forces to operate and win on the nuclear battlefields of tomorrow, which itself enhances the effectiveness of the US nuclear deterrent.

Conclusion

In conclusion, the Nuclear Deterrence Staff Ride is a vital educational initiative that significantly contributes to the strategic preparedness of future Joint Force officers. By providing a comprehensive and immersive experience, the Nuclear Deterrence Staff Ride ensures that participants develop a thorough understanding of nuclear deterrence, from US policy and strategy to the operational and manufacturing processes. This knowledge is essential for maintaining a credible and effective nuclear deterrent,

underscoring the need for continued emphasis on nuclear education in military training programs.

Additionally, the expanded format of the Nuclear Deterrence Staff Ride, incorporating both DoE and DOD perspectives, enriches the overall learning experience and highlights the collaborative efforts required to sustain the US nuclear enterprise and US nuclear deterrence. To assess the impact of this program, students completed both pre- and post-assessments. The significant improvement in scores following the Staff Ride demonstrated the efficacy of the immersive, multi-disciplinary approach in enhancing nuclear literacy among participants.

As future military leaders, the attendees leave better equipped for future military conflicts, especially against those with nuclear arsenals or ambitions. The Nuclear Deterrence Staff Ride teaches and demonstrates the critical importance of nuclear literacy in modern military education and the ongoing commitment required to maintain a credible US nuclear deterrent. ■

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Units Train for Nuclear Forensics Missions During Exercise Prominent Hunt in New York

By: Walter T. Ham IV



ABOVE: Capt. David Manzanares, a health physicist with the 20th Chemical, Biological, Radiological, Nuclear, Explosives (CBRNE) Command's Nuclear Disablement Team, scans the area for radiation during Exercise Prominent Hunt in Schenectady, N.Y., Jan 27–31. U.S. Army photo by Marshall R. Mason.

Highly specialized units from the U.S. Department of Defense's premier Chemical, Biological, Radiological, Nuclear, Explosives (CBRNE) Command trained for nuclear forensics missions during Exercise Prominent Hunt in Schenectady, New York, Jan. 27–30 this year.

U.S. Army 20th CBRNE Command Soldiers from a Nuclear Disablement Team (Infrastructure) and a CBRNE Response Team validated their readiness to serve on the National Technical Nuclear Forensics Ground Collection Task Force.

Since 2012, Nuclear Disablement Teams and CBRNE Response Teams have participated in Exercise Prominent Hunt.

Led by the Federal Bureau of Investigation (FBI), the National Technical Nuclear Forensics Ground Collection Task Force gathers and packages samples of radioactive fallout that enable partner agencies to determine the source of the radiation.

In addition to the experts from the FBI and Soldiers from the 20th CBRNE Command, airmen from the Patrick Space Force Base, Florida-headquartered Air Force Technical Applications Center (AFTAC) also participated in the task force during the exercise. Outside of participating on the task force, AFTAC provides national authorities with technical measurements to monitor nuclear-treaty compliance.

Maj. James C. Sealock, a nuclear exercise planner and the lead coordinator and senior controller and evaluator for Exercise Prominent Hunt, said the exercise prepares the units to support the FBI-led task force.

"This exercise contributes to deterring threats by validating the ability to identify perpetrators of such an event," said Sealock. "Nuclear forensics enables the United States Government to hold fully accountable any state, terrorist group or other non-state actor that supports terrorist efforts to obtain or employ nuclear devices."

Sealock said teamwork is critical for success during the high stakes mission. “This is an interagency mission and requires a lot of coordination and cooperation between all members of the task force,” said Sealock. “It demonstrates a capability to deter any potential bad actors, and in the event they are not deterred, assists national command authorities in attributing actions to those responsible by collecting forensic evidence that is processed by appropriate laboratories to provide data that would facilitate those attributions.”

Sealock, who grew up in an Army family, attended Virginia Tech and graduated from North Georgia College and State University with a degree in political science with a focus in terrorism and political violence. He also earned a graduate certificate in nuclear weapons effects, policy, and proliferation from the Air Force Institute of Technology.

Before becoming a Nuclear and Counter Weapons of Mass Destruction officer, referred to as a Functional Area 52, Sealock served as an Explosive Ordnance Disposal (EOD) officer for 14 years. He deployed to Afghanistan twice and served in South Korea for three years.



ABOVE: Sgt. First Class Rigoberto J. Olmeda-Flecha (center), a senior health physics noncommissioned officer with the 20th Chemical, Biological, Radiological, Nuclear, Explosives (CBRNE) Command's Nuclear Disablement Team, discusses the sample collection process with 1st Lt. Derek R. Canlas (right), a Chemical Corps officer with the 22nd Chemical Battalion, 48th Chemical Brigade, during Exercise Prominent Hunt in Schenectady, New York, Jan 27–31. U.S. Army photo by Marshall R. Mason.

Sealock said the highlight of the annual exercise was overcoming the cold and snowy weather in New York to successfully train for the mission designed to protect the homeland and safeguard the American people.

“Prominent Hunt exercises are conducted in a variety of geographic areas, and at different times of the year to test readiness against various conditions, and this one enabled the task force to test against collections in the snow, and our Department of Energy partners to process samples intermixed with snow and water,” said Sealock. “These conditions are not ideal but may be an eventuality in this mission and are very difficult to recreate in a simulated environment.”

In addition to the Nuclear Disablement Teams, the Aberdeen Proving Ground, Maryland-based 20th CBRNE Command is home to 75 percent of the active-duty U.S. Army's EOD technicians and Chemical, Biological, Radiological, Nuclear (CBRN) specialists, as well as the 1st Global Field Medical Laboratory, CBRNE Analytical and Remediation Activity and Weapons of Mass Destruction Coordination Teams. From 19 bases in 16 states, Soldiers and Army civilians from the 20th CBRNE Command deploy to confront and defeat the world's most dangerous hazards during joint, inter-agency and multinational operations around the world.

Walter T. Ham IV

is the Deputy Public Affairs Director for the 20th Chemical, Biological, Radiological, Nuclear, Explosives (CBRNE) Command, the U.S. Department of Defense's premier multifunctional and deployable CBRNE formation. A retired U.S. Navy Chief Journalist with a master's degree in nonfiction writing from Johns Hopkins University, he previously served as a Pacific Stars & Stripes reporter and a civilian public affairs officer for the U.S. Navy, U.S. Air Force, U.S. Coast Guard and U.S. Department of Defense.

Ecological Considerations for Operating in the Contaminated Biological Warfare Environment

By: Lt. Col. Michael A. Washington

Introduction

Biological warfare (BW) agents consist of bacteria, viruses, or toxins that attack the biological (rather than the physical) integrity of the target.¹ While the use of BW is not associated with the wide-spread physical and structural effects that result from the use of explosives and radiological weapons, they are inherently complex and capable of long-lasting environmental contamination and area denial.² The development of tactics, techniques, and procedures that can ensure survivability in a future BW environment will be necessary to deter the future use of these weapons and to maintain freedom of movement in a biologically contaminated environment.³ This is particularly true in the case of bacteriological agents which are capable of both rapid replication and simultaneously maintaining a state dormancy in the operational environment. It will thus be necessary for planners to understand the complexity of the agents involved, their interactions with the abiotic and biotic elements of the contaminated environment, and the potential of these interactions to degrade combat power.⁴

Bacteria are unique among BW agents in that (unlike viruses and toxins) they show properties that are typically associated with life such as metabolism and response to the environment. Importantly, they are unlike chemical agents in that they are capable of reproduction, a property that allows the potential for amplification deployment.⁵ This should be taken into consideration when evaluating the persistence of these bacteriological BW in the environment and when selecting among the available set of remediation procedures. This should also be

considered when planning for counter-weapons of mass destruction (CWMD) operations, especially when those operations are intended to prevent or respond to a bacteriological attack.⁶

Amplification of Bacteriological Agents in the Environment

BW agents have historically been developed for delivery through the aerosol route. This is particularly true of *Bacillus anthracis* (*B. anthracis*) which was successfully developed as an aerosol weapon by both the United States and the former Soviet Union.⁷ Since aerosol dispersal is potentially the most efficient method by which agent can be dispersed over a large area, it is likely to be employed in a future BW attack. Data obtained from field tests and from observations made during accidental release described in Alibek's research indicated that bacterial spores can both survive and germinate in diverse outdoor environments and that this can occur within the context of potentially hostile physical conditions. Bacterial agents are amplified in the environment through the process of replication. The most common mechanism for bacterial replication is binary fission. During this process, each individual bacterial cell is divided into two daughter cells and each daughter cell divides into two additional daughter cells.⁸ Under optimal conditions, a single bacterial cell that has been deposited onto an environmental surface (soil or sand or in an open pond, lake, river, or stream) can replicate in an exponential fashion resulting in the production of millions of progeny within a relatively short time period.⁹ In fact, the total number of bacterial cells will increase as a function of time and the number

of bacterial cells after a given unit of time will be directly proportional to the initial number of cells that were deposited in the environment and the replication rate of the bacterial strain (Figure 1). It can be readily shown that the deposition of a single bacterial cell with a doubling time of 30 minutes can result in the production of one trillion (1×10^{12}) progeny in a period totaling less than 20 hours. This corresponds to an increase in total bacterial mass from one picogram (1×10^{-12} grams) to

a mass of one gram in a single day. It should be noted that these kinetics primarily represent ideal conditions and do not always occur in the outdoor environment where ideal conditions rarely exist.^{10,11} However, the fact that biological agents tend to multiply over time (and are capable of an exponential increases in mass) clearly differentiates them from chemical or radiological agents which tend to decay over time and suggests that novel methods are needed to deal with them.

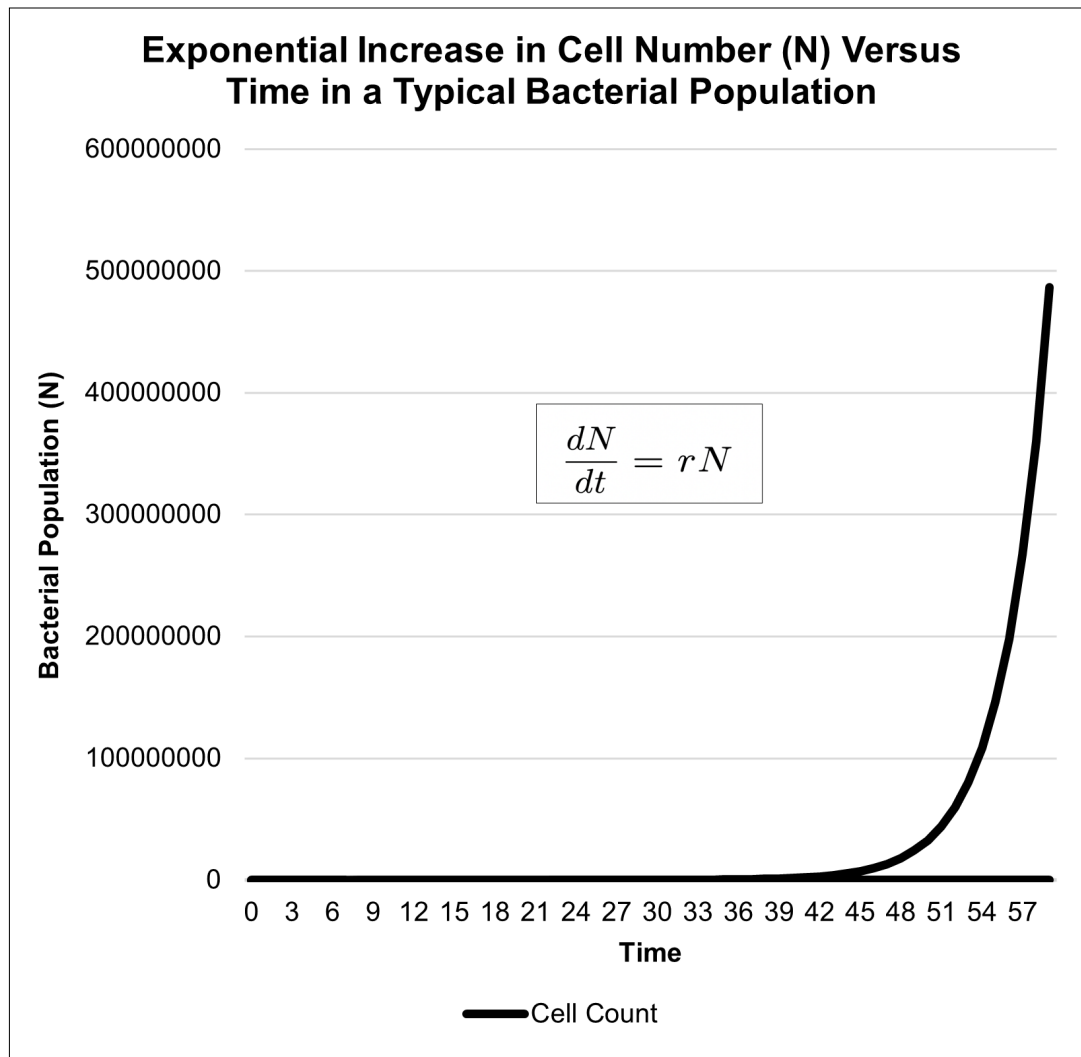


FIGURE 1: The increase in number individual cells in a bacterial population is proportional to the product of the replication rate and the starting number of cells. Here (N) indicates the number of cells, (t) indicates time, and (r) indicates the replication rate of the population. This is type of growth is known as exponential or logarithmic growth. It represents the potential for bacterial pathogens to dominate an environment that is characterized by ideal physical conditions and an unlimited nutrient supply.

Factors Limiting the Amplification of Bacterial Agents in the Environment

Bacterial agents require consistent sources of energy from growth and reproduction.¹²⁻¹⁶ Continued metabolic activity also requires a steady supply of organic material for essential anabolic processes and the availability of these materials in the environment is a key limiting factor of bacterial replication.¹⁷ These requirements have been described in the ecological principle known as Lieberg's Law of the Minimum which states that growth and reproduction of living organisms tend to be limited by the least abundant nutrient in the environment.¹⁸ This concept is related to Shelford's Law of Tolerance which states that bacterial agents are sensitive to any factor in the environment present in excess.¹⁹ The combination of these laws suggest that the presence or absence of multiple factors in an environment may influence the success and spatial distribution of living organisms that environment.²⁰ This has been formalized in a linear relationship known as the Monod Equation which describes the interactions between the growth rate of a bacterial population and the concentration of a limiting substrate in the environment (Figure 2).²¹⁻²² Specifically, this equation illustrates that the specific growth rate of a bacterial population

in a resource limited environment is dependent upon the limiting substrate (nutrient) concentration. It was developed during a series of investigations looking into the rate of change of bacterial mass with respect to time. The equation is significant in that the slope can be determined and compared between one or more groups of bacterial agents when they are grown on different substrates at different time points to determine the factors that are necessary for maximal or minimal growth.²³ It is therefore possible that data analyzed by this equation (with refinement) might be used in the operational setting to determine the growth kinetics of the most likely bacteriological agents before they are deposited in the environment. It might also be effective in determining whether the intentional depletion of nutrients from the environment can be used as a form of bioremediation following a biological attack. The rates at which a bacteriological agent is removed from the environment (decay) either by remediation or by natural processes can be modeled by a first order equation (Figure 3). Following a biological attack, bacterial decay rates can be determined by repeated sampling of the affected area and real-time evaluations can be made to determine when the levels fall below the danger threshold for protective posturing.²⁴

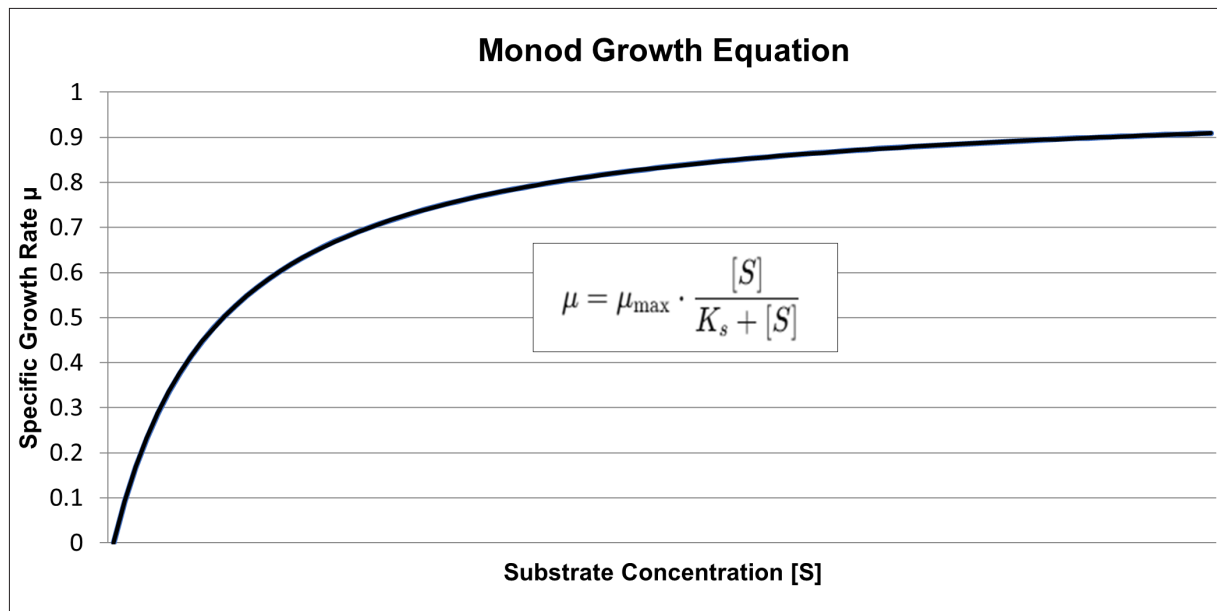


FIGURE 2: Sample plot of the Monod Equation showing that the specific growth rate or growth per unit biomass of a bacterial population is proportional to the product of the maximum possible growth rate (and the substrate saturation function ($[S]/K_s + [S]$)). This figure illustrates the dependence of bacterial growth on the limiting substrate (S) concentration.

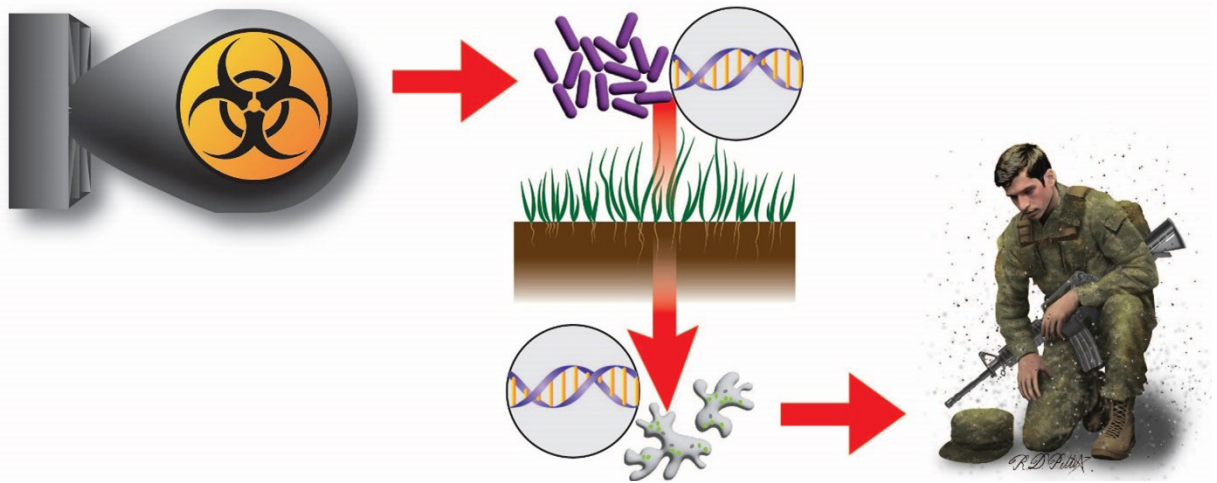


FIGURE 3: Biological agents such as *Bacillus anthracis* can be deposited into the environment by a biological munition or other dispersal mechanism. These agents can be amplified by interacting with the biological environment where they may replicate in the soil amoeba populating local root systems. Military personnel working in the area can become infected by exposure to the increased numbers of spores and vegetative cells in the affected area.

Abiotic Factors that Influence the Stability of Bacterial Agents in the Environment

Abiotic factors are those non-living aspects of the environment that can influence the physiology (growth and replication) of a bacterial agent.²⁵ These factors include temperature, light, salinity (salt concentration), pH (acidic versus basic environment), water availability, and the presence or absence of inorganic nutrients.²⁶ To illustrate the roles of these factors in the physiology of BW agents, the properties of *B. anthracis* can be used as a case study. This agent is often considered to be a prototypical biological weapon due to its historical use in both state and non-state programs.²⁷ The genetics, metabolism, and physiology of this agent give it all the features necessary for effective weaponization.²⁸ *B. anthracis* is relatively easy to produce in a laboratory setting, it can be readily converted from a vegetative (growing and metabolically active) form into an environmentally stable and easily dispersible spore (dormant and metabolically inactive form) which can be stored for long periods of time and rapidly reconstituted for offensive use.²⁹ Once the agent is released into the environment, it can contaminate large areas of land and remain viable for decades.³⁰ Indeed, the half-life of the spore form of *B. anthracis* has been estimated to be more than 100 years and remediation of affected areas is notoriously difficult.³¹⁻³²

In the outdoor environment, the survival, germination, and amplification of bacterial spores results from the

combined action of several abiotic factors. Laboratory studies have shown that *B. anthracis* is capable of replicating in sterilized soil under conditions of adequate moisture and temperature if the pH of the soil reached acidic levels.³³ Significantly, *B. anthracis* spores tend to be more abundant in alkaline soils than acidic soils.³⁴ This suggests that certain soil locations can serve as “incubator areas” where the pH, moisture level, and presence of inorganic and organic nutrients are such that they support robust bacterial growth. Indeed, the spores of multiple species of bacteria can colonize the root surfaces of plants where they tend to germinate (transformation into vegetative cells) and undergo vegetative growth, replication, and resporulation.³⁵ Plants growing in soils rich in calcium and organic matter seem to be ideal for this purpose.³⁶ Large anthrax outbreaks have been associated with agricultural areas overlaying calcium-containing limestone deposits and calcium has been found to be involved in the formation of the bacterial spore structure, in the activation of toxin production and in spore germination according to research by Griffin et al. The genetic composition of *B. anthracis* and soil chemistry correlate, as observed in the Kruger National Park in South Africa where certain subtypes accumulate in areas of higher calcium and pH levels than other areas.³⁷ Additional abiotic factors that are correlated with focal anthrax outbreaks include a temperature at or around 15.5°C, warm weather after a period of rain, and the presence of low-lying areas where the hydrophobic bacterial spores may accumulate due to surface runoff.³⁸

Biotic Factors that Enable the Replication of Bacterial Agents in the Soil Microenvironment

Biotic factors include all living systems and their products that are capable of influencing bacterial physiology.³⁹ These include the presence of other microorganisms such as bacteria, fungi, or single-celled eukaryotes such as the amoeba and the paramecium.⁴⁰ Biotic factors also include plants and organic compounds that are derived from living systems such as carbohydrates (sugars), lipids, and proteins. Much research has been conducted on the impact of environmental biotic factors on the growth and replication of *B. anthracis*. In nature, this agent undergoes an enzootic (animal-mediated) cycle in which spores are picked up by grazing animals, germinate into vegetative cells and replicate within the tissues of the animal resulting in local amplification.⁴¹ Once the animal dies, the process of sporulation occurs resulting in a set of metabolically inactive spores that can be gradually released into the environment as the animal carcass decays.⁴²

B. anthracis has also been found to be capable of germination and replication within the cytoplasm of soil-dwelling amoeba (small single-celled organisms that live in moist environments).⁴⁴ Specifically, the amoeba known as *Acanthamoeba castellanii* has been shown to have the ability to engulf *B. anthracis* resulting in local amplification in the soil microenvironment.⁴⁴ These amoebae might serve as a selective biological amplification mechanism for spores harboring specific sets of genes. Indeed, the time-dependent selection for cells harboring the pX01 plasmid (which harbors one of the virulence genes) has been used to verify the fact that amoebae are capable of amplifying *B. anthracis* in the environment by comparing genetic material from a set of samples collected in anthrax contaminated areas over a multi-year period. This analysis revealed stable maintenance of the pX01 and loss of the related pX02 plasmid that would be expected if pX01 was needed for amplification in the protozoal host.⁴⁵

Plants are often overlooked in discussions concerning the etiology of infectious disease and biological defense. However, they are a ubiquitous component of the biotic environment, and they have been implicated as reservoirs of numerous bacterial pathogens.⁴⁶ Bacteria tend to be more tightly associated with the zone of soil surrounding a plant root system than they are with root-free soil. This may be due to the nutrient-rich ecosystem that is formed by the release of amino acids, carbohydrates, vitamins, and organic acids by the roots. Concerningly, it has been found that

genes can be transferred between the bacterial cells that are attached to root surfaces.⁴⁷ This is significant because multiple species of bacteria colonize the root surfaces of most plant species, the inoculation of either a natural or genetically modified antibiotic resistant bacterium into the root system can potentially lead to the spread of novel or modified antibiotic resistant genes into the environment.⁴⁷ If this type of gene transmission were to occur in the operational environment it may increase the risk of a decrease in the effectiveness of the antibiotic countermeasures that are administered to exposed personnel.

Spatial Distribution of Bacterial Agents in the Post-Event Environment

Prior to release, bacterial spores are typically milled to form particles between two to six microns in diameter.⁴⁸ After dissemination by a point source such as an explosive munition, a stationary spraying device, or a non-explosive mechanical release device, it is expected that some of the particles will be carried away by prevailing winds, but most of the particles will eventually settle onto the local environmental surface in a Gaussian (bell shaped) distribution at a rate that is dependent on the size, density, and shape of the particle and the temperature and viscosity of the air.⁴⁹ Since biological agents are capable of replication, they can increase in numbers after dissemination and move from one location to another by various means.⁵⁰ Biological agents will most likely be distributed on environmental surfaces in one of several distribution patterns. These are known as the uniform, random, and clumped distributions.⁵¹ In a uniform distribution, the individual members of the population are evenly distributed within the area with equal space between each member. In a random distribution, the members of the population are distributed without any discernable pattern, and in a clumped distribution, the members of the distribution tend to aggregate into clusters. *B. anthracis* will most likely form random or clumped distributions of spores that will be expected to germinate in the weeks, months, and years following agent release. Furthermore, if each of the limiting growth and germination factors can be assumed to be normally distributed in the environment then the resulting infections will most likely occur in an ellipsoidal pattern resulting from the spatial interaction of multiple Gaussian curves.⁵²

Conclusion

B. anthracis is a prototypical BW agent that has been developed into an aerosol weapon by both state and non-state actors. This article discussed the bacterial pathogen agents that are most likely to be employed in future battlefields. Based on open-source data, it can be assumed that bacteriological agents can be utilized as both tactical and strategic weapons. They can be tactical in that they can generate casualties within 1 to 7 days following exposure and they are strategic in that they can act as area denial weapons with a half-life of up to 100 years.⁵³⁻⁵⁴ The case study of anthrax suggests that unlike chemical agents, ecological factors should be considered when developing mitigation strategies and force protection plans against bacteriological weapons. These factors include the combined actions of the abiotic and biotic elements of the physical environment and especially those factors that can lead to the amplification of the agent and the formation of localized and spreading regions of contamination. The literature reviewed here suggests that personnel will be at the highest levels of danger when they are operating in a potentially BW contaminated area during warm weather, especially following a period of rain when the temperature is greater than 15.5°C in regions in which there are soils with underlying limestone deposits or high calcium levels and an alkaline pH.²⁴⁻³⁵ This danger will be further increased in grasslands or in areas with scattered plants and moist soils that are capable of supporting the growth of amoeba (Figure 3).⁴⁴⁻⁴⁵ With respect to mitigation strategies, it is recommended that lime, quicklime, or hydrated lime should not be used to disinfect or treat contaminated soils as the calcium in the lime can protect spores and reduce the effectiveness of other agents.⁵⁴⁻⁵⁵ Rather, formaldehyde solutions or sodium hypochlorite solutions should be employed for soil decontamination.⁵⁶⁻⁵⁷ A detailed understanding of the physical environment in the area of operations and how the abiotic and biotic factors in those environments interact with potential bacteriological weapons will be essential for protecting the force and for providing a credible deterrence against the use of these agents. ■

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The Emergence of Global Antimicrobial Resistance: A Formidable Biodefense Consideration?

By: Capt. Rebecca Melendez

Abstract

The emergence of antimicrobial resistance (AMR) represents a global health crisis with profound implications for public health, national security, and biodefense readiness. This study explores the confluence of AMR, water disinfection techniques, and biofilm-mediated resistance in opportunistic bacterial isolates from US wastewater treatment plants. Using phenotypic assays, antimicrobial susceptibility testing, chlorine minimum inhibitory concentration (MIC) studies, and whole-genome sequencing, the authors evaluated and characterized 92 genetically distinct Gram-negative isolates for multidrug resistance (MDR), biofilm formation, and disinfectant tolerance.

Findings reveal that 58% of moderate or strong biofilm-forming isolates exhibited MDR profiles, and over 90% tolerated chlorine concentrations commonly used in water treatment. Chlorine-resistant isolates frequently harbored virulence genes (indicating pathogenicity), suggesting selective pressure linked to disinfection protocols. Biofilm capacity significantly increased the odds of chlorine resistance, demonstrating the resilience of these waterborne bacterial isolates in treated water systems.

Given AMR's potential to threaten total force readiness and even enable adversarial use of biotechnology to target critical infrastructure, these findings highlight a direct threat to force health protection and overall mission readiness. The 2023 Department of Defense *Biodefense Posture Review* identifies AMR as a naturally occurring biological threat with the potential to overwhelm response capacities. This research reinforces that view and highlights the need for integrated medical countermeasures, updated disinfection strategies, and enhanced countering weapons of mass destruction (CWMD) preparedness to mitigate AMR as an emergent weaponizable threat in the 21st century.

Introduction and Relevance

Antimicrobial agents have been used to treat bacterial infections in humans, animals, and plants for the past 90 years. They have also been exploited for diverse purposes including animal husbandry and animal production as preventive measures in many underdeveloped and developing countries for decades. Antimicrobial therapy represents one of the most prescribed class of medications to manage both chronic and acute bacterial diseases in clinical settings.¹

However, the emergence of AMR has escalated into a crisis for modern medicine, representing a critical threat to global health. This rising problem is characterized by a significant increase in the number of antimicrobial resistant strains, "Super Bugs," many of which confer significant levels of multidrug resistance (MDR). In 2019, 569,000 deaths were attributed to AMR-associated infections.² The prevalence of antimicrobial resistant bacteria has gained a high level of attention worldwide and threatens global public health as a silent pandemic, necessitating urgent intervention.

AMR

AMR is an inevitable evolutionary phenomenon resulting from bacteria adapting to the therapeutic antibiotics and developing defenses against the mechanistic effects of antibiotic molecules. Bacterial microbes develop resistance against antibacterial drugs, rendering them ineffective. This adaptation occurs by either chromosomal gene mutations or the acquisition of foreign deoxyribonucleic acid (DNA) through horizontal gene transfer that codes for resistance determinants. AMR proliferates as bacteria acquire resistance genes, often through conjugation, and through internal evolving mechanisms of resistance to withstand and persist despite antibiotic treatments. This phenomenon is also documented with disinfecting agents.³ Bacteria evolve to evade the effects of antimicrobial agents, utilizing various mechanisms such as enzymatic breakdown, chemical modifications rendering agents functionless, inhibition of uptake, activation of efflux pumps, and modifications of target sites. Furthermore, with the aid of biofilm, bacteria are protected from disinfection agents, and biofilms can support communities of clinically relevant bacteria, posing public health threats. The biofilm mode of growth enhances bacterial resistance to antimicrobial agents, presenting a challenge in treating infections.

In the Department of Defense (DOD) 2023 *Biodefense Posture Review* (BPR), the emergence of antimicrobial resistance is described as “a naturally occurring biological threat that can overwhelm response capacities...” The BPR includes novel infectious diseases, the resurgence of once-localized diseases, and zoonotic diseases as growing challenges to outbreak control.⁴ As a result, there is a considerable danger that emerging AMR poses to the health and readiness of the Total Force, as well as the broader US population. While Force Health Protection (FHP) measures are usually effective in countering endemic diseases, some emerging infectious disease threats, such as MDR bacteria, risk rendering medical counter-measures ineffective. Naturally occurring diseases, that

hinder mission effectiveness, pose potentially serious harm to the Total Force and associated missions. Detection and surveillance capabilities addressing emerging disease threats are required for a resilient, ready force, which enables the Total Force to operate through biologically threatened environments.

Increased AMR prevalence, particularly from multi-drug-resistant “super bugs,” exacerbates public health challenges by limiting therapeutic options, leading to heightened morbidity, mortality, and financial burden. AMR hinders treatment efficacy of microbial infections and threatens the scope of treatment for many common illnesses, resulting in treatment failure, permanent disability, or even death. The emergence of AMR bacteria makes major medical procedures like chemotherapy and organ transplants significantly riskier, as patients become more vulnerable to infections that are increasingly difficult to treat without new antimicrobials. Perhaps more concerning are the evolving advancements in biotechnology that could exacerbate and exploit the AMR challenge. The 2023 BPR notes that developments in biotechnology, including but not limited to gene editing and whole genome sequencing, are driving an increase in the scope and diversity of biothreats. Similarly, the 2023 Department of Defense *Strategy for Countering Weapons of Mass Destruction* points out that the global availability of dual-use technologies, particularly in biotechnology, may erode traditional barriers to proliferation and



ABOVE: (Left to right) Sgt. Monika MacDonald and Private 1st Class Miguel Arroyo, members of the 753rd Quartermaster Company water purification team, drain the cyclone separator as they shut down operations at the Reverse Osmosis Water Purification Unit Rodeo at Fort Story, Virginia, June 15, 2017.

reduce opportunities to deny or disrupt the development of offensive weapons of mass destruction (WMD) programs.⁵ This point implies the potential of adversarial engineering or enhancing of pathogens with increased virulence and antimicrobial resistance using advanced biotechnological techniques.

As advancements in biotechnology expand the potential for pathogen manipulation, the high prevalence of MDR among waterborne pathogens presents a particularly concerning vector of exploitation, posing a threat to both public health and critical infrastructure, especially through the contamination and disruption of water purification and distribution systems. While the vulnerability of water systems to deliberate contamination has long been recognized—from ancient military campaigns to 20th-century conflict—the modern threat posed by AMR pathogens introduces a new dimension of complexity. The possibility of adversarial exploitation of waterborne pathogens (conferring AMR) to disable infrastructure or target military populations raises critical concerns for biodefense planning and strategy. To address this, experimental research has become essential for characterizing resistance profiles in real-world environments. This study, which examined 92 genetically distinct bacterial isolates from US wastewater treatment facilities, revealed a significant overlap between water disinfection resistance (chlorine and ultraviolet (UV) irradiation), biofilm formation, and multidrug resistance. By incorporating phenotypic assays, chlorine minimum inhibitory concentration (MIC) studies, and whole-genome sequencing of *Klebsiella* spp., the findings demonstrate how persistent, biofilm-forming MDR strains could potentially evade standard water disinfection protocols. These results not only highlight the weaknesses in current water treatment defenses but also provide empirical evidence to inform CWMD and FHP strategies as well as biosecurity risk assessments.

Methods and Materials

SAMPLE COLLECTION AND BACTERIAL ISOLATION

Between April and December 2023, surface ($n = 82$) and effluent water samples ($n = 96$) were collected from four wastewater treatment plants (WWTPs) in North Carolina—two employing UV irradiation disinfection and two using chlorination disinfection techniques. Surface swabs were enriched in buffered peptone water and incubated at 37°C for 18–24 hours. Effluent (post-disinfection) water samples were processed using

the World Health Organization Tricycle Surveillance Protocol, plated on MacConkey agar, and incubated for 24 hours at 37°C.⁶ Up to three colonies per plate were isolated and further sub-cultured on blood agar plates. Genera of cultured isolates were identified using Matrix-Assisted Laser Desorption Ionization- Time of Flight Mass Spectrometry (MALDI-TOF MS).⁷



ABOVE: Melendez's research produced cultured bacterial colonies isolated on MacConkey agar. Pictured bacterial growth was collected from UV-irradiation WWTP effluent discharge point.⁸

BIOFILM FORMATION ASSAY AND ANTIMICROBIAL SUSCEPTIBILITY TESTING (AST)

Opportunistic isolates identified as *Enterobacter* spp., *Klebsiella* spp., *Citrobacter* spp., *Aeromonas* spp., *Pseudomonas aeruginosa*, *Escherichia coli* (*E. coli*), and *Acinetobacter baumannii* ($n = 137$) were assessed for *in vitro* biofilm potential using a crystal violet microtiter plate assay. Cultures adjusted to 0.5 colony forming unit/milliliter (CFU/mL) McFarland standard (1.5×10^8 CFU/mL) were incubated in M9 minimal media supplemented with casamino acids and glucose. Biofilm formation was quantified by crystal violet staining, solubilized in ethanol, and read at Optic Density (OD) value of 570 nanometers. Biofilm potential classification followed established OD cut-off values as described by Leoney et al., 2020.⁹ Isolates identified as moderate ($n = 40$) or

strong ($n = 52$) biofilm producers were subjected to AST using Sensititre Gram-negative plates containing 14 antimicrobials (Thermo Scientific, Fisher). MICs were interpreted according to Clinical and Laboratory Standard Institutes 2023 breakpoints, and MDR was defined as resistance to ≥ 3 antimicrobial classes.¹⁰

CHLORINE SUSCEPTIBILITY AND IMPACT ON BIOFILM FORMATION INVESTIGATION STUDIES

The 92 bacterial isolates previously classified as moderate or strong biofilm formers were evaluated for chlorine susceptibility. A two-fold serial dilution of 12 ppm sodium hypochlorite was prepared in 96-well plates, and approximately 7.5×10^5 CFU/mL of each strain was added per well. Plates were incubated aerobically at 37°C, and the MIC was defined as the lowest chlorine concentration that inhibited visible bacterial growth. Furthermore, each strain was tested for biofilm formation in the presence of sodium hypochlorite using a crystal violet microtiter plate assay. Chlorine dilutions were prepared as described above and added to the wells. After a 24-hour incubation, biofilm biomass was quantified at OD570. OD cut-off values were determined using *E. coli* BAA 2469 as the negative control, and the MIC impacting biofilm was defined as the lowest chlorine concentration where OD570 fell below twice the cut-off threshold.



ABOVE: The 96-well plate with Mueller-Hinton Broth supplement added to columns 2-12 and chlorine solution (sodium hypochlorite solution, 12%) added to columns 1 & 2. Serial dilutions from columns 2-11. *Column 12 left without chlorine disinfectant (growth control). Bacterial suspension added to each well (final concentration 7.5×10^5 CFU/mL). ***ROW E demonstrates strong biofilm potential (in response to chlorine exposure) of pathogenic, multidrug-resistant *P. aeruginosa* strain cultured from chlorine-treated water sample.**¹¹

GENOMIC ANALYSIS OF CHLORINE-TOLERANT ISOLATES

DNA was extracted from selected from MDR *Klebsiella* spp. isolates ($n = 25$) and quantified via Qubit 4.0. Whole genome sequencing was performed on the Illumina MiSeq platform (2 × 250 bp, v3 chemistry).¹² Contigs ≥ 200 bp were assembled using the Bacterial and Viral Bioinformatics Resource Center (BV-BRC), BV-BRC Genome Assembly Services, and analyzed through the Amazon Machine Image Pipeline for antimicrobial resistance and virulence factors.¹³ Phylogenetic analysis was visualized using Interactive Tree of Life (iTOL) software.

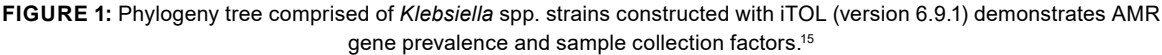
Results

Over a 9-month sampling period, 178 environmental samples were collected from four North Carolina WWTPs, with 77% of surface samples and 28% of effluent samples yielding culturable isolates. *Enterobacter* spp. and *Klebsiella* spp. were predominant among surface isolates, while *Aeromonas* spp. represented the preponderance of effluent water samples. Notably, 68% of all surface isolates exhibited moderate or strong biofilm formation potential *in vitro*, with *Klebsiella* spp. showing the highest prevalence of strong biofilm production. AST assays revealed that 58% of moderate and strong biofilm formers were MDR, with chlorine-treated isolates accounting for the majority of MDR phenotypes. Importantly, all isolates remained susceptible to ciprofloxacin, suggesting limited fluoroquinolone resistance and demonstrating its potential anti-biofilm properties. Repetitive-polymerase chain reaction fingerprinting confirmed high genetic diversity across isolates with no clear clustering by treatment type, indicating a heterogeneous bacterial population. These findings

TABLE 1: PHENOTYPIC AST RESULTS

Treatment/ Sample Type	MDR Count	MDR Percent
Chlorine/Surface	24	45%
Chlorine/Water	10	19%
UV/Surface	19	36%
UV/Water	0	0%

TABLE 1: Phenotypic AST Results: 58% ($n = 53$ of 92) of the moderate and strong biofilm potential strains displayed MDR.¹⁴



Minimum inhibitory concentration (MIC) assays revealed that over half of isolates from both chlorine-treated (57%) and UV-irradiated (50%) WWTPs maintained a chlorine tolerance of 12 ppm. No significant differences were observed in MIC values across treatment types, sample sources, or initial biofilm classifications. However, follow-up biofilm assays under chlorine stress showed statistically significant associations: isolates from UV-treated WWTPs and surface samples exhibited higher chlorine MICs and biofilm formation capacity ($p < 0.05$). Strong biofilm producers had notably higher odds (OR = 5.42) of chlorine tolerance. These findings suggest that biofilm-forming capacity may confer increased resistance to disinfection, with UV-treated isolates potentially compensating for lack of residual disinfectant by enhancing biofilm development. Interestingly, several effluent water isolates from chlorine WWTPs—despite showing high MICs—failed to produce biofilm under chlorine exposure, indicating potential chlorine-induced damage beyond recovery thresholds.

Whole genome sequencing of 25 *Klebsiella* spp. MDR strains (with moderate or strong biofilm potential) confirmed high genetic diversity and widespread resistance. All strains harbored at least one AMR gene, and 84% amplified AMR gene presence, with β -lactam and fosfomycin resistance being most prevalent. Virulence genes (indicating pathogenicity) such as *fimH*, *iutA*, *iroN*, and *traT* were commonly detected, with chlorine-derived strains showing higher frequencies of *fimH* and *traT*, suggesting possible selection pressures from residual chlorine. The presence of *fimH*, associated with adherence and biofilm initiation, was significantly higher among chlorine-treated isolates (80%) versus UV-treated (30%). These results strongly indicate the potential role of disinfectant type in shaping both phenotypic resistance and virulence traits in *Klebsiella* spp. and necessitate the need for targeted biofilm-disruption strategies and deeper molecular characterization in water disinfection management.

The threat posed by AMR is no longer a distant theoretical risk; it is an emerging biological reality with profound implications for military readiness, national security, and global public health. AMR has demonstrated the potential to erode the effectiveness

of current medical countermeasures, compromise water safety, and effectively overwhelm healthcare and logistical systems during conflict or humanitarian crisis.

The findings from the recent experimental studies further reinforce the urgency of this challenge. Characterization of chlorine-resistant Gram-negative bacteria isolated from US wastewater treatment plants revealed a concerning prevalence of multidrug resistance (58%) among moderate-to-strong biofilm-forming isolates, with 92% demonstrating resistance to chlorine concentrations commonly employed for water disinfection. The persistence of these bacterial populations under chemical disinfection pressures, compounded by enhanced survival through biofilm formation, highlights a dual challenge: the erosion of existing public health defenses and the growing opportunity for adversarial exploitation of resilient pathogens.

These experimental outcomes emphasize that combating the AMR threat demands not only innovation in medical countermeasure development, but also a fundamental re-examination of disinfection protocols, surveillance strategies, and biodefense frameworks. Bridging laboratory evidence with operational preparedness is essential. Only through integrated research, policy adaptation, and proactive security environment-focused initiatives can we preserve operational resilience and effectively mitigate the evolving biological threats of the 21st century as WMD threats expand beyond the nuclear domain. ■

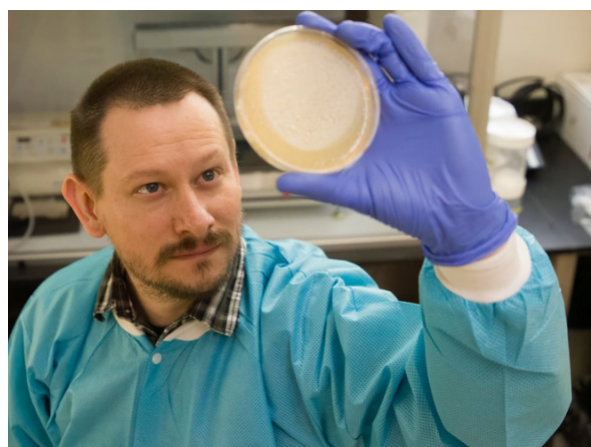


FIGURE 2: Matthew Henry, a research associate with the Naval Medical Research Command's (NMRC) Biological Defense Research Program (BDRD) Directorate, observes signs of successful infection of bacterial agents by isolated bacteriophage products during a separate study. (U.S. Navy photo by Mike Wilson/Released)

Key Points:

AMR is a critical factor in the DOD's biological defense efforts due to its potential to:

- Undermine the health and readiness of the Total Force as a naturally occurring threat.
- Possible exploitation as a characteristic of biological weapons targeting water systems and populations.
- Enhancement or engineering through advancing biotechnology, leading to more complex and resistant biothreats.
- Influencing by current disinfection practices and lack of antibiotic stewardship, exacerbating the problem.
- AMR necessitates the development of novel and adaptive medical countermeasures.
- AMR require ongoing advanced research and development to anticipate and counter future AMR threats.

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Our Fireball Radius Height of Burst Calculation is 75 Years Old: is it Time for an Update?

By: Col. Jeff Kendellen

Introduction

For every single nuclear weapon considered for employment—starting with Japan in 1945 to today at US Strategic Command—planners will address the question: will this detonation produce fallout? The 'yes' or 'no' answer to this question has traditionally relied upon an equation with known significant limitations. I propose a paradigm shift. It is time to update the way our community calculates heights of burst when considering fallout. In this paper, I will present important aspects of the nuclear fireball, a historical review of fireball radii calculations, and introduce some fireball and ground interaction errors and misconceptions. And finally, I challenge the Functional Area (FA)52 community to start a 'scientific revolution' to critically evaluate and update our current paradigm to serve our community better.¹

Before we dissect the fireball, we must reflect on important and meaningful variations on the question asked above: will this detonation produce fallout? Another similar question is: will the fireball touch the ground? Perhaps more important to Army FA52 Officers and military planners is the question: will this detonation produce *militarily significant* fallout?

The Fireball

With the foci and questions above in mind, I will describe the very basic elements of the fireball creation. When a nuclear weapon detonates, the fissile material

is consumed in small fractions of a microsecond creating extreme temperatures (millions of degrees Celsius) and pressures (hundreds of gigapascals). These extremes generated by the bomb energy are caused primarily by x-rays radiating outwards, interacting with and depositing energy in the bomb casing materials and with the nitrogen, oxygen, and other minor components of the atmosphere. The elements and molecules in the vicinity essentially fully disassociate and ionize. These close-by x-ray interactions contribute to fireball growth through the generation of a quasi-spherical volume of incredibly hot and dense plasma. This is the "initial fireball."² "The fireball subsequently expands through a combination of radiative and hydrodynamic processes."³ The expansion of the initial volume of plasma occurs because of the drastic increase in mean free path of follow-on x-rays.⁴ These x-rays travel farther extending the sphere outwards because of more atmospheric interactions and energy deposition thereby creating the later growing hot and dense ionized plasma fireball.

The expansion of the fireball generates a "intense shock wave (thin region of highly compressed air)... formed at the surface, which expands outwards at a high velocity."⁵ Eventually, the x-rays stop radiating and the hot dense plasma, the excited molecules, and the environment around the fireball all must find balance and equilibrium. Part of this process as the fireball stops expanding creates a transition from a radiation wave to a shock wave—referred to as "hydrodynamic separation."⁶ This is the breakaway point and aligns approximately with the fireball's maximum size.⁷

Historical Review of Radius Calculations

With the basic elements of the fireball discussed, the next section will review various sources, equations, and calculations for fireball radius. Sir Geoffrey Taylor's initial fireball radius work started in 1941 when he was tasked by the British Civil Defence Research Committee to mathematically evaluate a "bomb in which a very large amount of energy would be released by nuclear fission."⁸ He completed his task in June 1941, but his work was classified until 1950.⁹ In 1950, he published his report with the addition of an application of his mathematical work applied to the 1945 Trinity test.¹⁰ Sir Taylor took film and still photographs released by the US Atomic Energy Commission (and printed by Life Magazine) of the growing fireball radius over time and was able to fine tune his assumptions and novel constants within his previous work to calculate a yield of 16.8 kilotons.¹¹ His yield estimate is surprisingly close to the actual yield of 21 kilotons.¹² Taylor's Equation is given as equation (1) where W is yield in kilotons, and ρ_0 is atmospheric pressure in kg / m³.

$$Radius = \left(\frac{t}{0.926} \right)^{2/5} \frac{W^{1/5}}{\rho_0^{1/5}} \quad (1)$$

The next equation is perhaps the most well-known. Glasstone's (and later Dolan's) equation provided another classic equation within *The Effects of Nuclear Weapons* to determine where the height of burst "ceases to be a serious problem."¹³ This is given by equation (2) where W is yield in kilotons.

$$\begin{aligned} \text{Height of Burst (ft)} &= 180 * W^{0.4} \\ \text{Height of Burst (m)} &= 54.9 * W^{0.4} \end{aligned} \quad (2)$$

It is important to stress, given the focus of this paper, that Glasstone and Dolan provide a substantial +/-30% error and specifically state heights of burst calculated using equation (2) may still produce fallout, but the fallout will "be small enough to be tolerable under emergency conditions."¹⁴

With respect to Glasstone's (and Dolan's) work, there are important editorial observations that should be pointed out. Equation (2) within the 1977 'third edition' exists in the same form within earlier versions of *The Effects of Nuclear Weapons* which were released in February 1964 and April 1962 (which were solely attributed to Glasstone as editor). Interestingly, Glasstone's 1957 version has the same equation (2), but the '+/-30%' and follow-on discussion regarding 'emergency conditions' was not mentioned. Instead, Glasstone provides a separate equation, only published in the 1957 version, for

the "maximum size of the luminous fireball" which is given by equation (3) where W is yield in kilotons.¹⁵

$$\begin{aligned} \text{Radius(ft)} &= 230 * W^{0.4} \\ \text{Radius(m)} &= 70 * W^{0.4} \end{aligned} \quad (3)$$

The reader should immediately note any output from equation (3) will be greater than equation (2) which perhaps speaks to Glasstone's 1957 view of 'negligible fallout.' Glasstone's (and the broader community's) data at the time was limited to sixty-one non-contact above ground nuclear weapon tests starting with Trinity in 1945 and ending with Operation Redwing in 1956.¹⁶ Because equation (3) did not survive for future printings of *The Effects of Nuclear Weapons*, and future prints did include the '+/- 30%,' it is likely Glasstone reevaluated their stance to view fallout more conservatively when considering heights of burst and resultant fallout.

An interesting mathematical point about equation (3) is it is 27.8% more than calculations in equation (2). It is likely that while equation (3) was eliminated after the 1957 version, it was effectively retained in the 1962, 1964, and 1977 versions by rounding up and utilized as the '+30%' within the '+/- 30%.' A graph view of this point is clearly visible in Figure 1.

Another equation for fallout safe heights of burst is presented within the Staff Officers Field Manual *Nuclear Weapon Employment* FM 101-31-1 published in February 1963. Within this text, the authors provide two equations based on yield.¹⁷ They can be seen below. Again, W is yield in kilotons.

$$HOB_{is}(m) = 30 * W^{1/3} \text{ when the yield is } \leq 100 \text{ kilotons.} \quad (4)$$

$$HOB_{is}(m) = 56 * W^{1/3} \text{ when yield is } > 100 \text{ kilotons.} \quad (5)$$

Equation (5) tracks reasonably well with the -30% version of equation (2) as seen in Figure 1. Between 105 kilotons and 500 kilotons the two never diverge more than 6%. Equations (4) and (5) are added to the buffer distance which is shown in equation (6) and accounts for weapon delivery inaccuracy. The inaccuracy in this case is not concerned with circular error probability, but instead the probable error in height (PE_h).

$$\text{Buffer distance} = 3.5 * \text{Probable Error Height} = 3.5 * PE_h \quad (6)$$

$$HOB = HOB_{is} + 3.5 PE_h \quad (7)$$

PE_h is a weapon system dependent error that also differs for how weapons are employed. Meaning, a certain nuclear capable artillery may have a different

PE_n for targets at different ranges which would be given in a table. The buffer distance creates assurances that 99% of the time, the weapon will detonate above the height calculated in equation (7).¹⁸ Later versions of this Field Manual stopped using equations (4) and (5) and instead used heights of burst and PE_n from classified tables. Operators were expected to add the two together when planning weapon employment.

The next equations are primarily different from the previous equations in that they consider atmospheric pressure. The first is equation (8) contained within Ernest Bauer's 1990 report "Physics of High-Temperature Air – Part II Applications."¹⁹ Radius is in kilometers, W is yield in megatons, ρ is air density at burst altitude, and ρ_0 is air density at sea level.

$$Radius(t) = 0.85 W^{0.33} \left(\frac{\rho_0}{\rho}\right)^{0.3} + 0.037 W^{0.3} \left(\frac{\rho_0}{\rho}\right)^{0.2} (t - t^*) \quad (8)$$

Equation (9) comes from Dr. Hans Bethe's report "Theory of the Fireball" published in 1964.²⁰

$$Radius = 0.78 W^{\frac{1}{3}} (T')^{-0.1} \rho^{-\frac{1}{4}} \quad (9)$$

W is yield in megatons, ρ is pressure in bars, ' T ' is equal to 0.92, and radius is in kilometers. Dr. Bethe's equation, as shown in Figure 1, is a fair approximation to the output of equation (2).

Due to Distribution Statement restrictions, fireball equations from John Northrop's *Handbook of Nuclear Weapon Effects* and Charles J. Bridgman's *Introduction to the Physics of Nuclear Weapon*

Effects, cannot be recorded here. It can be said that Northrop's equation plotted is less than the plot created using equation (2). And Bridgman's equation plotted out is less than the -30% Glasstone and Dolan plot. Both curves are of similar shape.

Aside from equations in books, manuals, and reports, there are number of tools that calculate fallout safe heights of burst. One notable software program in use currently is called *Hazard Prediction and Assessment Capability* which models nuclear weapon detonations by creating a source term and using real-world weather to model the lofting of material and subsequent fallout deposition in the local area. This software simply uses Glasstone and Dolan's equation (2) with no error applied.²¹ US Strategic Command planners also use tools and software based on Glasstone and Dolan's classic equation (2). Planners use equation (2) as the "analytic starting point" while also incorporating weapon delivery system errors.²²

The last tool I will mention, is somewhat of a collector's item—the *Nuclear Bombs Effects Computer*—often call a whiz-wheel which was included at the back of some copies of Glasstone and Dolan. Naturally, the calculations for fireball and ground interaction are based on Glasstone and Dolan's equation (2).

Misconceptions and Errors

These sources and equations should offer a perspective on how important the fireball/fallout

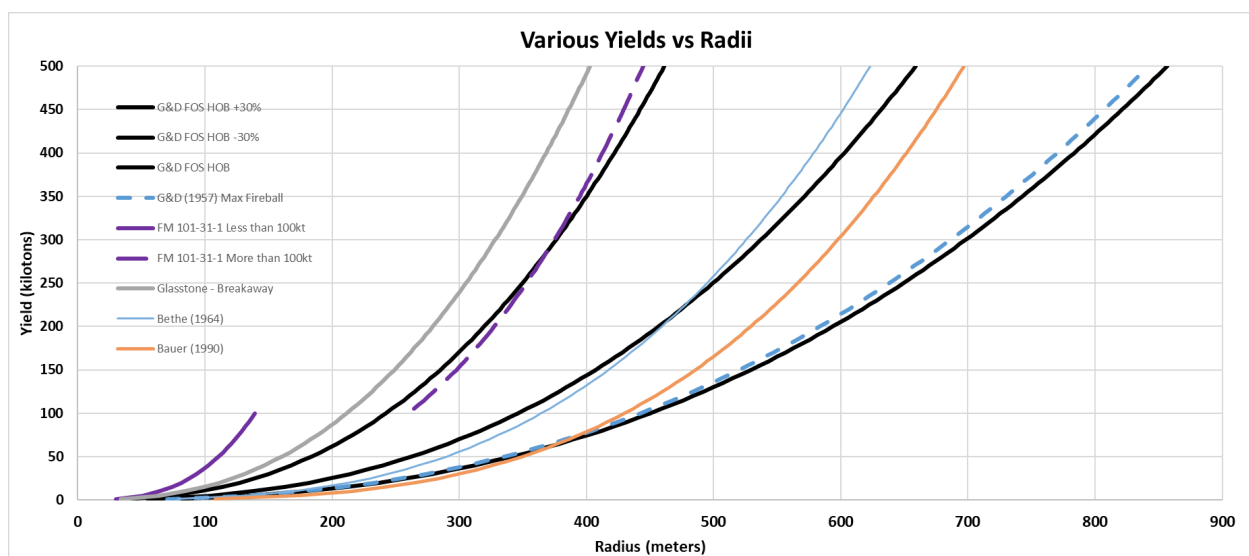


FIGURE 1. is a representation of the equations listed in the section above.

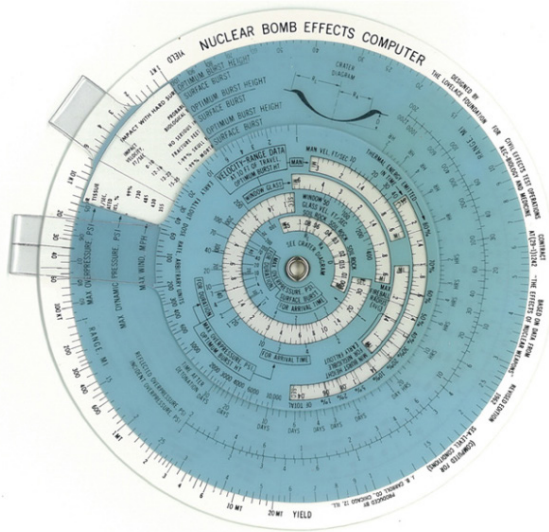


FIGURE 2. The Nuclear Bomb Effects Computer.

question was and is. Yet the lack of fidelity – both in the number of ways to calculate fireball radius, the errors given for these ways, and the changes over time—should cause concern. In the next section, the paper will attempt to add to that concern by providing some obvious, and perhaps not-so-obvious challenges, misconceptions, and errors when answering the fireball/fallout question(s).

Nuclear Weapons are Not Perfect

No nuclear weapon detonation is ever perfect. Films of fireballs prove this with what Harold Brode describes as fireballs that “lack symmetry or sphericity. There are frequently relatively large blisters and bright spots associated with the glassy looking shock front. These are evidence of the non-ideal nature of fireballs or rather features of nuclear explosions, which are thrown at high velocities against the back of the shock wave.”²³ Brode elaborates further by stating radius calculations should be based “on x-ray yield fraction rather than total yield, since this fraction can change both as a function of the yield and as a function of the bomb design.”²⁴ Glasstone goes even further on this idea by writing: “numerical values given in this book are not—and cannot be—exact. They must inevitably include a substantial margin of error. Furthermore, two weapons of different design may have the same explosive energy yield, but the effects could be markedly different.”²⁵ Beyond the fireball itself, the associated delivery systems have errors as articulated above within the discussion on buffer distances and probable error in heights.

Reentry Body/Vehicle Body Velocity

Another consideration is the distinction between testing and real-world employment. In practice, heights of burst calculations were developed using films from above ground tests where the nuclear detonation was nearly stationary (e.g., using towers or balloons). Obviously, in these test detonations, collecting data was important which necessitated a quasi-static test subject. These stationary detonations are the antithesis of the extreme velocities experienced by reentry bodies or reentry vehicles. The reentry bodies/vehicles attain maximum velocities ranging from 4,500 to 6,000 m/s which equates to Mach 13 to 17.²⁶ By assuming Mach 17 as a worst case, what do back of the envelope calculations tell us about this velocity on fireball creation?

First, it is important to note the energy released from the fissile fuel occurs very quickly, on the order of ~100 shakes or ~1 microsecond.²⁷ Even when traveling Mach 17, the weapon only moves about ~1 centimeter while the weapon fuel is expended. X-rays and gamma rays are traveling at the speed of light (2.99×10^8 m/s) outward, but the fireball itself grows over seconds to about 10 seconds for megaton explosions.²⁸ An interesting consideration is: does the fireball have inertia and does the inertia contribute to fireball deformations and/or fireball/fallout errors?

The author had hopes that nuclear weapon testing would elucidate what happens during a detonation at ~Mach 17, but the single above ground test of a ballistic missile was not associated with an “experimental program.”²⁹ This one test of a ballistic missile with a nuclear warhead was during Operation Dominic under test Frigate Bird. During Frigate Bird, a 200-1000 kt weapon was launched the USS Ethan Allen on May 6, 1962.³⁰ Unclassified videos of the detonation do not contain sufficient detail to discern any visual conclusions about the fireball or deformation.

To answer the question above, the fireball will certainly not travel ~6,000 meters over one second along the flight path of the reentry body/vehicle. Similarly, the velocity/path will not appreciably deform the fireball or the fireball development. The reasoning is as follows: The x-rays are traveling at the speed of light which dwarfs the speed of the reentry vehicle/body. In other words, the bomb will detonate and the x-rays will be deposited generating the growing fireball with little to no impact from the Mach 17 velocity.

Does a Mushroom Cloud = Radioactive Fallout?

Within this topic, I will address both a misconception and a factor that further burdens fireball/fallout error. Here, the question is: does the creation of a mushroom cloud mean the fireball touched the ground? As presented, this is a trick question that happens to be well covered by in a 2020 article entitled “Fallout Cloud Regimes.” The question above is a trick question because there are cases where the fireball does not touch the ground (verified visually during above ground tests), but “dust [can be] drawn into the fireball during the initial fireball phase, quenching part of the plasma in the fireball.”³¹ The authors list multiple tests where the fireball does not touch the ground, yet created local and downwind radioactive fallout. These tests are summarized below with comparisons to Glasstone and Dolan’s equation (2) along with +/-30%.

What Figure 3 shows is the Buster Jangle Easy test which was done at a height of burst 85% more than Glasstone and Dolan’s equation (2) value of 217 meters, and yet still produced fallout. Grable, where the height of burst was nearly equal to the equation (2) calculation, also produced fallout. The authors also stress in their paper: “fallout-free height of burst, as frequently included in discussion of nuclear effects literature ... [is] something of a misnomer.”³²

Fireball Deformation

In this section, I will elaborate on a special aspect of the previous errors concerning mushroom clouds. It is the case that the fireball can also be deformed by the reflected ground shock. FM 101-31-1 states: “If the weapon is burst in the air close to the ground, the blast wave will reach the ground, be reflected, and flatten the bottom of the fireball before it has reached maximum size.”³³

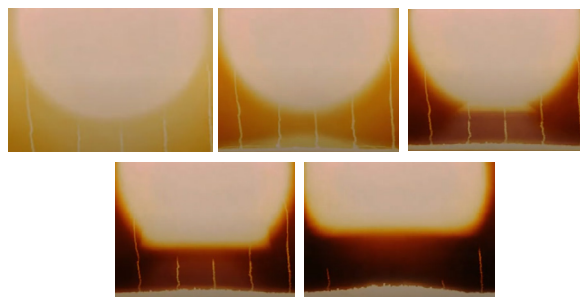


FIGURE 4. Still images from an unidentified nuclear weapon test near the ground. Extracted from the Nuclear Science and Engineering Research Center video: *A Review of Prompt Effects*.

This effect, when seen visually in Figure 4, is assistive in helping one understand fireball growth, development and breakaway. At this point, the fireball is no longer growing appreciably, but can still be influenced by strong forces, such as from *its own* reflected shockwave.

Conclusion and a Challenge

Given the history, the stated errors, and potential misconceptions—Is it time for to update Glasstone and Dolan’s classic equation?

Yes!

To make this point simply, imagine the new car you just bought had a manufacturer stated highway range of 400 miles. Only for it to run out of gas at 280 miles. Where else in life or work do we just accept a +/- 30% error? Furthermore, an accurate starting point *is the source term* feeding follow-on tremendously important analysis. Without an accurate source term, our processes are ‘garbage in, garbage out.’

	Yield (kt)	Test HOB (m)	SHOB Calculations			(m/kt ^{1/3})
			G&D HOBfs (m)	G&D HOBfs + 30% (m)	G&D HOBfs - 30% (m)	
Buster Jangle Easy	31	401	217	282	152	128
Tumbler Snapper Dog	19	317	178	232	125	119
Grable	15	160	162	211	113	65
Priscilla	37	213	233	302	163	64

FIGURE 3. The summaries of relevant test data, equation (2) calculations, and Scaled Height of Burst calculations from Spriggs, et al.

I challenge the FA52 community to revolutionize how we—as a community charged with keeping our fellow Soldiers safe from contaminated fallout—calculate fallout safe heights of burst at US Strategic Command and the US Army Nuclear and Countering Weapons of Mass Destruction Agency. This will start with gathering the best data offered within Effects Manual-1, the research community, and recent scholarly work on nuclear weapon testing film digitization. Whatever the equation ends up being, and it might not even be much different from Glasstone and Dolan's equation (2), at least we will know the new answers have a solid foundation based on extensive data. Or we—as a community—can continue obdurately employing an equation from 1977, that was carried over from 1967 and 1962; copied from the 1957 version; and adopted from a 1950 manuscript 75 years ago after only five air burst nuclear weapons tests.³⁴ ■

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Laser-Driven X-Ray Source Array Design for Nuclear Weapons Effects Testing

By: Cadet Ryan Brockington, Cadet Owen Koster, Cadet Roberto Lopez, Cadet Joshua Santana, Mr. Michael Tobin, and Dr. John Davis

Abstract

This research develops nine x-ray source arrays to simulate nuclear weapons effects through stimulated x-ray emission. The sources are low-density foams that are irradiated by intense short-pulse laser beams to stimulate x-ray emissions. Each array is designed to adhere to a specific target fluence given a set number of source targets and a total amount of incident energy. Furthermore, each source array is required to maintain 80% surface x-ray fluence uniformity at the desired average fluence. The specifications provided for each source array are designed to simulate significant nuclear weapons x-ray effects, specifically Blowoff Impulse and Shock (BIS), Thermo-Structural Response (TSR), and System Generated Electro-Magnetic Pulse (SGEMP). Results generated using Monte Carlo N-Particle (MCNP6) simulations identify the potential for enhancing as well as filling in gaps in current United States (U.S.) radiation effects testing capabilities.

Introduction

Xcimer Energy Corporation (XEC) is a private company focused on achieving both Inertial Fusion Energy (IFE) and an x-ray effects test capability using their novel krypton fluoride (KrF) laser. XEC is building on the impressive recent fusion ignition and gain results at Lawrence Livermore National Laboratory's (LLNL) National Ignition Facility (NIF) to propose a pathway to fusion energy. LLNL has now achieved a total of eight results on NIF exceeding fusion ignition with the largest yield being 8 MJ. The XEC proposal to the Department of Energy earned them a nine-million-dollar award from the Office of Fusion Energy Sciences for the first year of a multi-year milestone-driven agreement. After successfully demonstrating their KrF laser concept in an XEC facility called Phoenix by March 2026, XEC intends to design and build a 4-MJ KrF laser and chamber to achieve a 200-MJ inertial confinement fusion (ICF) yield demonstrating economic breakeven for IFE. A second chamber would be added for nuclear weapons effects testing (NWET) using the same KrF laser.

The U.S. began voluntarily adhering to the Comprehensive Test Ban Treaty in 1992, which stopped all nuclear weapon underground tests – including x-ray effects testing. Since then, the U.S. has been limited in its ability to conduct testing for BIS, TSR, and SGEMP effects on full systems. This inability to conduct comprehensive radiation effects testing for full systems has left a shortfall for U.S. national security – a shortfall that the XEC KrF laser has been designed to address.

The development of XEC's target arrays is an interdisciplinary project that combines multiple lines of effort to develop a system for simulating NWET. Through a Cooperative Research and Development Agreement (CRADA) between the United States Military Academy (USMA) and XEC, this research applies computational analysis using the radiation transport code, MCNP6, to accurately design source arrays for a radiation effects testing chamber.¹

Theory

The XEC KrF laser utilizes novel laser scattering effects to achieve very high energy flux without solid-state media. This substantially reduces the overall cost of this laser compared to a solid-state laser like the NIF. The laser utilizes Stimulated Brillouin Scattering (SBS) to compress laser pulses as well as Stimulated Raman Scattering (SRS) to combine laser pulses together. These two processes enable the KrF laser to combine multiple laser pulses into a single, short, high-intensity pulse.² XEC is currently building a research KrF laser to demonstrate the successful application of these techniques. The KrF laser design has unique dual use capabilities for ICF and NWET and can service two chambers.

The XEC KrF laser creates a unique opportunity for x-ray effects testing with the potential to significantly increase x-ray source outputs with distributed source arrays. Intense laser energy deposition into specific low-density materials can be used to generate x-rays as electrons de-excite (typically from L- or K-shells).³ These “underdense” sources have shown useful high laser-to-x-ray conversion efficiencies.⁴ The NIF houses the world’s largest laser, which generates two megajoules of laser energy, yet this is not sufficient to cover the relevant spectrum of energies for NWET at the desired fluence-area products as shown in Figure 1. The XEC KrF laser at energies of 4- to potentially 12-MJ or higher will be able to address this gap.

Methodology

This research focuses on designing nine unique source arrays that successfully simulate the full range of nuclear weapon x-ray effects. The nine cases are detailed in Table 1 and focus on simulating three relevant radiation effects – BIS, TSR, and SGEMP.

In Table 1, each test case calls for a different laser energy as well as a different number of x-ray sources to optimize for that specific x-ray effect. The source type describes the metallic foam target material – which is selected for its x-ray emission energy – that emits characteristic x-rays in the relevant energy range for the desired radiation effect (BIS, TSR, or SGEMP). Furthermore, the source type efficiency describes the percentage of incident energy that is maintained in the emission of relevant x-rays. The spectra produced by each source type was provided by XEC and is filtered using Kapton® or aluminum to remove low energy x-rays that are not relevant for the desired radiation effects testing. The specifications provide a desired fluence for each array based on client expectations to simulate NWET at relevant distances for each radiation effect from the point of detonation. The array design must have a minimum of 80% uniformity across the test article surface to align with design requirements. Lastly, the test article material is important for energy deposition analysis and serves to mimic relevant components that would be studied for the given radiation effect (for example SGEMP is significant for electronic components that are commonly made of silicon).

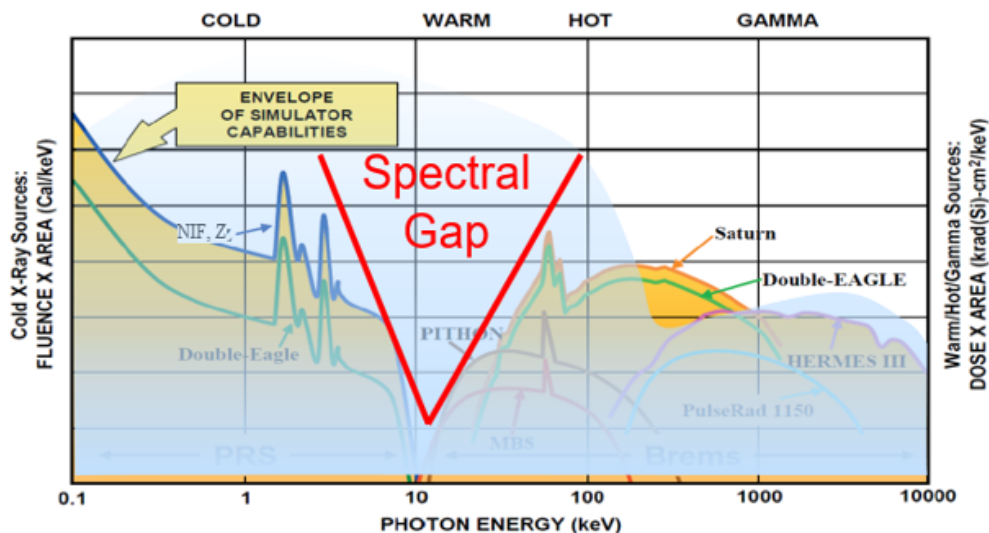


Figure 1: Graphical depiction of current radiation effects testing capabilities over different photon energies. The spectral gap highlights the region that current capabilities cannot sufficiently simulate.⁵

TABLE 1: NINE LASER-DRIVEN X-RAY SOURCE ARRAY SPECIFICATIONS

X-Ray Source Spectrum	Laser Energy (MJ)	Number of X-Ray Sources	Energy Per Source (kJ)	Source Type (Efficiency)	Filtering Material and Thickness (um)	Goal Fluence (cal/cm ²)	Test Article Material
Cold-BIS	4	8	500	Ag (35%)	-	10	Carbon
	12	24	500	Ag (35%)	-	10	Carbon
	24	24	1000	Ag (35%)	-	20	Carbon
Warm-TSR	4	4	1000	Cu (3.5%)	Kapton (37.5)	5	Aluminum
	12	12	1000	Cu (3.5%)	Kapton (37.5)	5	Aluminum
	24	12	2000	Cu (3.5%)	Kapton (37.5)	10	Aluminum
Warmer-SGEMP	4	4	1000	Cu (1%)	Aluminum (37.5)	3	Silicon
	12	8	1500	Cu (1%)	Aluminum (37.5)	3	Silicon
	24	12	2000	Cu (1%)	Aluminum (37.5)	5	Silicon

This research assumes that x-ray sources in an array can be modeled as “n” isotropically emitting points located some distance from a test object. Figure 2 depicts the assumed x-ray output using isotropic sources compared to a uniform wall of x-rays, which can be used to model x-ray effects from a nuclear weapon on some system in line-of-sight, yet far from the point of detonation, of that weapon.

This design mimics a planar wall of photons. Point sources cannot perfectly recreate a planar wall with sufficient fluence; so, arrays are designed to meet

The fluence for test objects must meet the specifications provided in Table 1, while maintaining 80% uniformity at the surface. These strict requirements demonstrate the fidelity of utilizing XEC’s KrF laser design to provide the U.S. with a comprehensive means of assessing critical nuclear weapons effects. With an optimal arrangement of sources and sufficient laser energy, it is possible to simulate radiation effects on full systems for NWET. This would eliminate any future need to resume underground NWET by providing a facility for understanding x-ray radiation effects on any desired system.

Laser Sources X-Rays



Uniform X-Ray Arrangement

VS

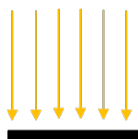


Figure 2: Laser source x-rays compared to a uniform x-ray photon arrangement.

the 80% uniformity specification. To determine the uniformity of simulated results, the minimum fluence is divided by the maximum fluence for a given simulation. Arrays are modelled in MCNP6, where tallies can be added with the appropriate multiplier to assess the total energy deposited across the area to ensure sufficient uniformity is maintained. It is important to note that these calculations can be computed analytically by assessing the flux of specific points from all associated sources; however, this analytical process becomes much more challenging when the number of sources significantly increases and the number of critical points on the test article significantly increases. Thus, MCNP6 offers a unique means of overcoming this problem set.

To determine x-ray fluence uniformity, a lattice mesh design is constructed to model the desired test article. This base design utilizes a square plate for the desired test article. A 15 x 15 lattice is used for 4- and 8- source arrays and a 23 x 23 lattice for the 12- and 24-source arrays to provide greater resolution for more complex designs. Energy fluence uniformity incident on the top surface of the plate is only dependent on the amount of incident energy, and not on the specific energy spectra applied, so 3-keV photons are applied to all surface uniformity calculations to simplify MCNP6 inputs and associated tally multipliers for the initial analysis. Regarding uniformity, the photon flux (F4 tally) is evaluated for each lattice cell. By dividing the minimum fluence from the maximum fluence, the uniformity across the test article can be easily determined. This procedure is then repeated for each array design to identify the optimized target array for NWET.

This research applies two fundamental methods for optimizing array designs: progression and revision. Progression focuses on starting with the most basic array design (4-source arrays) and taking the lessons learned from that design into the more complex arrays progressively (8-, 12-, and 24- source arrays). Revision requires utilizing a multitude of designs to assess the optimal array for each source case. One example

of revision is provided in Figure 3 for the 24-source model. The array on the top utilizes the same basic design as the 8-source array iterated three times to maximize distance; however, this creates a significant minimum to occur along the edges between the middle and corner sources. Because the edges are typically where a minimum is most likely to occur, the array on the bottom is developed. This array accounts for the significant role of the edges in determining uniformity by placing more sources along the edges and using fewer sources to balance out the center of the array. Aside from all sources needing to be in the same x-y plane, there are no restrictions on how sources can be arrayed, so each array has an infinite orientation of sources that all yield a distinct target size that meets the design requirements. For this reason, array designs should be constantly reassessed and built upon to identify unique array orientations that may yield a larger target size. Ultimately x-ray source arrays will be standardized to control costs and to allow well understood environments to be repeated as needed.

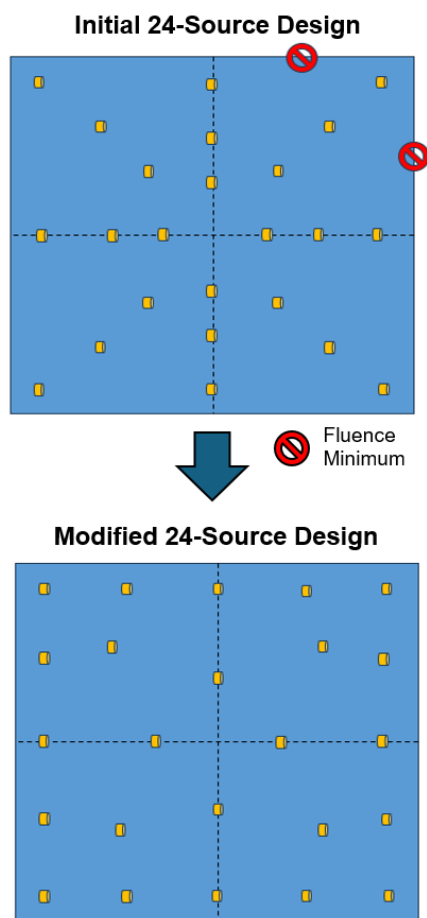


Figure 3: Revision utilizing the 24-source model.

The critical design requirement to make the optimal array for NWET across all array designs is ensuring that minimum surface uniformity is maintained. Most designs require a constant process of revision and reassessing source orientation to maintain uniformity while maximizing target size. If fluence did not meet the goal fluence with 80% uniformity, then the source array is modified until all design requirements are met. Once all design specifications are met, the area of the test object is then increased until it fails the uniformity requirement, and then the array geometry is adjusted again. This process continues until the maximum area that maintains all design requirements is identified. A concept drawing of the 8-source SGEMP source array is depicted in Figure 4. Flux uniformity plots, such as Figure 5 are generated utilizing a Julia parser script that compiles the results from the F4 mesh tallies across each cell from MCNP6.

Once the optimal array design is identified, the given source arrays can be applied using the appropriate photon energy with the given target material (as specified by Table 1) to analyze x-ray energy deposition. For x-ray energy deposition, the lattice mesh used to analyze surface uniformity is iterated into 15-layers that progressively increase in size to measure both the deposition (F6 tally) within the x-y plane and the z-axis of the test article. Similar to uniformity calculations, tallies are placed in each cell. To generate meaningful results for energy deposition calculations, it is critical to understand that the mean free path of x-rays in the relevant materials is small, so the most important regions within the material are those closest to the surface. Therefore, the total depth increases progressively from 1 nm to 3 nm, to 10 nm, then to 30 nm, and so forth until a total thickness of 1 cm is achieved. The thickness of each lattice increases with depth.

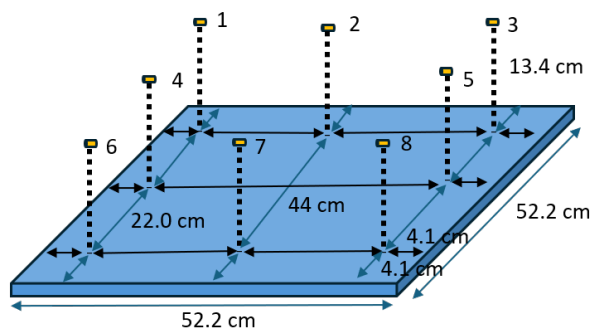


Figure 4: Full-sheet initial concept drawing of the 8-source SGEMP 12 MJ source array.

TABLE 2: SUMMARIZED RESULTS FOR EACH CASE

X-Ray Source Spectrum	Laser Energy (MJ)	Number of X-Ray Sources	Energy per Source (kJ)	Source Type (Efficiency)	Goal Fluence (cal/cm ²)	Area for 80% Uniformity (cm ²)	Max Fluence (cal/cm ²)	Surface Uniformity (%)
Cold-BIS	4	8	500	Ag (35%)	10	9,801	11.13±0.01	80.40
	12	24	500	Ag (35%)	10	46,785	11.19±0.01	81.35
	24	24	1000	Ag (35%)	20	46,785	22.37±0.02	81.35
Warm-TSR	4	4	1000	Cu (3.5%)	5	1,089	5.544±0.008	80.46
	12	12	1000	Cu (3.5%)	5	5,715	5.458±0.006	82.56
	24	12	2000	Cu (3.5%)	10	5,715	10.92±0.01	82.56
Warmer-SGEMP	4	4	1000	Cu (1%)	3	548	3.374±0.004	80.05
	12	8	1500	Cu (1%)	3	2,724	3.307±0.004	82.26
	24	12	2000	Cu (1%)	5	3,969	5.451±0.005	80.44

Results and Analysis

This research successfully identifies nine x-ray source arrays to simulate nuclear weapons effects. Once each source array is finalized, the results for each design present a generally consistent pattern as each array is developed to meet the required design specifications. For that reason, one source array design, the 8-source SGEMP 12-MJ array, is presented to discuss the general pattern from all results garnered. The surface uniformity, energy deposition, and energy deposition uniformity for the 8-source SGEMP 12-MJ array are presented in Figure 5, Figure 6, and Figure 7, respectively. Furthermore, Table 2 provides summarized results for all nine cases.

The simulation results demonstrate important increases in target size for current testing capabilities as well as the potential for novel testing capabilities that would fill the spectral gap. Table 2 identifies the maximum target areas that maintain the design specifications to be 9,801 cm² and 46,785 cm² for the Cold-BIS simulations. These values represent roughly a two order of magnitude increase from the current testing capabilities.⁶ Additionally, the Warm-TSR and Warmer-SGEMP capabilities illuminate the potential to use XEC's KrF laser to simulate relevant radiation effects that are beyond current U.S. testing capabilities. XEC's KrF laser design makes this testing feat possible by generating a laser beam with sufficient power to stimulate relevant x-ray emissions from a low-density copper source.

The MCNP6 results demonstrate that the surface uniformity requirement is maintained for all nine test cases. Moreover, the maximum energy deposition is proportional to the maximum fluence across the surface of the test article; however, it is also pertinent to discuss the uniformity at depth as depicted in Figure 7. Uniformity at depth should be distinguished from surface uniformity as uniformity at depth reveals how the mean free path of the aggregate x-rays move through the target, while surface uniformity merely describes the total amount of x-rays that pass through the surface of the target. The 80% uniformity requirement applies exclusively to surface uniformity, but uniformity at depth may also be relevant due to

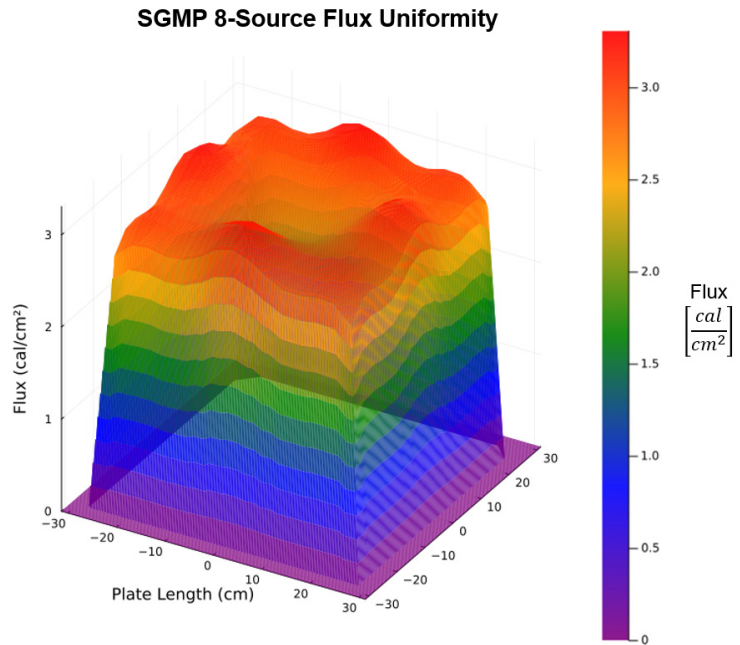


Figure 5: The results of x-ray uniformity across the test article for SGEMP with 8-sources and 12-MJ of incident energy.

Energy Deposition as a Function of Depth for SGEMP 8 Source (12 MJ) Case

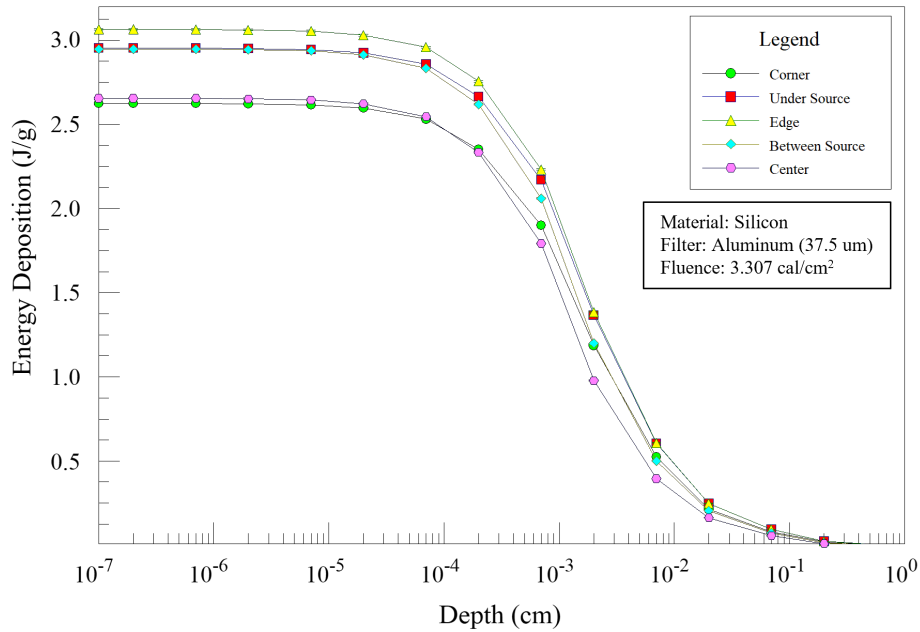


Figure 6: The results of energy deposition into the test article for SGEMP with 8-sources and 12-MJ of incident energy.

Energy Deposition Uniformity as a Function of Depth for SGEMP 8 Source (12 MJ) Case

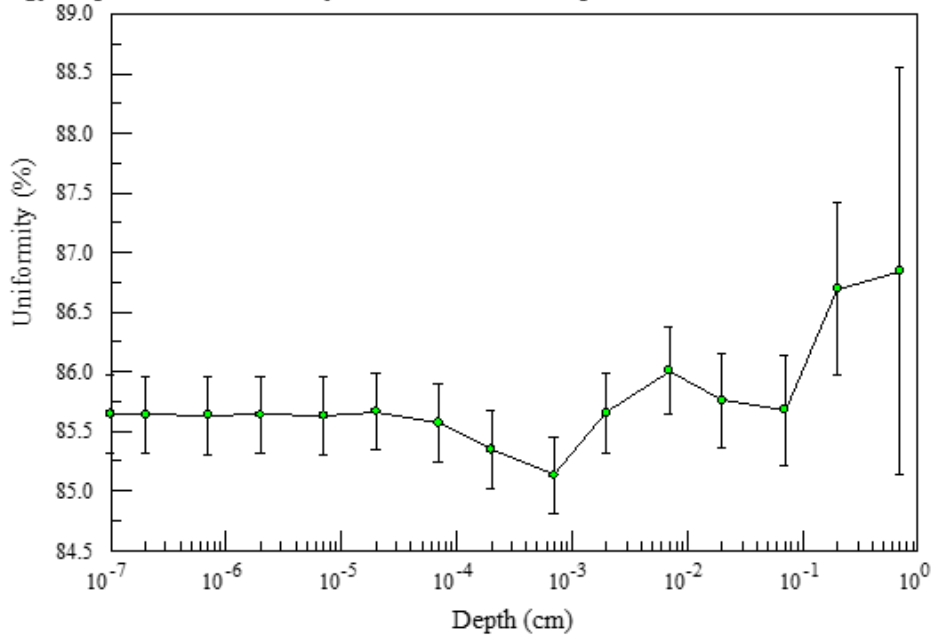


Figure 7: The results of energy deposition uniformity at depth for SGEMP with 8-sources and 12-MJ of incident energy.

the sharp incident angles of x-rays emitting from distinct point sources. From Figure 5 and Table 2, the SGEMP 8-source, 12-MJ array has a surface uniformity of 82.26%; however, Figure 7 reveals that there is greater uniformity at depth (closer to $\pm 85.6\%$).

The differences between surface uniformity and uniformity at depth are greatest for the 24-source BIS array. This is symptomatic of both the radiation effect being analyzed as well as the design strategy used to achieve 80% surface uniformity. SGEMP exhibits higher uniformity at depth given warmer x-rays, while BIS exhibits lower uniformity at depth. This difference is likely due to the angle of incident x-rays onto the larger BIS target becoming much more acute. X-rays that are incident at a sharp angle, rather than orthogonal to the target surface (like those from a planar source), penetrate the target less effectively and deposit more energy toward the surface with less energy deposition at depth. This contrast becomes starkly relevant at the edges and minimums of a given target because the edges and minimums generally have the most x-rays incident at acute angles. Furthermore, the larger the target, the more acute the incident angles become, causing greater variance in uniformity at depth for the 24-source arrays. From Table 2, the 24-source BIS array has the largest target area, and its uniformity at depth is only 70%. With increased areas, the effects of using isotropic sources become more pronounced. While surface uniformity may be met, not all x-rays may be depositing energy in a manner consistent with the nuclear weapon effects that are being simulated.

Ultimately, these results highlight the benefit of designing nine source arrays using low-density foam sources to simulate relevant radiation effects from a nuclear weapon. All nine designs meet the specifications requested by XEC, and each case greatly increases the testable target surface area relative to current radiation effects testing capabilities. These results demonstrate areas that are potentially one to two orders of magnitude higher than current testing capabilities. The results suggest using XEC's KrF laser for radiation effects would greatly increase U.S. capabilities in this area.

Conclusions

This research effectively designs nine x-ray source arrays to simulate nuclear weapons effects through stimulated x-ray emission. The designs are guided by strict requirements to maintain sufficient target surface uniformity with a desired influence. The successful

design of each array identifies the fidelity in utilizing XEC's KrF laser to greatly bolster current radiation effects testing, while expanding U.S. testing capabilities to become more comprehensive. These results demonstrate a first look into the usefulness of this novel laser design for NWET; however, many questions lie ahead before these simulation capabilities can be used to their fullest potential. In future research, target dimensions should be more complex to replicate systems that may be of interest for radiation effects testing. Complex designs will require significant modifications and add complexity to the optimization of area, as that process was completed manually for the square plate. Future studies may be able to utilize AI to analyze the optimal orientation and optimal number of sources to achieve sufficient fluence across a given target design. Additionally, BIS energy deposition profiles may be used in multi-physics codes such as ALE3D to investigate impulse compared to nuclear threat spectra results to further understand test fidelity. Ultimately, these source array designs operate as a powerful proof of concept to highlight the potential for XEC's KrF laser in expanding U.S. NWET capabilities. ■

Acknowledgments

This research would not have been possible without Xcimer Energy Corporation, specifically the guidance and mentorship of Mr. Michael Tobin and Dr. John Davis. Mr. Tobin and Dr. Davis provided invaluable insight to the methodology and operation of XEC's KrF laser design. Furthermore, their communication highlighted the expectations and design specifications for the arrays designed throughout this research. Additionally, Lt. Col. Peter Nelson and Dr. Kenneth Allen from USMA provided guidance and leadership to generate credible results and establish effective project management skills to facilitate this research.

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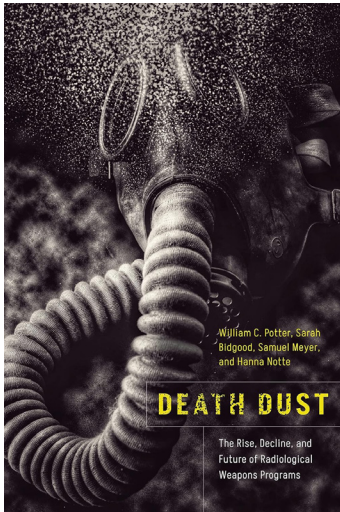
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Book Review

Death Dust: The Rise, Decline, and Future of Radiological Weapons Programs

William C. Potter, Sarah Bidgood, Samuel Meyer, and
Hannah Notte,
Stanford University Press, 2023

Reviewed by: Maj. Parker Stevens

The United States (US) Army has made major efforts to address Weapon of Mass Destruction (WMD) use, with investments towards upgrading capabilities for responding to Chemical, Biological, Nuclear, and Radiological (CBRN) threats, chemical use featuring prominently in exercises, and releasing new strategies for operating in nuclear and biological weapon-affected battlefields. What about the letter 'N' in CBRN? Despite worries over what terrorists could do with a Radiological Weapon (RW), states have not employed them on the battlefield—why? In *Death Dust: The Rise, Decline, and Future of Radiological Weapons Programs*, authors William C. Potter, Sarah Bidgood, Samuel Meyer, and Hannah Notte address these questions by walking through the historic state RW programs, discussing how they started, why they stopped, and what this portends for the future.

The paradox the authors examine is, with the supposed ease of making RWs, why have states with actual industrial bases not employed them? They highlight other questions that follow: Are they a thing of the past? Is the focus on use by terror groups misplaced? Are new tools like updated arms control instruments needed to limit future dangers? The authors address these questions with a comparative case study analysis of known state programs, using a common matrix of questions informed by a general model of military technological innovation. The authors define RWs as weapons designed to disperse radioactive material in the absence of a nuclear detonation, excluding nuclear weapons with enhanced payloads, they did not include nonstate actor attempts.

The body of *Death Dust* consists of the country case studies, emphasizing the main factors in starting and stopping the RW programs. Of these, the US and Soviet programs get the most coverage as they made the greatest progress and are used as the archetypes of contrasting approaches to military innovation. Starting shortly after World War II, the US program embodied a “bottom up” approach, with individual scientists and military personnel advocating for RW development. This devoted but narrow base of support only kept the program running until the late 1950s. Results of early weapons tests were promising, but RWs never enjoyed the broad, cross-service support of nuclear weapons and did not seem to justify their cost, especially with the absence of intelligence on a similar Soviet threat.

Meanwhile, the Soviet Union began developing RWs shortly after starting their nuclear program, a decision largely driven by intelligence on US RW efforts. The Soviet Union used a “top down” approach, with the senior military leadership pushing for rapid weaponization and live weapons reaching testing by the mid-1950s. The Soviets continued their program for longer than the US, but ultimately interest faded for similar reasons. Even with very different decision-making approaches, in both cases bureaucratic reasons won over technical factors. Testing showed there certainly were significant problems in employing RWs such as radiation affecting the delivery system, or managing munitions that decayed away in storage, but the real deciding factor was the greater pull of nuclear advocates.

The coverage of “minor” programs varies widely by available evidence. Postwar Britain only conducted academic studies throughout the 1950s, constrained by lack of funds. Never developing a deep bench of advocates, even this modest effort was cut for more focus on the nuclear program. Evidence on the Egyptian program is limited but points to concerns over Israel’s rumored nuclear developments and the desire for prestige as a “nuclear” power. It did not pass the exploratory phase, encountering severe difficulties producing radiological material and developing a suitable delivery system. The final program covered is Iraq. Alongside the more publicized WMD programs, the country managed to test live RWs by the late 1980s. Strong senior leader interest came from a search for advantages as the Iran-Iraq War dragged on and the lack of international reaction to chemical weapon use; however, their interest faded as costs increased and chemical weapons proved their value. The chapter about Iraq also highlights the difficulty that United Nations inspectors, prepared for other WMD programs, had in collecting evidence of RW development. Overall, the minor program chapters illustrate how top-down decision-making and associated lack of feedback can allow scientifically dubious schemes to survive and thrive.

The book concludes with a review of factors seen across case studies, the outlook for future RW development, and what may change with the war in Ukraine. The authors observe that while there are common traits among the known programs, there are no universal factors that explain why they began or ended. Most were conducted in great secrecy, so previous efforts rarely informed later decisions.

The outlook for the future is murky. RWs have severe technical limitations, but as arms control instruments tend to focus on more common WMDs they may offer a way around such measures. While the authors identify no actual wartime use of RWs, they point out that RWs are back in the news with Russia accusing Ukraine of developing them before the current invasion. The authors identify the specific Russian allegations as spurious at best and showing a surprising lack of technical merit, but this elevated profile may reinvigorate interest as an option for nations who would have difficulty acquiring nuclear weapons.

Overall, the approach of using common questions presented in an overarching innovation framework provides a well-organized decision-making study of novel weapons development programs, but those

expecting a more technical treatise may be disappointed. There is no dedicated background discussion on how RWs function, and what factors into potential military effectiveness is only brought up as a detail in the vignettes. This makes sense with how technical facts usually took a backseat to bureaucratic imperatives in program decisions, but providing some background on the technical details and the very real problems in employing RWs could have helped better define where these diverged. Finally, the conclusion thoroughly discusses situational factors likely to affect future RW development but may have benefited from also examining changing technical factors such as increasing proliferation of radioactive material for medicinal, industrial, and power applications.

I would recommend *Death Dust* for readers specifically studying RWs as it provides a consolidated background resource in an area where discussion is focused on nonstate actors. I would also recommend it to those looking at state WMD pathways who may be interested in learning about how internal dynamics can interact with external threats and scientific reality. Those involved with conducting inspections may find some interesting vignettes on being prepared for the unexpected. While this book was published in the midst of Russia’s invasion of Ukraine, just as with nuclear proliferation it remains to be seen if the concerns raised over loopholes in the international nonproliferation regime will prove prescient. ■

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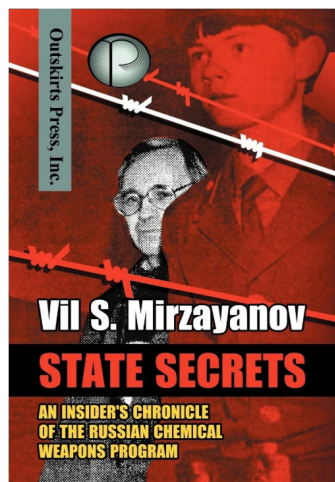
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Book Review

State Secrets: An Insider's Chronicle of the Russian Chemical Weapons Program

Vil Sultanovich Mirzayanov,
Outskirts Press, 2008

Reviewed by: Maj. Pucheng Ke, Lt. Col. Marisol S. Castaneto, Dr.
Ralph A. Stidham, Maj. Gregory D. Reynolds,
and Capt. Hieu T. Dinh

*State Secrets: An Insider's Chronicle of the Russian Chemical Weapons Program*¹ is Vil Mirzayanov's personal account of his experiences from being a country boy in the rural Tatar region in the Union of Soviet Socialist Republics (USSR) to a whistleblower to the Soviet's most advanced and covert chemical warfare nerve agent program. Mirzayanov, a member of the Turkic ethnic minority group in the Soviet Union, overcame systemic social discrimination due to his language barrier and the complicated relationship between Russians and Tartars over hundreds of years.

Upon graduating from high school and turning down a job offer with a unit at the local Committee for State Security, otherwise known as the KGB, Mirzayanov took a crowded train alone and arrived in Moscow on July 26, 1953, to explore opportunities for higher education. After homeless nights in Moscow, he was desperate to find a college that could immediately offer him shelter and a scholarship. As a result, he enrolled in the Department of Organic Synthesis with a focus on artificial gases and fuels at the Lomonosov Institute of Fine Chemical Technology. The Lomonosov Institute of Fine Chemical Technology is also known as MITKhT. Unlike the higher education system in the West, the USSR assigned college graduates to different research institutes and manufacturing plants based on the country's needs. In 1958, before he defended his undergraduate thesis, Mirzayanov received orders to the Institute of Organosilicon Compounds. At the time it was a secret plant producing boranes. Boranes are important chemicals for potential military use. They are made from boron and hydrogen and could be used as powerful rocket fuel. The Soviet military assigned the Institute of Organosilicon Compounds a

codename, "P.O. Box 4019," similar to how the United States (US) used "Project X" for the Manhattan Project.

However, two years of dangerous and labor-demanding work at the P.O. Box 4019 quickly eroded Mirzayanov's idealistic and romantic dream of unconditionally serving the Soviet motherland. He escaped out of the Box with a foMITrged document and managed to enroll in the Institute of Petrochemical Synthesis of the Academy of Sciences of the USSR as a graduate student in 1960, where he developed his lifetime passion in the research of chromatographic analysis. However, he was not able to earn his Ph.D. degree until more than twenty years later, because his study was soon interrupted by the financial crisis associated with his divorce. Without other choices, he had to suspend his academic research program and joined the State Research Institute of Organic Chemistry and Technology (GosNIIOKhT) in 1965. His main task was to develop gas chromatography testing methods to determine the concentrations of chemical warfare agents in different media. This officially marked the starting point of his 26-year scientific career in the Soviet chemical weapons development and testing program.

Under extreme secrecy, GosNIIOKhT was the umbrella organization for several satellite chemical weapons production facilities and testing sites. It conducted research and development activities with a wide range of chemicals with existing and potential military applications, from less toxic common riot control agents, CS and CR to highly lethal G- and V-series nerve agents. Later, GosNIIOKhT synthesized the Soviet analog of VX, code name Substance-33 (also known in the West as Russian VX), and turned it into

full scale production. In addition, Soviet chemists became interested in developing more advanced binary chemical weapons. According to the US Army Chemical Materials Activity, binary chemical weapons contain two non-lethal chemicals that mix together upon launch or in flight to form a lethal chemical agent. GosNIIOKhT later successfully developed a Russian VX-based binary chemical weapon rocket as the Soviet military's first approved binary chemical weapon system in 1990, arguably inspired by the FBI's Operation Shocker.² Meanwhile, GosNIIOKhT continued to conduct research and even publish results on legitimate organophosphate pesticides in open scientific journals to conceal its secret mission of chemical weapons development, very similar to the practice already adopted by the Soviet biological weapons program.

GosNIIOKhT invented the 4th generation nerve agents (also known in the West as Novichok agents and somehow mistakenly as A-series agents because several known A-series agents are not Novichok agents) which served as an exciting milestone for the Soviet military complex. In the early 1970s, scientists at GosNIIOKhT attempted to enhance the lethality of traditional German and British developed organophosphate-based nerve agents, e.g., sarin, soman, and VX. By replacing the isopropyl ether group on the phosphorus atom with different side chains, they discovered a nitrogen analog of sarin (Soviet code name: agent A-230) that was 5-8 times more toxic than Russian VX. Encouraged by the progress, GosNIIOKhT chemists later conducted more diverse functional group modification and synthesized a group of A-230 analogs (A-232, A-234, A-242, A-262, etc., all considered as Novichok agents). In comparison to the traditional G- and V-series nerve agents, Novichok agents exhibited either ultra toxicity or significantly improved resistance to extreme weather conditions commonly encountered in the field. More importantly, the synthesized group of A-230 analogs were not covered by the Chemical Weapon Convention (CWC) at the time and were easily hidden from the CWC inspectors due to the feature that some of them share the same moieties with legitimate commercial organophosphate pesticides.

The Soviet military conducted extensive field testing of Novichok agents containing munitions throughout the 1980s and Mirzayanov's team actively contributed by examining the effective ranges and penetrating abilities of Novichok agents by providing real-time gas chromatography analyses of agent concentrations in a broad range of natural and biological matrices. With the positive experimental results, A-230 was officially approved by the Soviet government to be

added in its chemical weapons stockpile in 1990 and other Novichok agents were included shortly after. After the USSR collapsed in December 1991, the Russian Federation carried on the USSR's secret binary weapons project and successfully developed an A-232-based binary chemical weapon system (Novichok-5), despite the fact that the Wyoming Memorandum of Understanding signed by the Soviet government in 1989 was still in effect at the time.

Mirzayanov's later career at GosNIIOKhT was to serve as the head of the Foreign Technical Counterintelligence Department. The purpose of establishing this separate sector was to keep the Soviet chemical weapons development program secret by eliminating any trace of chemical weapons research and production that could potentially be discovered outside GosNIIOKhT by foreign intelligence agents. GosNIIOKhT naturally relied on Mirzayanov's specialty to establish and maintain a chemical analysis and monitoring program to ensure the concentrations of the chemical warfare agents that escaped out of the laboratory buildings remained below the limits of detection by Western intelligence.

The urgency of the work drastically increased in 1989 when the Soviet government finally agreed to allow American inspectors to visit several industrial plants that were suspected by the West of involvement in chemical weapons production. GosNIIOKhT capitalized on the urgent status and convinced the Soviet government to provide a large amount of hard currency for purchasing Western-made advanced analytical instruments. Ironically, most of the instruments were later used for further Soviet and Russian chemical weapons development.

In Chapters 6 through 12 of *State Secrets*, Mirzayanov shifted from a personal narrative to a more technical, yet equally gripping, exposé of the Soviet chemical weapons program. While challenging for the average reader, the level of reading in this section is appropriate for personnel who work in chemical defense. The text offers a chilling glimpse into the scientific heart of a morally bankrupt system. With his chemist's background and expertise, Mirzayanov detailed the properties of various nerve agents, including the terrifying potency Novichok agents. He did not shy away from chemical formulas and technical jargon, making these chapters dense but essential reading for those seeking to comprehend the deadly ingenuity behind these weapons.

Beyond the science, Mirzayanov revealed the shockingly casual attitude toward safety and environmental concerns within the Soviet chemical weapons program. He described a system where production quotas and secrecy trumped human life and ecological well-being. This section demanded close attention and provided a sobering and essential understanding of the real-world consequences of unchecked scientific ambition. Mirzayanov began questioning the legality and necessity of chemical weapons in 1982, the year that coincides with the period when his team was deeply involved in the development of Novichok agents. His firsthand experience with these highly toxic substances, combined with the evidence of contamination he observed, likely played a significant role in his decision to eventually expose the program. Shortly after the collapse of the USSR, he started revealing the secret Soviet chemical weapons program and confirmed the existence of Novichok agents to Russian and foreign news media including *Moscow News* and *Baltimore Sun*. As a result, he was arrested by the KGB on October 22, 1992, assumedly for disclosing the state secrets. Without knowing the exact charges, he was detained at the notorious Lefortovo Prison for several months and later periodically harassed by interrogations organized by the Federal Security Service of the Russian Federation until his case was dismissed in March 1994 due to lack of adequate evidence. Mirzayanov later moved to the United States and continued to condemn the Russian chemical weapons program.

Good: The author offered highly valuable insights into many previously secret or little-known Soviet chemical weapons research, development, and production activities at GosNIIOKhT from the 1960s to the early 1990s. Mirzayanov's account revealed the three top priority tasks of the Soviet chemical weapons program: (1) mass production of traditional organophosphate nerve agents and their more lethal analogs, e.g., sarin, soman, and Russian VX; (2) development, production, and fielding of ultra-lethal nerve agents that can survive harsh field environment, defeat the North Atlantic Treaty Organization personnel protection equipment, and remain undetectable by international inspections performed under the CWC, e.g., Novichok agents; and (3) creation of highly mobile binary nerve agent weapon systems, e.g., munitions that can carry less toxic chemical precursor agents inside and rapidly generate Russian VX and A-232 upon launch.

In the book, Mirzayanov disclosed the detailed chemical structures of five Novichok agents. Although the structures for A-230, A-232, and A-234 were later

challenged by Hoenig and Ellison, which led to once extensive debates on whose version was correct, to date, Mirzayanov's description has received far more recognition based on the information published in the Russian internal investigation reports indirectly verified by the West.³ Mirzayanov also explains the different physical and toxicological properties of military significance among Novichok agents, including that A-230 easily freezes at temperatures below -10°C; A-232 is more volatile than A-230, the solid version of A-232; and A-262 is remarkably more lethal than A-232.

The book described how chemists at GosNIIOKhT manipulated the physical properties and biological activities of the nerve agents by deliberately modifying functional groups on the phosphorous center or side chains. It also provided the details on how the Soviet military took advantage of more sophisticated Western analytical instruments for testing the penetrating abilities of nerve agents against personnel protective gear and facilitating the development of binary weapons. In addition, Mirzayanov described extremely poor safety practices inside the Soviet military chemical complex including noncompliance with chemical hygiene standards. Mirzayanov discussed how the USSR deliberately transferred technology and instruments for chemical weapons production to its allies under the pseudo environment monitoring and protection projects since the 1970s. He claimed that the USSR delivered chemical weapons to the Mideast in the 1980s and Iraq most likely acquired Russian VX production technology directly from the USSR. Although not confirmed by US officials, the author suggested that US military and intelligence might have overlooked the presence of Russian VX in the Iraqi chemical weapons stockpile during the Gulf War, because Russian VX was not on the United Nations testing panel at the time. Since the Russian Federation directly inherited the chemical weapons doctrine and practice from the Soviet era without much change, the information in the book could be useful when the United States establishes strategies and policies to prevent the future proliferation of chemical weapons from Russia to other countries or terrorist organizations. The book also described the unnecessarily redundant and disorganized security measures and highly politically influenced personnel management in the Soviet military chemical complex. This could offer a unique opportunity for the West to utilize information warfare to deliberately create obstacles for Russia's illegal activities in chemical weapons development and production.

Mirzayanov discussed delayed symptoms associated with long-time exposure to nerve agents at low

concentration levels, which may contribute to Gulf War Syndrome. He offered a novel opinion that did not draw adequate attention from toxicologists or medical professionals in the West at the time when the book was published. The annexes of the book also presented many detailed Russian court documents that either quoted or cited information on the Soviet chemical weapons development program from official and classified Soviet records.

Bad: Out of the total 23 chapters in the book, only seven chapters (Chapters 6-12) are directly related to the Soviet chemical weapons program. The rest are merely lengthy and redundant descriptions of the author's trial and unfair treatment, government corruption, and political power struggles he had encountered in the USSR and Russia. It seems that better editing would make the book much more readable. Some parts of the book are particularly hard to follow for the readers who do not possess the knowledge of how Russian first, patronymic, and last names are used by Russian native speakers. Without explanation notes, the reader would easily become confused about whether the three different names the author just mentioned correspond to three distinct persons or the same one. In addition, it seems that the translation of chemistry terminologies was provided by poor-quality online translation software. Neither the software designer nor the translator had sufficient knowledge of organic chemistry vocabulary in English. Therefore, the resulting product is often inaccurate and confusing. The book has very limited discussion on the chemical synthesis of the novel nerve agents or how the Soviet scientists conducted structure-activity studies for designing new agents or modifying existing agents. This may be because either the relevant information is unavailable in the West or the author's specialty was analytical chemistry, not organic synthesis.

Overall: In *State Secrets*, Vil Mirzayanov, a former Soviet chemist, takes readers inside the secretive and deadly world of the Soviet Union and Russia's chemical weapons programs. More than just a memoir, the book recounts Mirzayanov's journey from an idealistic scientist to a disillusioned whistleblower as he witnesses firsthand the devastating environmental and human costs of developing nerve agents like Novichoks.

Mirzayanov's account is both a personal story of courage and a damning indictment of government secrecy and scientific complicity in creating weapons of mass destruction. The book offers a rare glimpse into a shadowy world, exposing the dangers of unchecked power and the moral dilemmas faced

by those who dare to speak truth to it. This chilling exposé, especially chapters 6-12, is also a must-read for anyone interested in Cold War history, arms control, or the ethical responsibilities of scientists. ■

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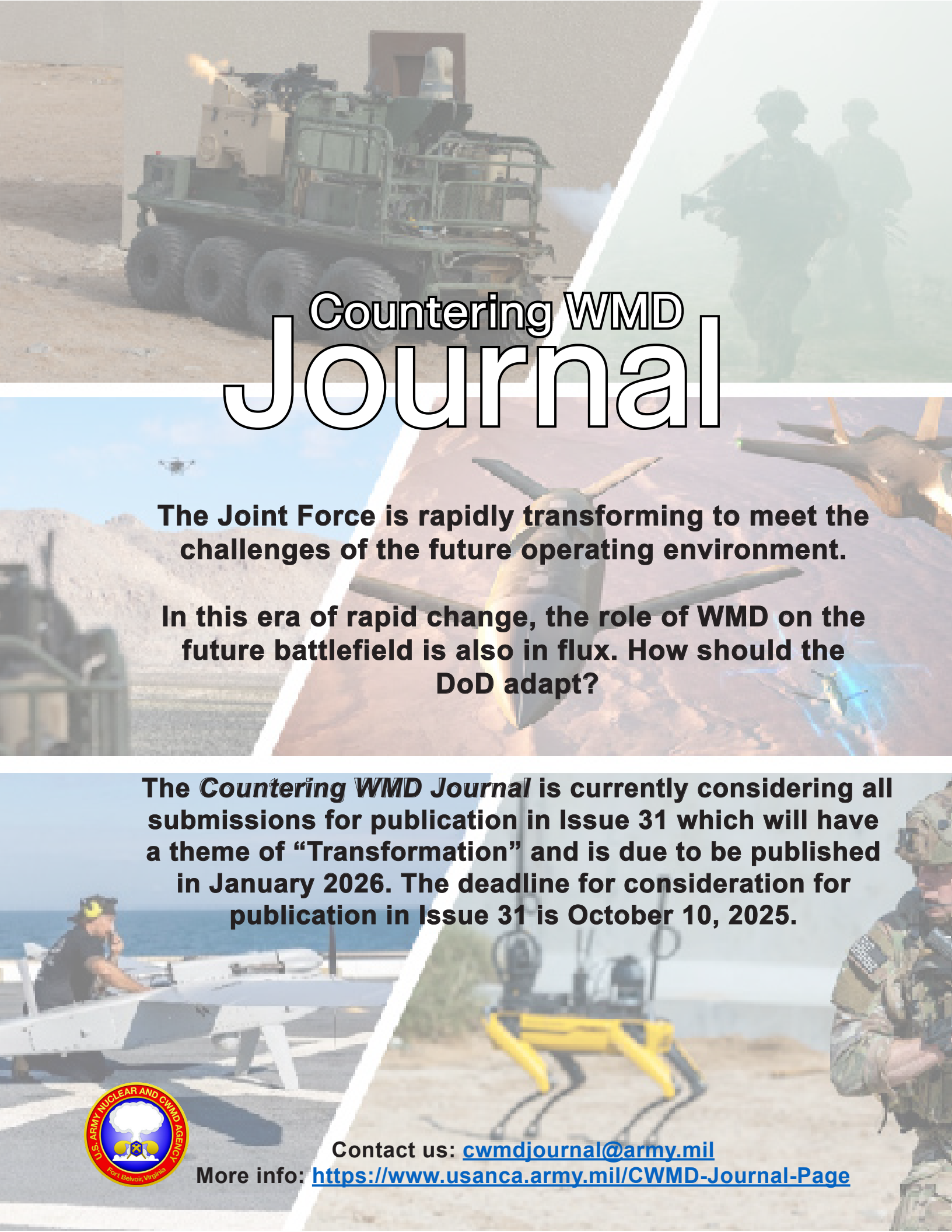
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Countering WMD Journal

The Joint Force is rapidly transforming to meet the challenges of the future operating environment.

In this era of rapid change, the role of WMD on the future battlefield is also in flux. How should the DoD adapt?

The *Countering WMD Journal* is currently considering all submissions for publication in Issue 31 which will have a theme of “Transformation” and is due to be published in January 2026. The deadline for consideration for publication in Issue 31 is October 10, 2025.



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